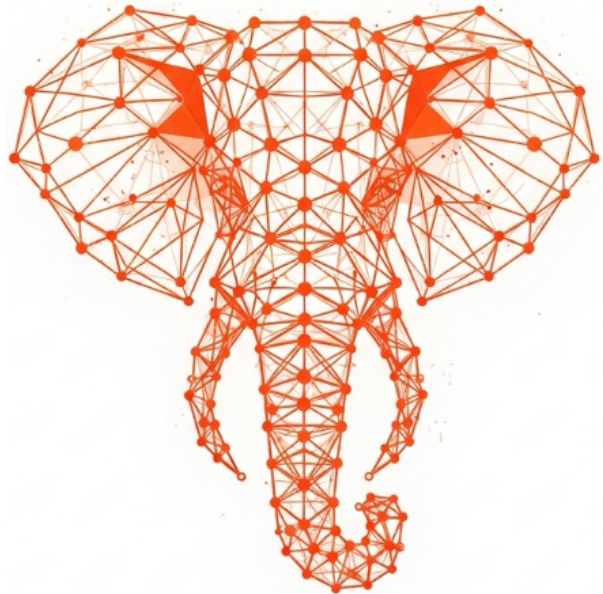




Uni
ct **DEEP LEARNING**
ADVANCED MODELS AND METHODS

Deep Learning

Advanced Models & Methods

Prof. Antonino Furnari (antonino.furnari@unict.it)

Corso di Laurea Magistrale in Informatica

Dip. di Matematica e Informatica

Università di Catania

Self-Supervised Learning

Problem: Supervised Learning is Expensive!

Assume you want to label 1M images. How much will it cost?

(1,000,000 images)

× (10 seconds/image)

× (1/3600 hours/second)

× (\$15 / hour)

= \$41,667

(Small to medium sized dataset)

(Fast annotation)

(Low wage paid to annotator)

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

Problem: Supervised Learning is Expensive!

Assume you want to label **1B** images. How much will it cost?

(1,000,000,000 images)

(Large dataset)

× (10 seconds/image)

(Fast annotation)

× (1/3600 hours/second)

× (\$15 / hour)

(Low wage paid to annotator)

= **\$41,666,667**

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

Problem: Supervised Learning is Not How We Learn

Babies don't get supervision for everything they see!



[Baby image](#) is [CC0 public domain](#)

Solution: Self-Supervised Learning

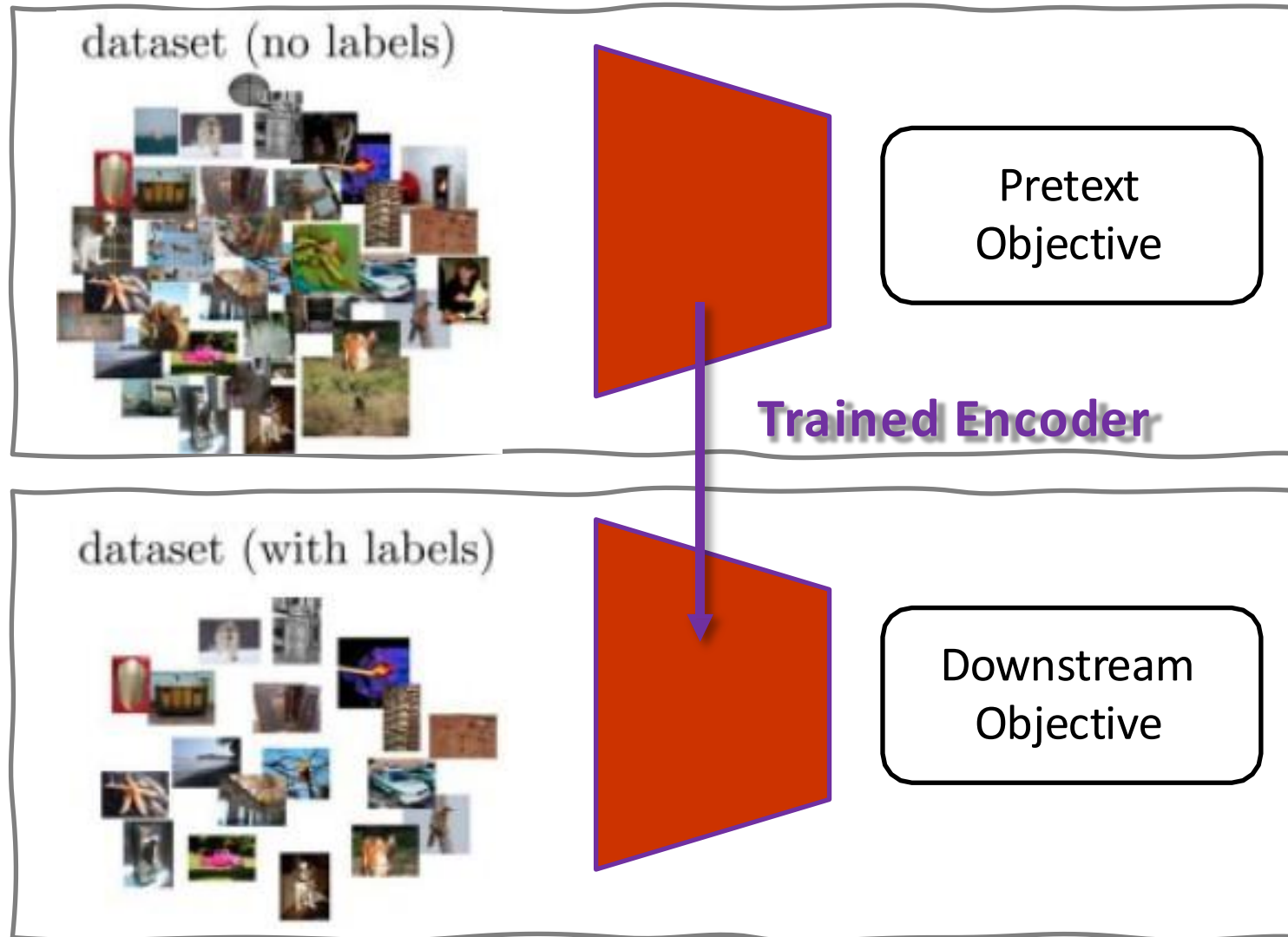
Lets build methods that learn from "raw" data – no annotations required

Unsupervised Learning: Model isn't told what to predict. Older terminology, not used as much today.

Self-Supervised Learning: Model is trained to predict some naturally-occurring signal in the raw data rather than human annotations.

Semi-Supervised Learning: Train jointly with some labeled data and (a lot) of unlabeled data.

Self-Supervised Learning



Pretext Task

- Define a task based on the data itself
- No manual annotation
- Could be considered an **unsupervised** task;
- but we learn with supervised learning objectives, e.g., classification or regression.

Downstream Task

- The application you care about
- You do not have large datasets
- The dataset is labeled

Self-supervised pretext tasks

A pretext task is a supervised task that uses automatically generated labels from raw data to learn useful representations — without requiring manual annotation.

Example: learn to predict image transformations / complete corrupted images

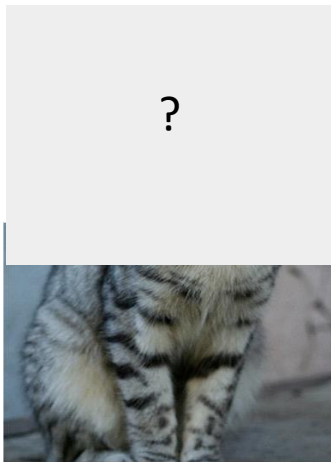
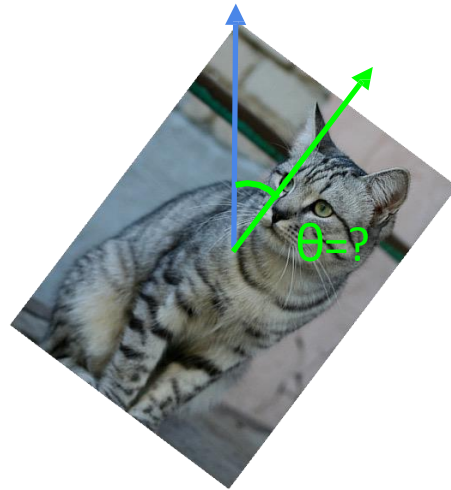


image completion



rotation prediction



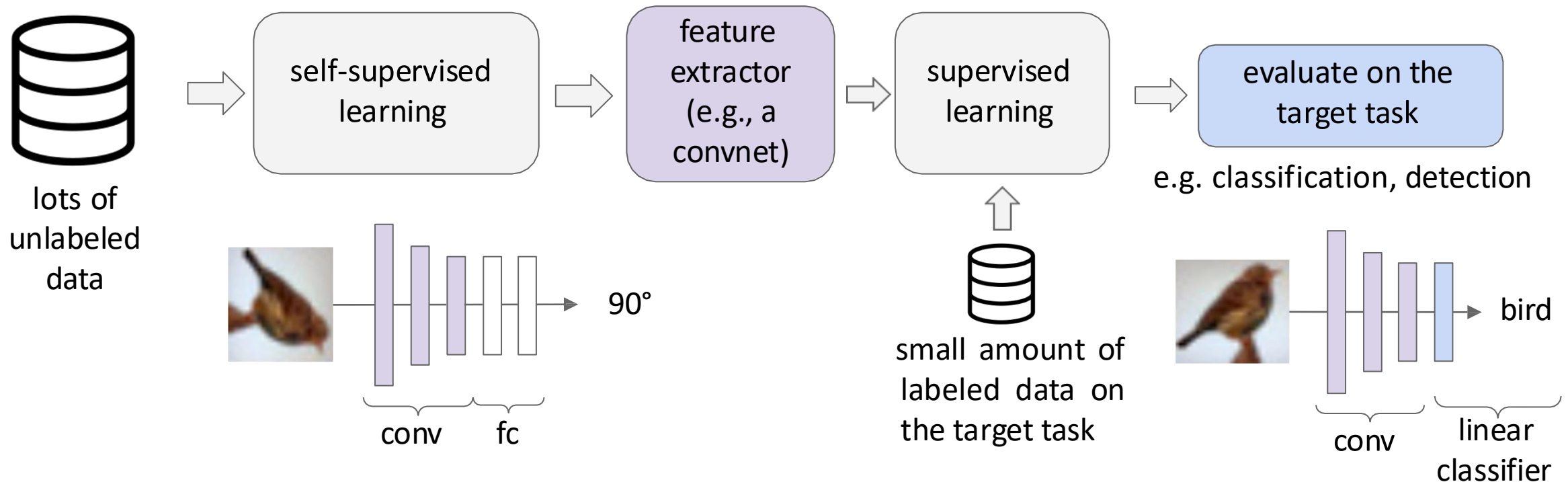
jigsaw puzzle



colorization

1. Solving the pretext tasks allows the model to learn good features.
2. We can automatically generate labels for the pretext tasks.

How to evaluate a self-supervised learning method?

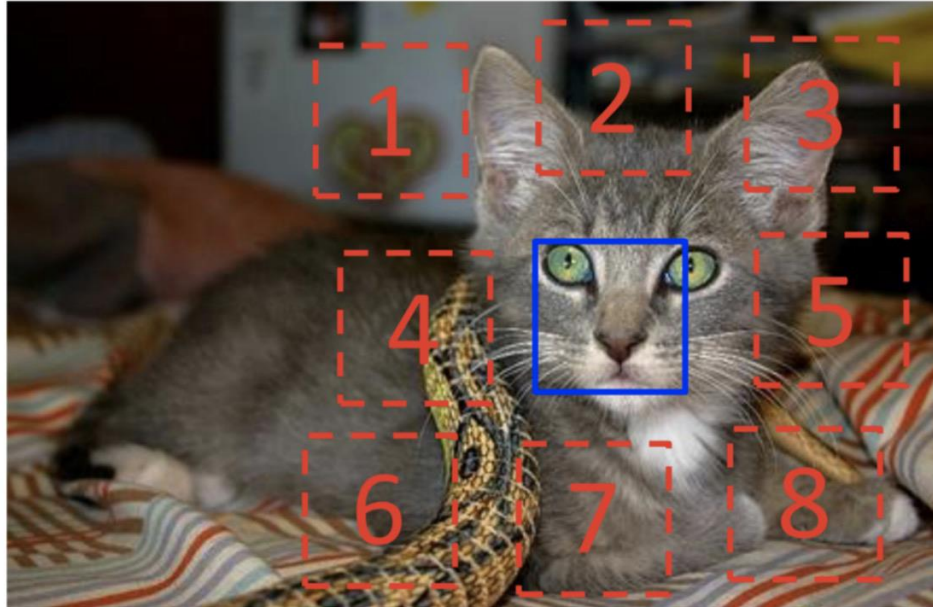


1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations

2. Attach a shallow network on the feature extractor; train the shallow network on the target task with small amount of labeled data

Pre-text Tasks

Pretext task: predict relative patch locations



$$X = (\text{patch 4}, \text{patch 5}); Y = 3$$

Example:



Question 1:



Question 2:



(Image source: [Doersch et al., 2015](https://arxiv.org/abs/2003.03931))

Classification over 8 positions

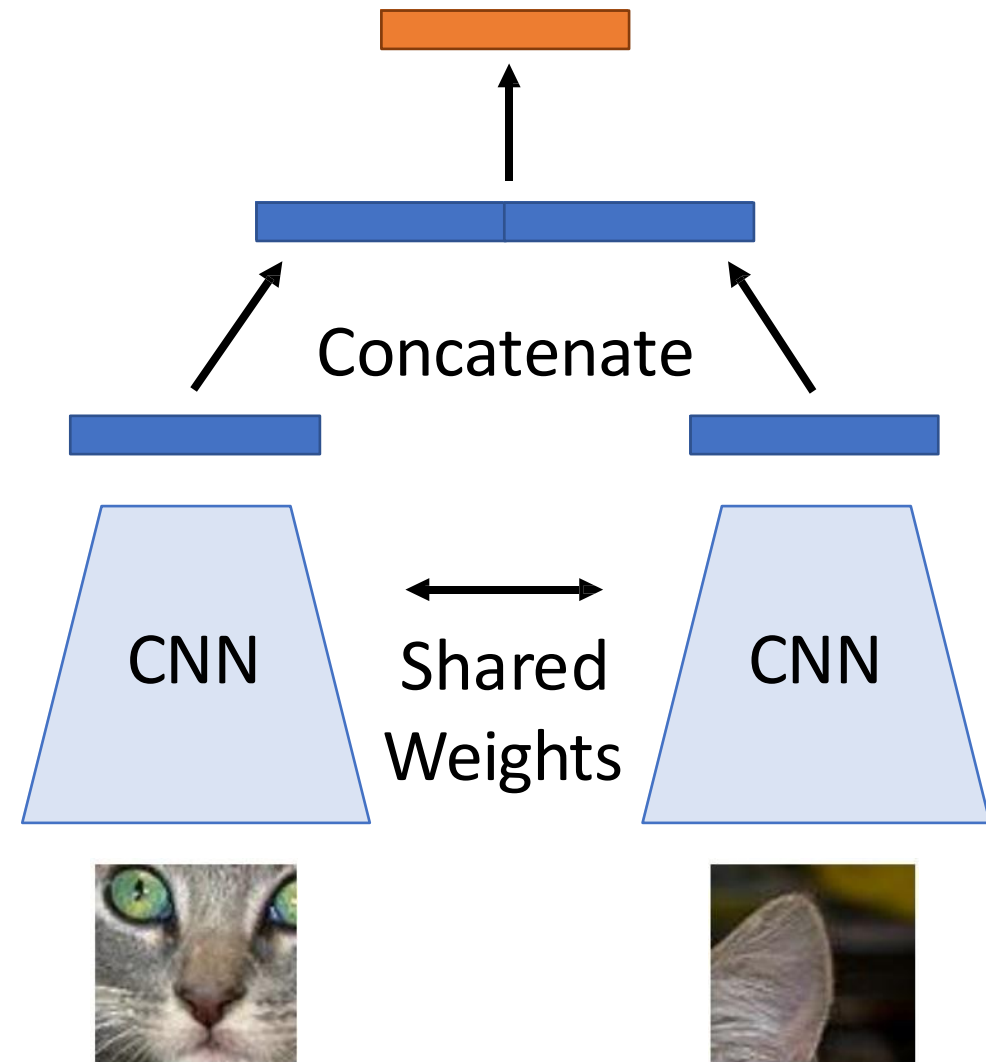
Context Prediction

Model predicts relative location of two patches from the same image.

Discriminative pretraining task

Intuition: Requires understanding objects and their parts

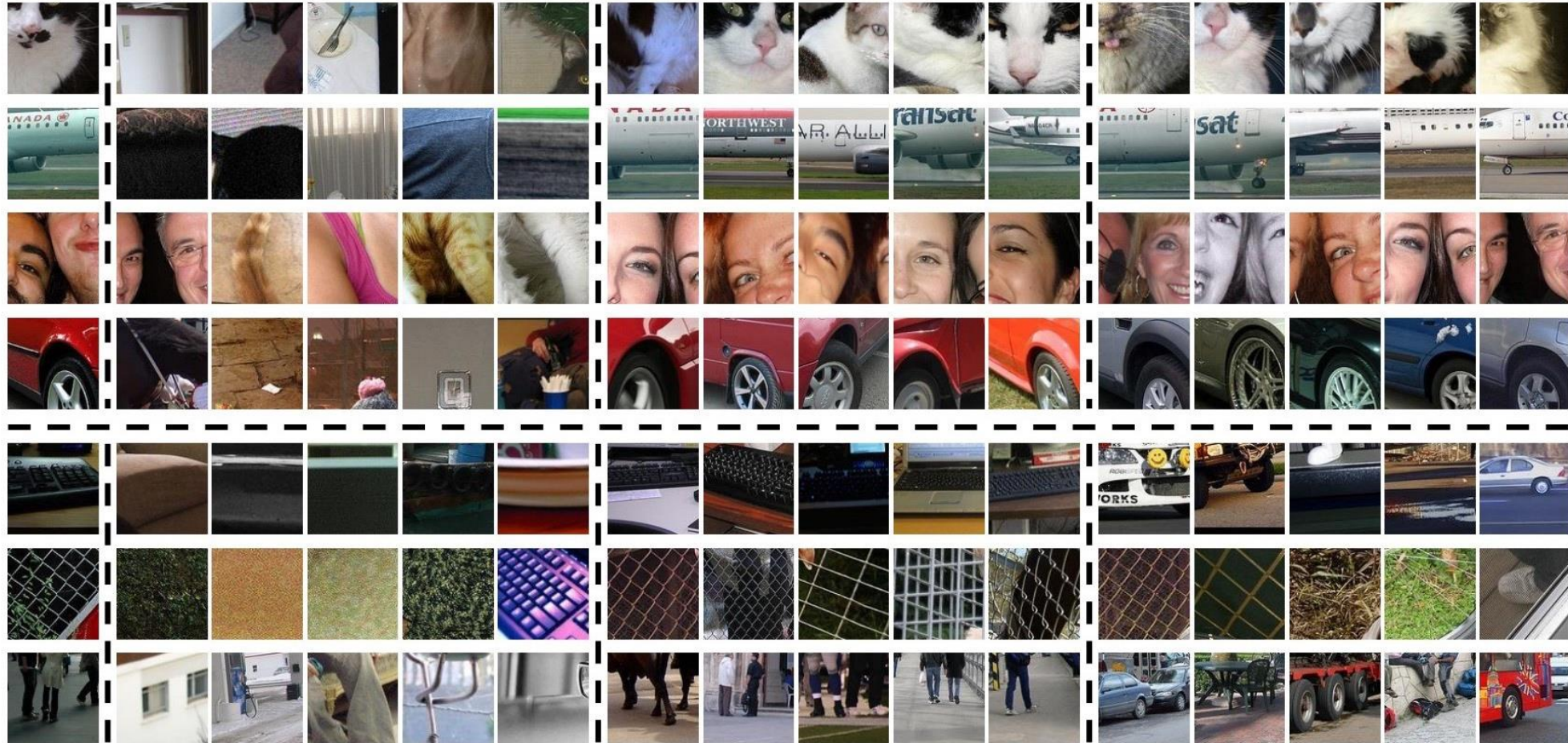
Two networks with shared weights is often called a "Siamese network"



Doersch et al, "Unsupervised Visual Representation Learning by Context Prediction", ICCV 2015

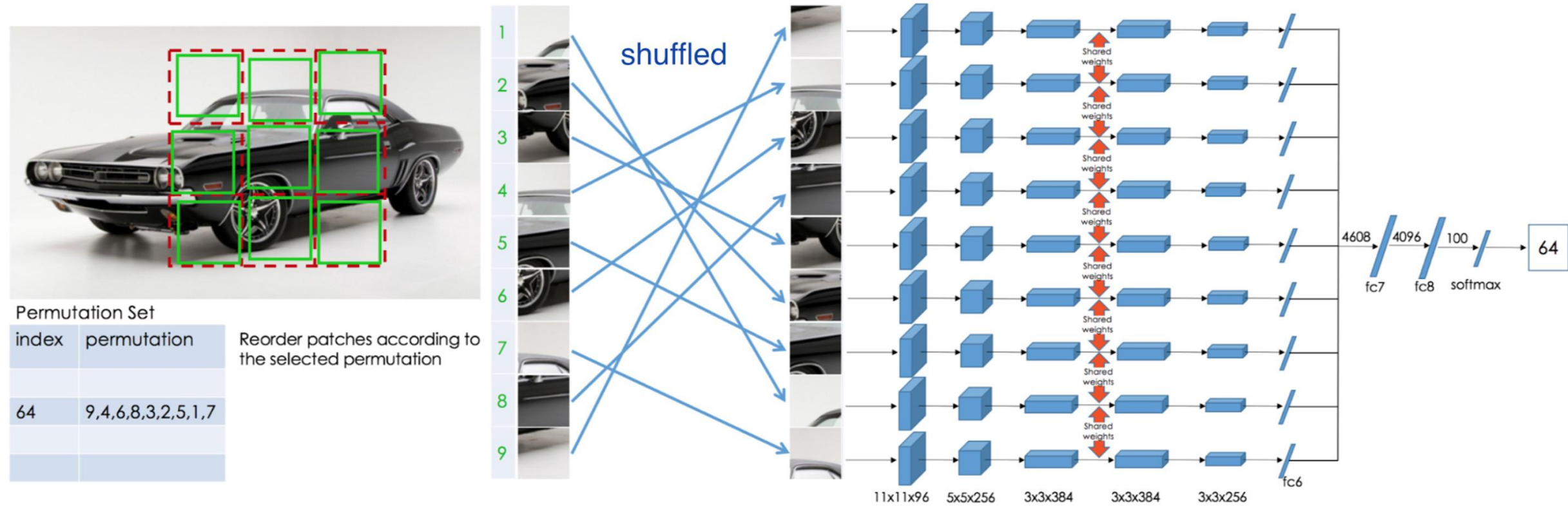
Context Prediction: Nearest Neighbors in Feature Space

Input Patch Random Init Supervised AlexNet Their Features



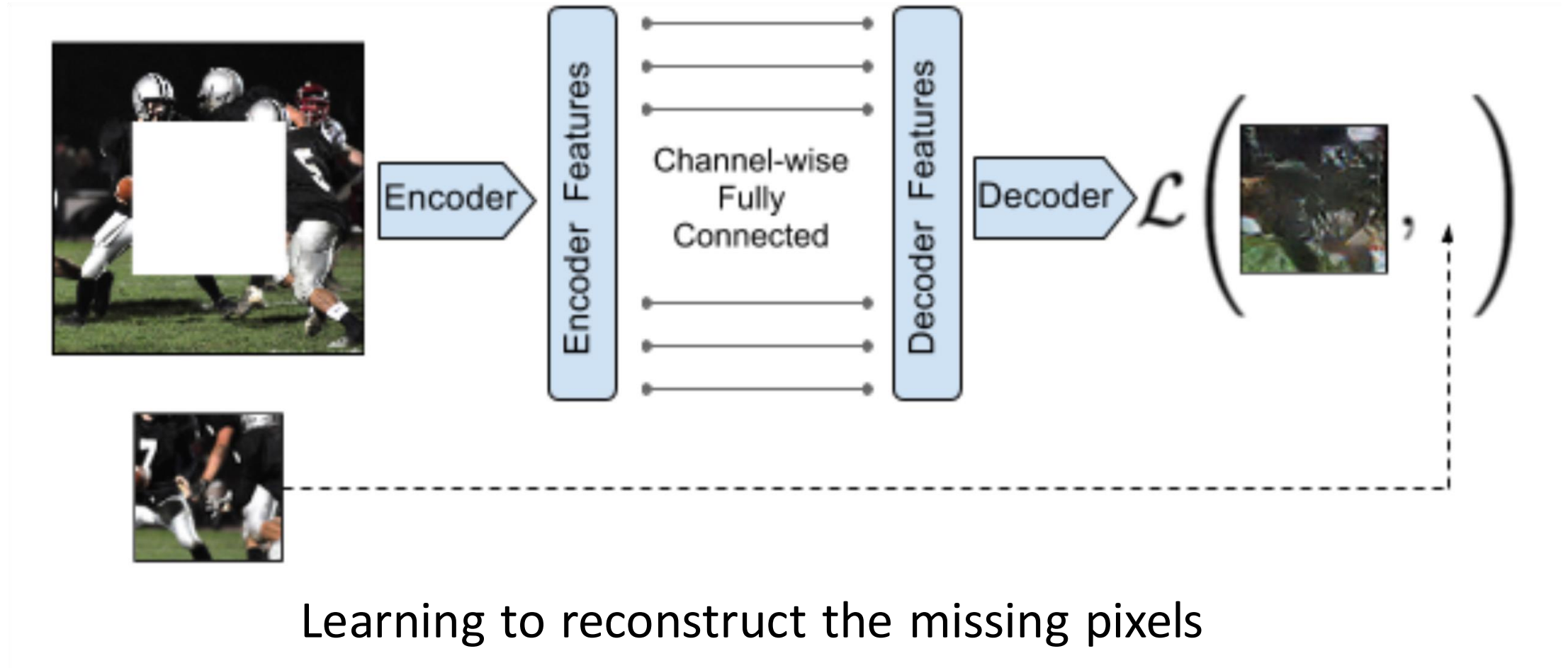
Doersch et al, "Unsupervised Visual Representation Learning by Context Prediction", ICCV 2015

Pretext task: solving “jigsaw puzzles”



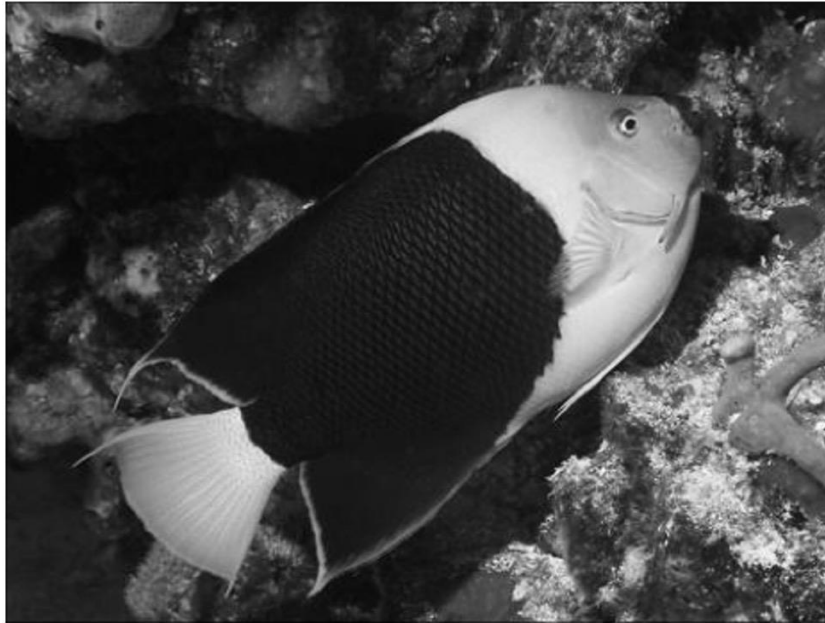
(Image source: [Noroozi & Favaro, 2016](#))

Learning to inpaint by reconstruction



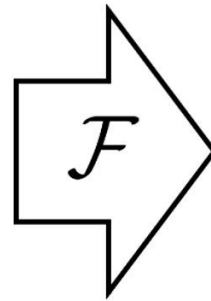
Source: [Pathak et al., 2016](#)

Pretext task: image coloring



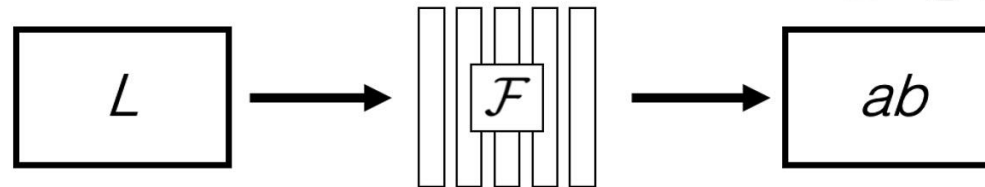
Grayscale image: L channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Color information: ab channels

$$\hat{\mathbf{Y}} \in \mathbb{R}^{H \times W \times 2}$$



Source: Richard Zhang / Phillip Isola

Pretext task: image coloring



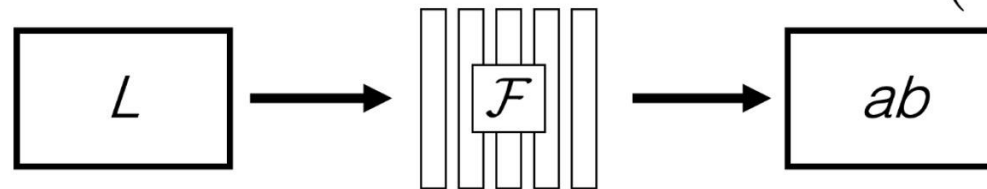
Grayscale image: L channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Concatenate (L, ab) channels

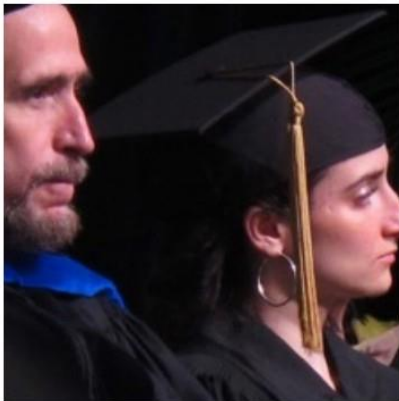
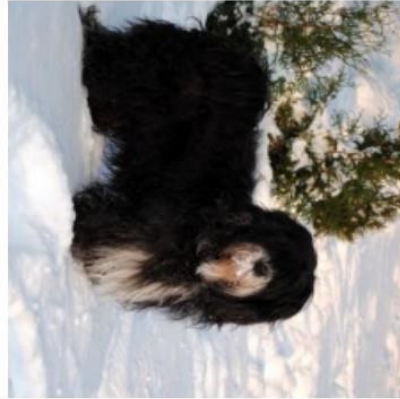
$$(\mathbf{X}, \hat{\mathbf{Y}})$$



Source: Richard Zhang / Phillip Isola

RotNet: Predict Rotation

4-way classification task: How much was each image rotated? (0, 90, 180, or 270 degrees)



90

270

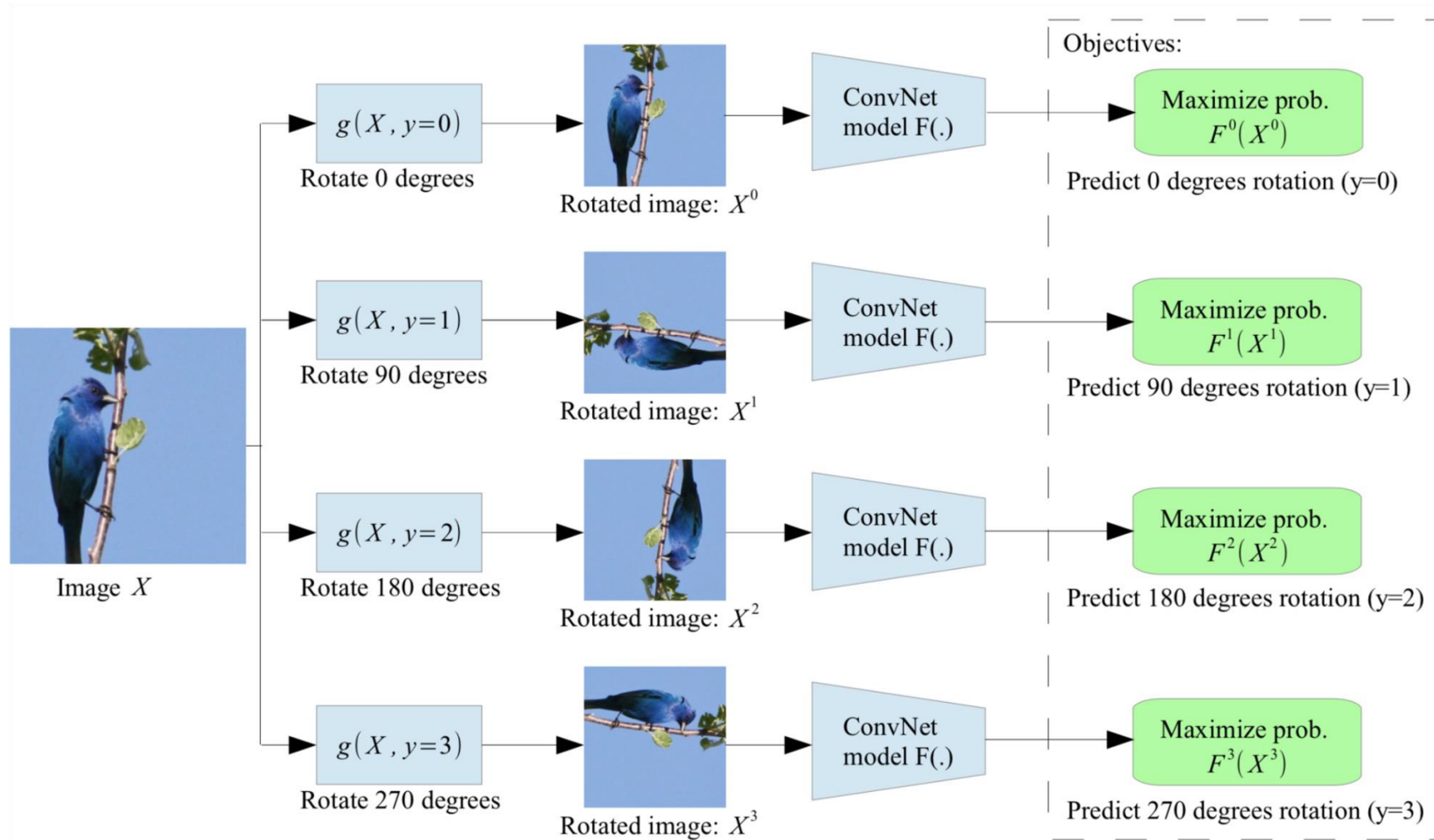
180

0

270

Gidaris et al, "Unsupervised representation learning by predicting image rotations", ICLR 2018

Pretext task: predict rotations



Self-supervised learning by rotating the entire input images.

The model learns to predict which rotation is applied (4-way classification)

(Image source: [Gidaris et al. 2018](#))

Transfer learned features to supervised learning

	Classification (%mAP)		Detection (%mAP)	Segmentation (%mIoU)
Trained layers	fc6-8	all	all	all
ImageNet labels	78.9	79.9	56.8	48.0
Random		53.3	43.4	19.8
Random rescaled Krähenbühl et al. (2015)	39.2	56.6	45.6	32.6
Egomotion (Agrawal et al., 2015)	31.0	54.2	43.9	
Context Encoders (Pathak et al., 2016b)	34.6	56.5	44.5	29.7
Tracking (Wang & Gupta, 2015)	55.6	63.1	47.4	
Context (Doersch et al., 2015)	55.1	65.3	51.1	
Colorization (Zhang et al., 2016a)	61.5	65.6	46.9	35.6
BIGAN (Donahue et al., 2016)	52.3	60.1	46.9	34.9
Jigsaw Puzzles (Noroozi & Favaro, 2016)	-	67.6	53.2	37.6
NAT (Bojanowski & Joulin, 2017)	56.7	65.3	49.4	
Split-Brain (Zhang et al., 2016b)	63.0	67.1	46.7	36.0
ColorProxy (Larsson et al., 2017)		65.9		38.4
Counting (Noroozi et al., 2017)	-	67.7	51.4	36.6
(Ours) RotNet	70.87	72.97	54.4	39.1

Pretrained with full ImageNet supervision

No pretraining

Self-supervised learning on ImageNet (entire training set) with AlexNet.

Finetune on labeled data from Pascal VOC 2007.

Self-supervised learning with rotation prediction

source: [Gidaris et al. 2018](#)

Pre-text Tasks (Video Based)

Leveraging Video Characteristics

A temporal sequence of frames



What can we use to define a proxy loss?

- Nearby (in time) frames are strongly correlated, further away may not be
- Temporal order of the frames
- Motion of objects (via optical flow)
- ...

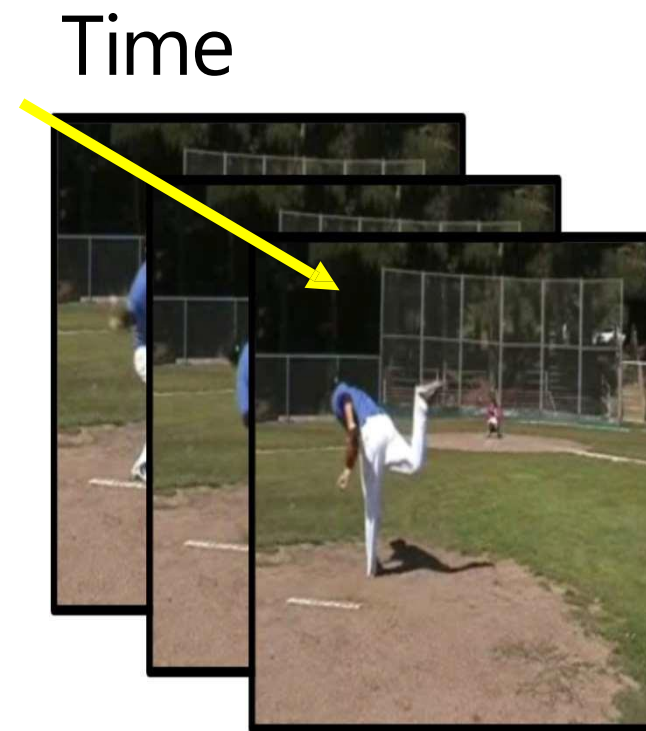
Shuffle and Learn

Unsupervised Learning using Temporal Order Verification

Is this a valid sequence?



Sun and Giles, 2001; Sun et al., 2001; Cleermans 1993; Reber 1989
Arrow of Time - Pickup et al., 2014

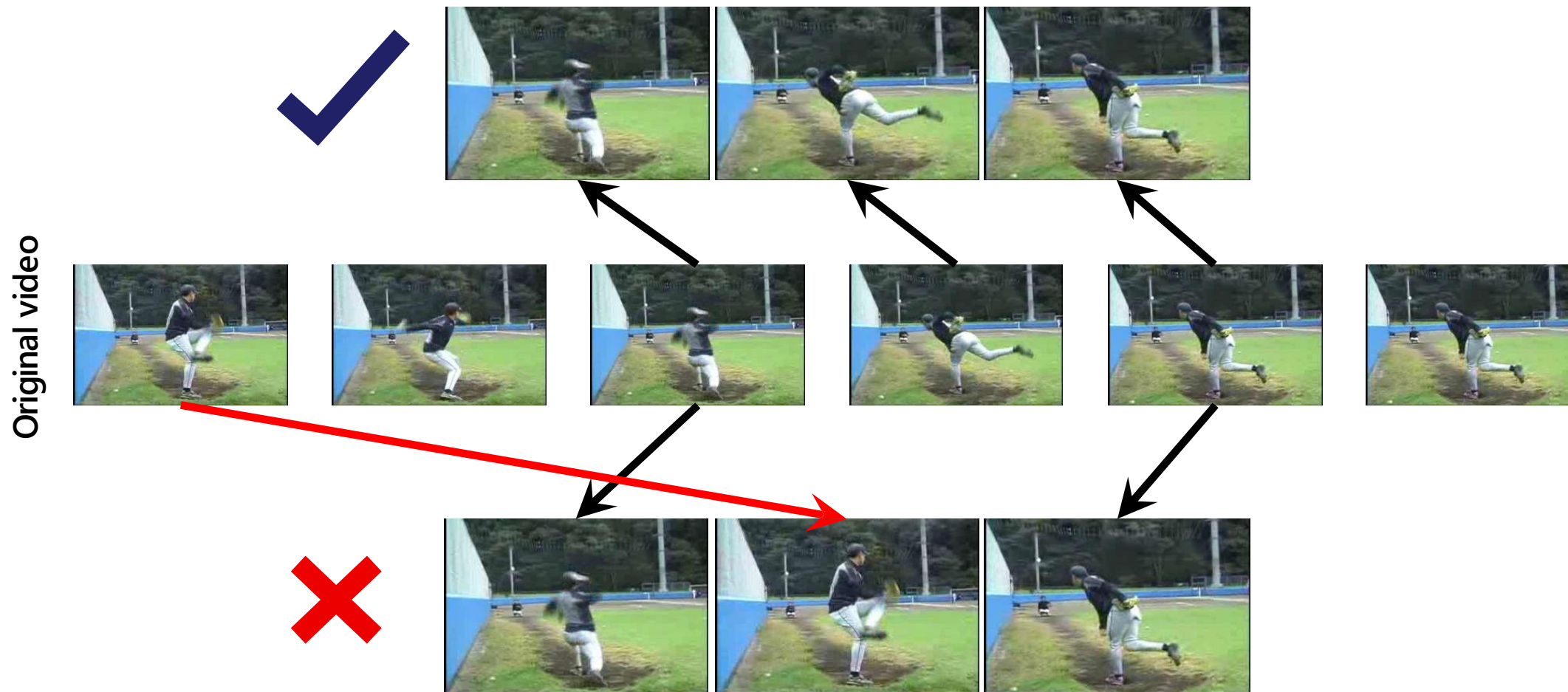


"Sequence" of data

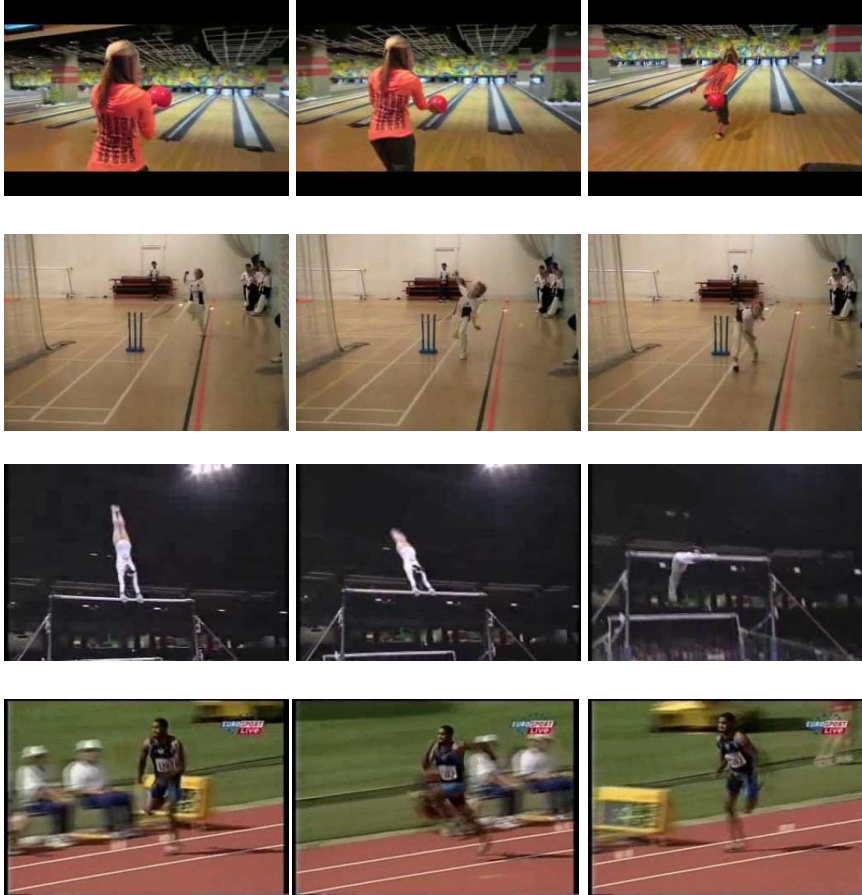
Misra, I., Zitnick, C. L., & Hebert, M. (2016, October). Shuffle and learn: unsupervised learning using temporal order verification. In European Conference on Computer Vision (pp. 527-544). Springer, Cham.

Shuffle and Learn

Temporally Correct order



Positive Tuples



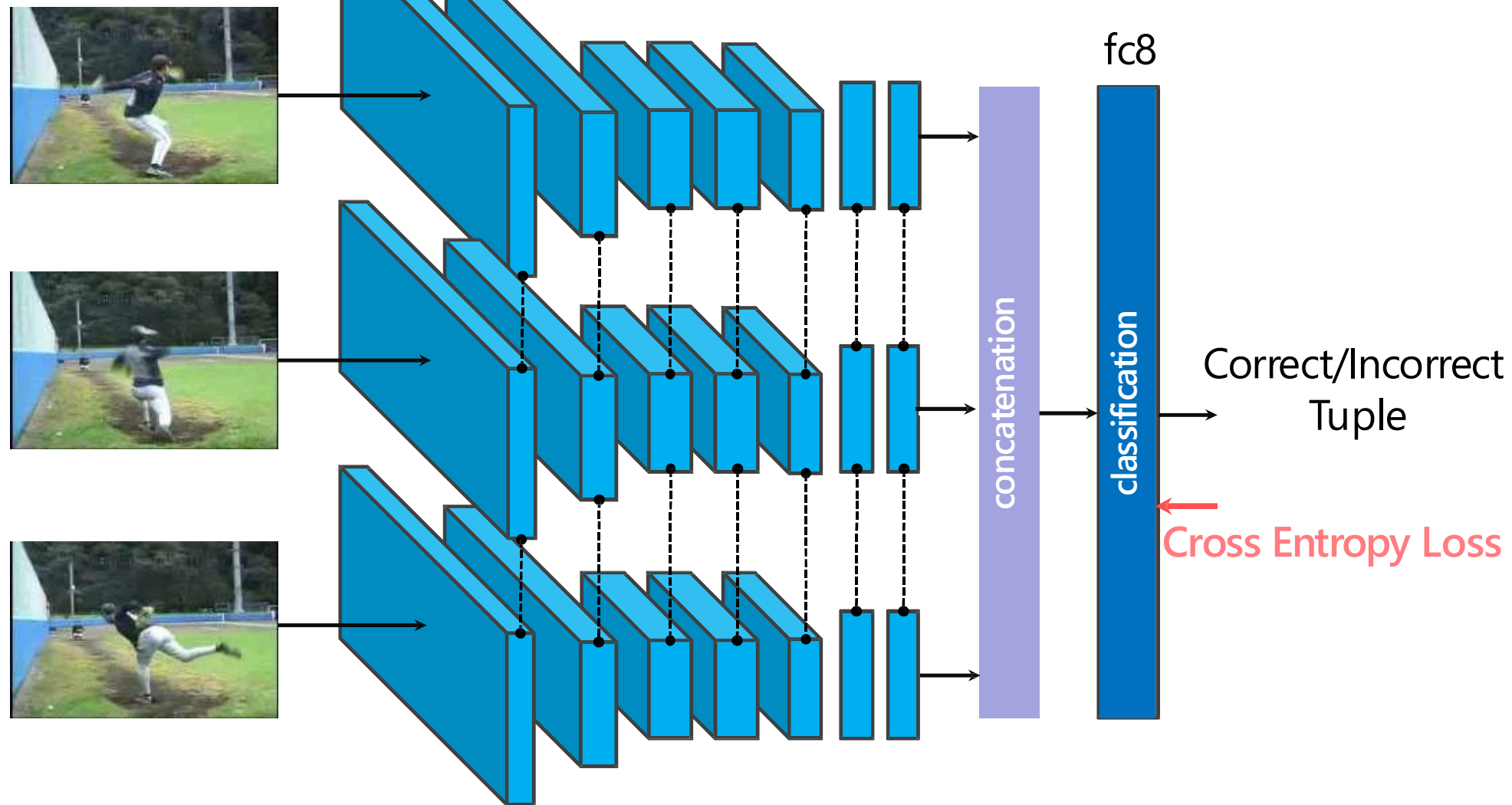
Negative Tuples



~900k tuples from UCF-101 dataset (Soomro et al., 2012)

Architecture

Input Tuple



Query



ImageNet



Shuffle & Learn



Random



Slide credit: Ishan Misra

Finetune on Action Recognition

Dataset	Initialization	Mean Classification Accuracy
UCF101	Random	38.6
	Shuffle & Learn	50.2
	ImageNet pre-trained	<u>67.1</u>

Finetune on Human Pose Estimation



Initialization	FLIC Dataset		MPII Dataset	
	Mean PCK	AUC PCK	Mean <u>PCKh@0.5</u>	AUC <u>PCKh@0.5</u>
Shuffle & Learn	84.9	49.6	<u>87.7</u>	<u>47.6</u>
ImageNet pre-train	<u>85.8</u>	<u>51.3</u>	85.1	47.2

Pretext task: video coloring

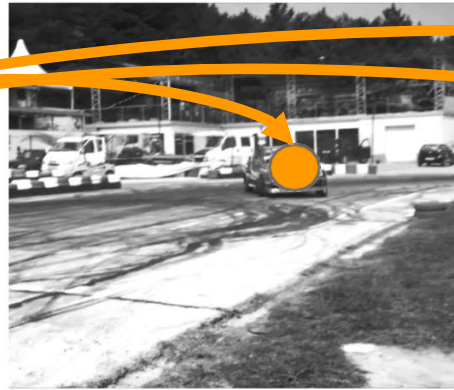
Idea: model the temporal coherence of colors in videos

reference frame

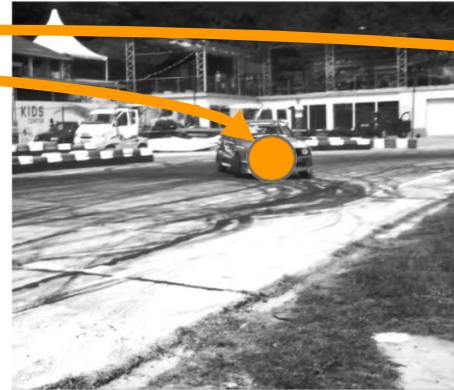
how should I color these frames?
Should be the same color!



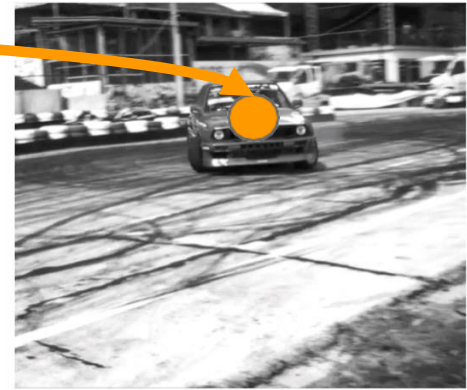
t = 0



t = 1



t = 2



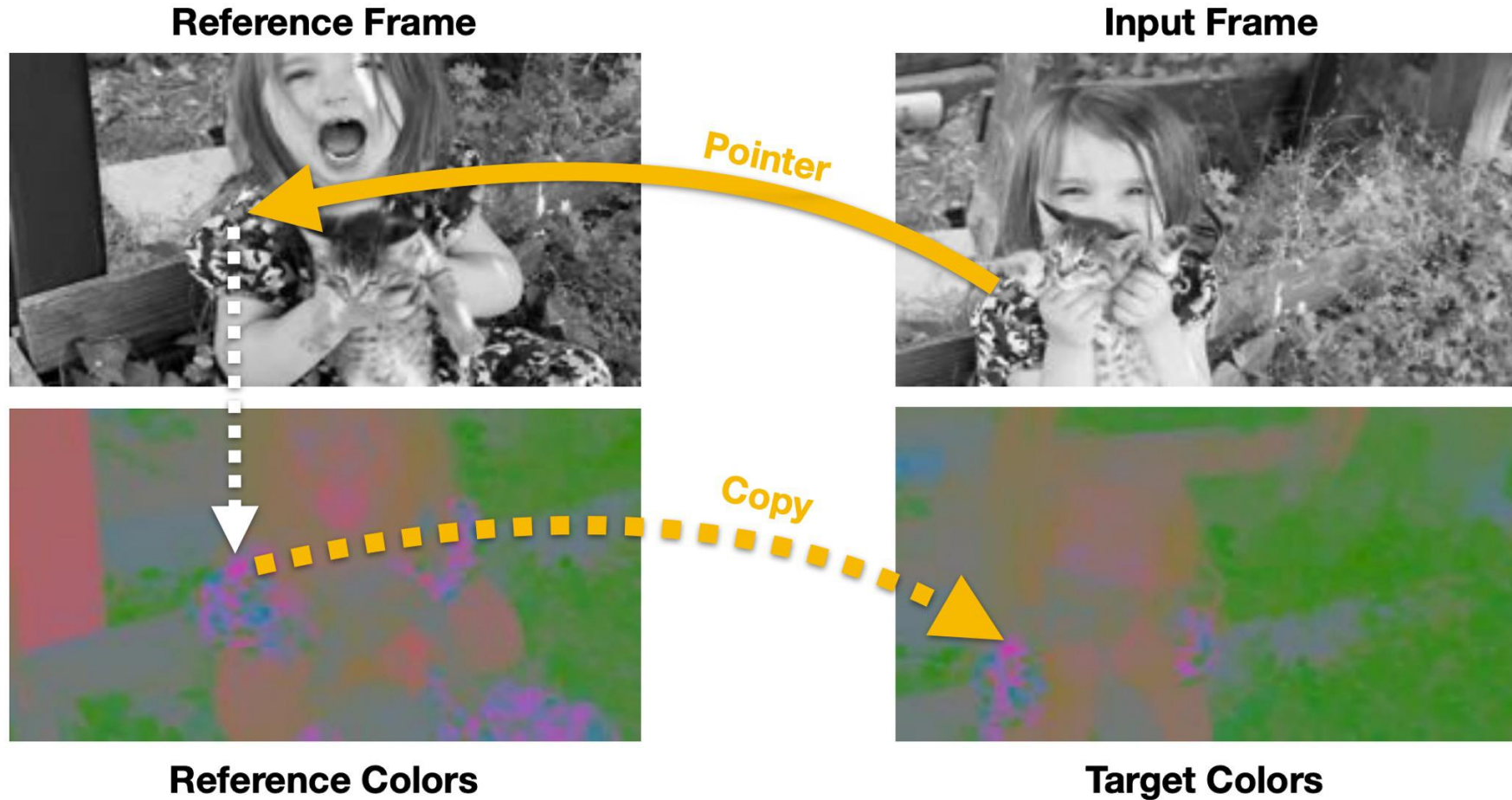
t = 3

...

Hypothesis: learning to color video frames should allow model to learn to track regions or objects without labels!

Source: [Vondrick et al., 2018](#)

Learning to color videos



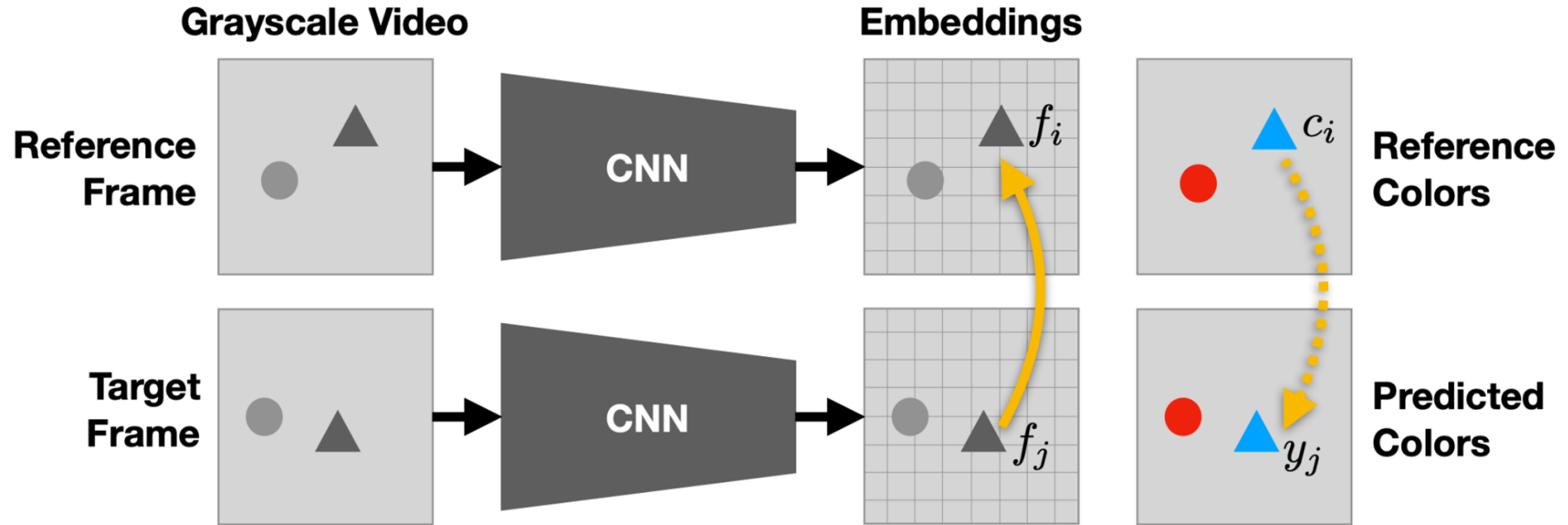
Learning objective:

Establish mappings between reference and target frames in a learned feature space.

Use the mapping as “pointers” to **copy** the correct color (LAB).

Source: [Vondrick et al., 2018](#)

Learning to color videos



attention map on the reference frame

predicted color = weighted sum of the reference color

loss between predicted color and ground truth color

$$A_{ij} = \frac{\exp(f_i^T f_j)}{\sum_k \exp(f_k^T f_j)}$$

$$y_j = \sum_i A_{ij} c_i$$

$$\min_{\theta} \sum_j \mathcal{L}(y_j, c_j)$$

Source: [Vondrick et al., 2018](#)

Colorizing videos (qualitative)

reference frame



target frames (gray)



predicted color



Source: [Google AI blog post](#)

Colorizing videos (qualitative)

reference frame



target frames (gray)



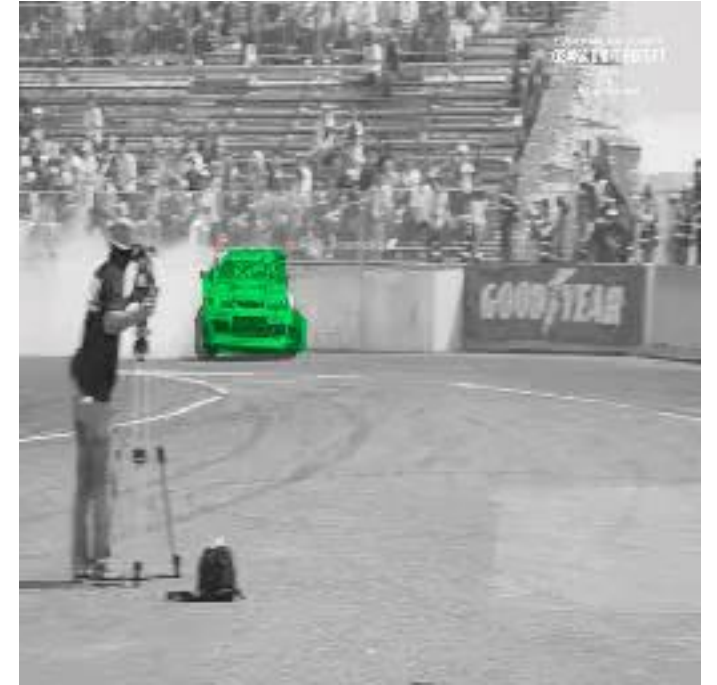
predicted color



Source: [Google AI blog post](#)

Tracking emerges from colorization

Propagate segmentation masks using learned attention



Source: [Google AI blog post](#)

Tracking emerges from colorization

Propagate pose keypoints using learned attention



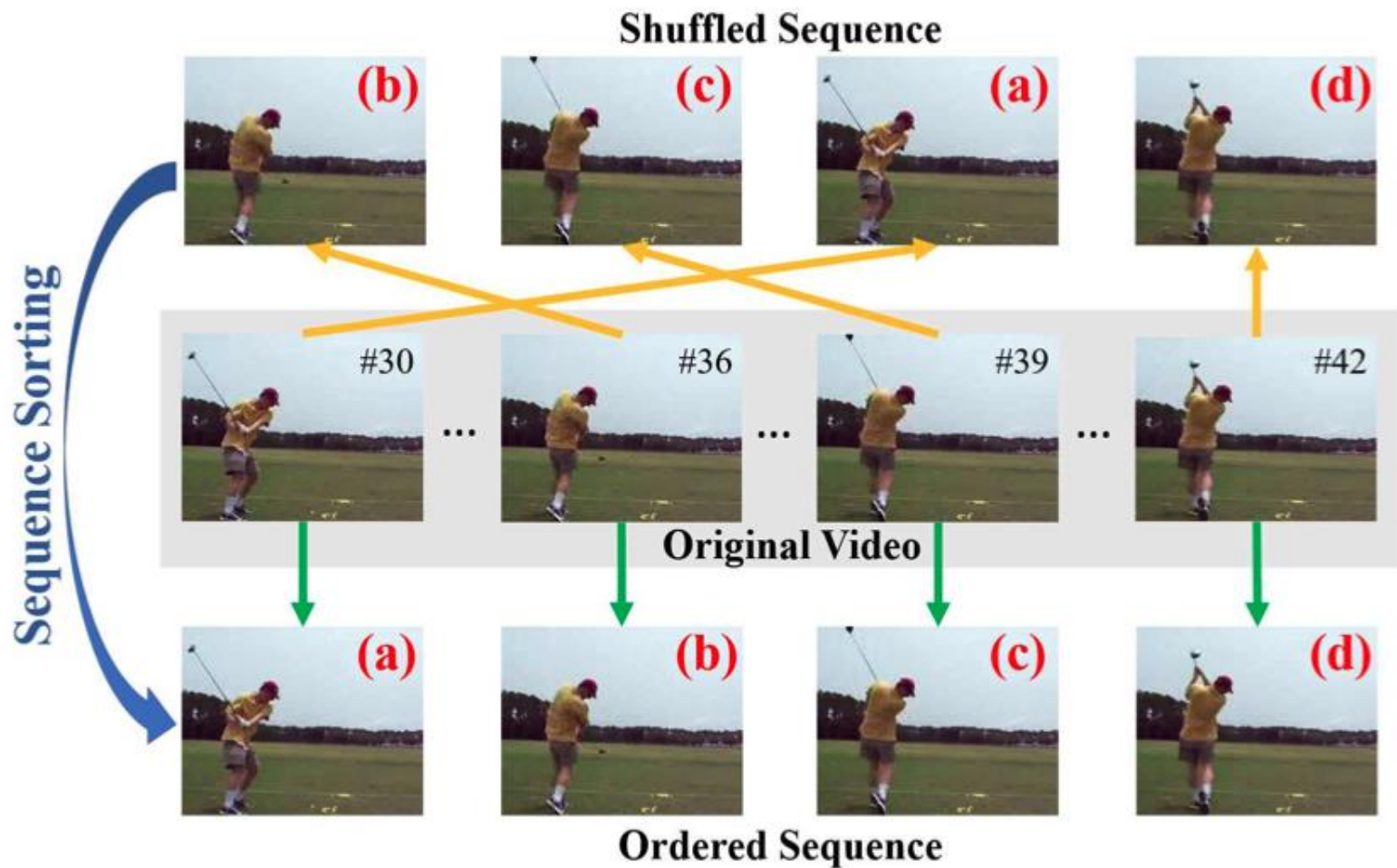
Source: [Google AI blog post](#)

Results – Action Recognition

Dataset	Initialization	Mean Classification Accuracy
UCF101	Random	38.6
	Shuffle & Learn	50.2
	Colorization	52.1
	ImageNet pre-trained	<u>67.1</u>

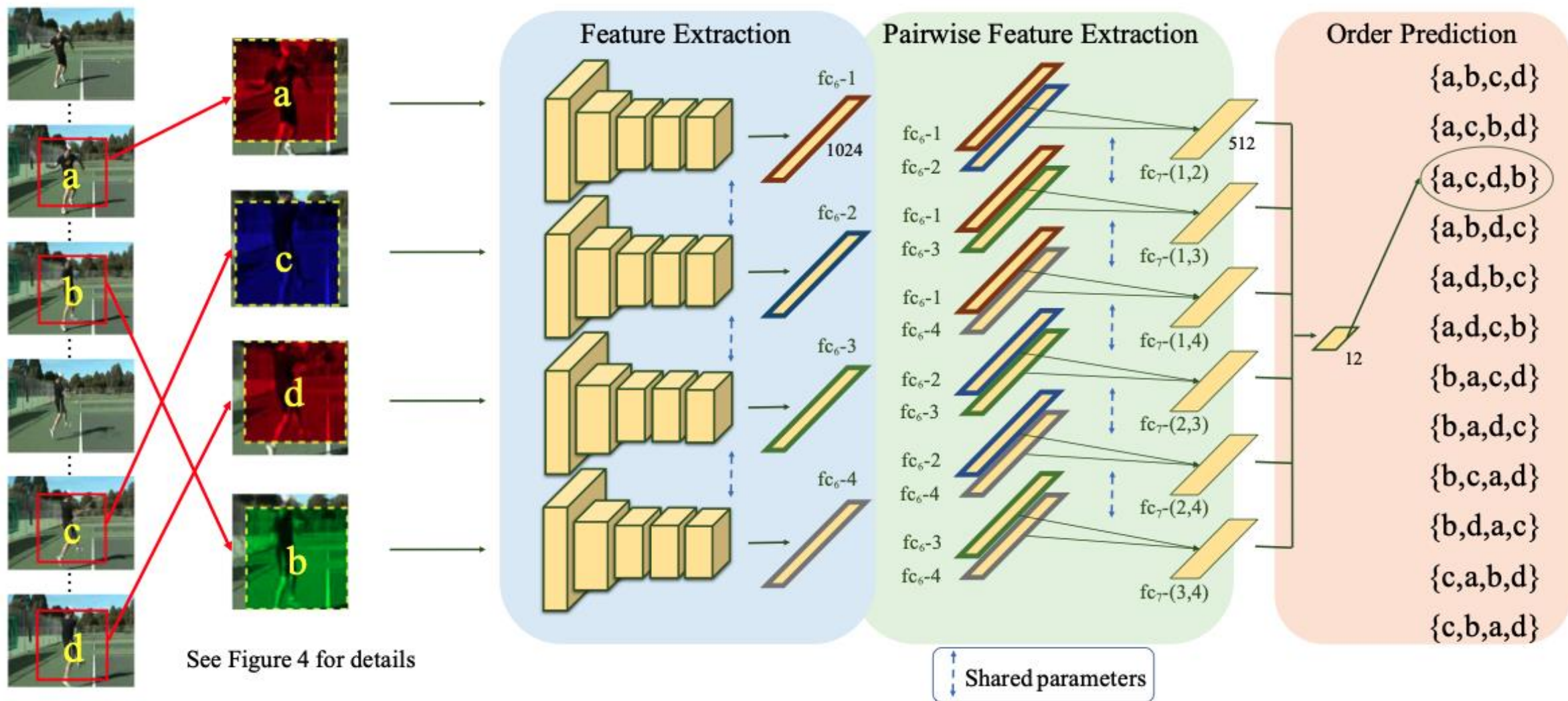
Unsupervised learning by Sorting Sequences

Use frame sorting as a pre-text task



Lee, H. Y., Huang, J. B., Singh, M., & Yang, M. H. (2017). Unsupervised representation learning by sorting sequences. In Proceedings of the IEEE International Conference on Computer Vision (pp. 667-676).

Unsupervised learning by Sorting Sequences



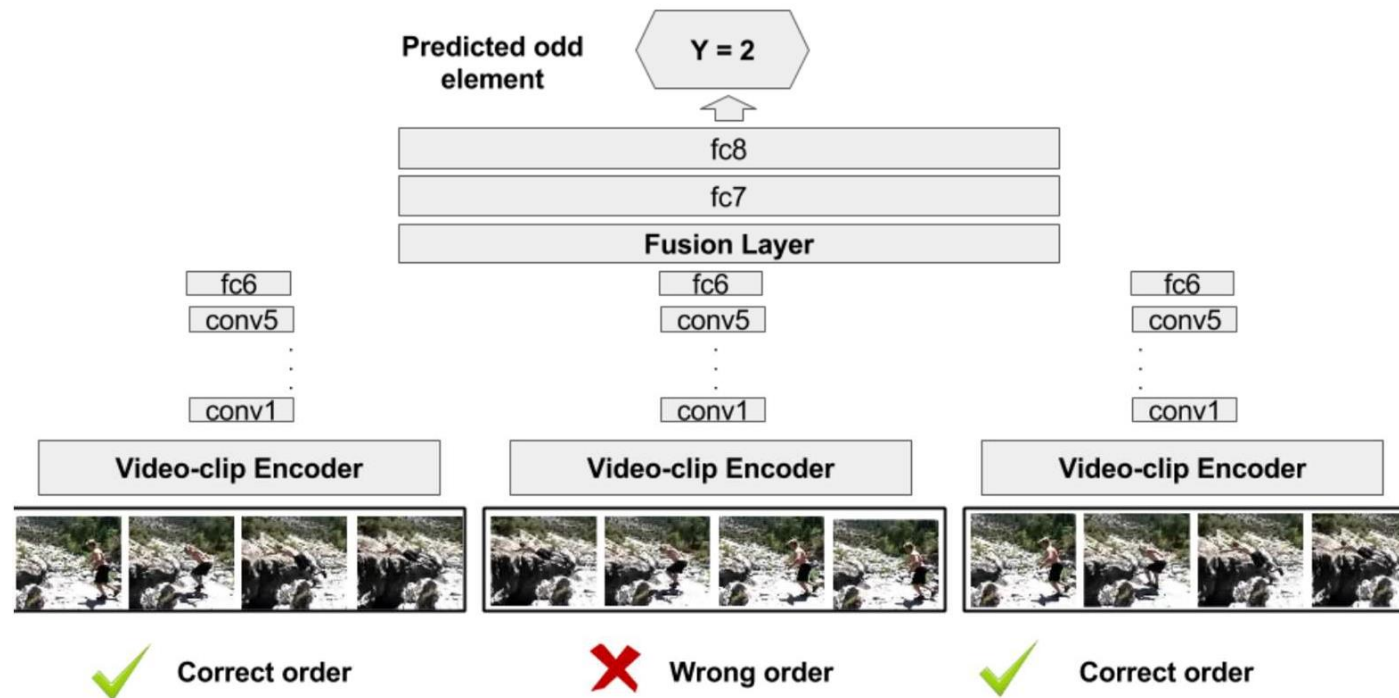
Lee, H. Y., Huang, J. B., Singh, M., & Yang, M. H. (2017). Unsupervised representation learning by sorting sequences. In Proceedings of the IEEE International Conference on Computer Vision (pp. 667-676).

Results – Action Recognition

Dataset	Initialization	Mean Classification Accuracy
UCF101	Random	38.6
	Shuffle & Learn	50.2
	Colorization	52.1
	Sorting Sequences	56.3
	ImageNet pre-trained	<u>67.1</u>

Self-Supervised Video Representation Learning With Odd-One-Out Networks

- The network is given N sequences of video frames.
- N-1 are correctly ordered, one is shuffled
- The model must identify the «odd one out»
- Encourages learning of motion features



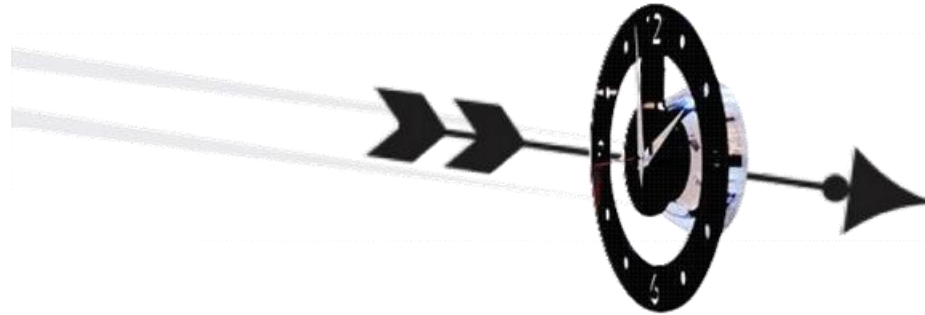
Fernando, Basura, et al. "Self-supervised video representation learning with odd-one-out networks." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.

Results – Action Recognition

Dataset	Initialization	Mean Classification Accuracy
UCF101	Random	38.6
	Shuffle & Learn	50.2
	Colorization	52.1
	Sorting Sequences	56.3
	Odd-One-Out	60.3
	ImageNet pre-trained	<u>67.1</u>

Learning the Arrow of Time

Task: predict if video playing forwards or backwards



Supervision:

Positive training samples: video clips playing forwards

Negative training samples: video clips playing backwards

Wei, D., Lim, J. J., Zisserman, A., & Freeman, W. T. (2018). Learning and using the arrow of time. In *Proceedings of the IEEE conference on computer vision and pattern recognition*(pp. 8052-8060).

Strong cues

Semantic, face motion direction, ordering,
physics, entropy, causality

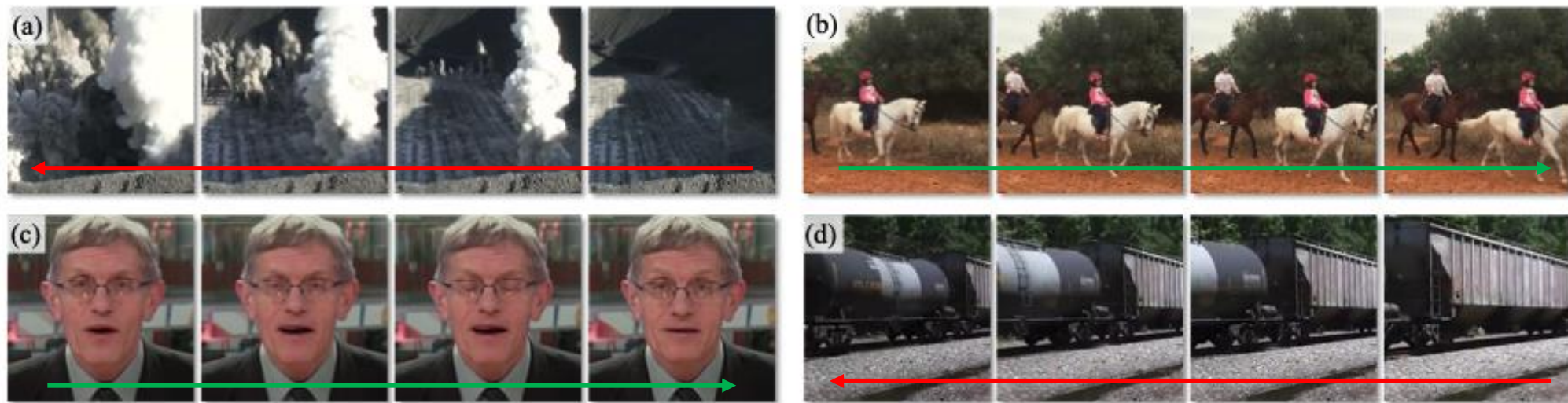
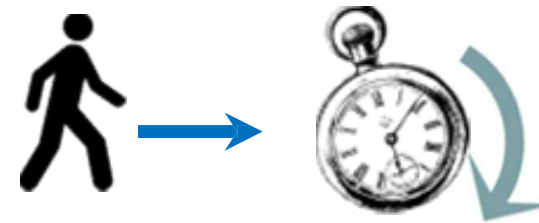
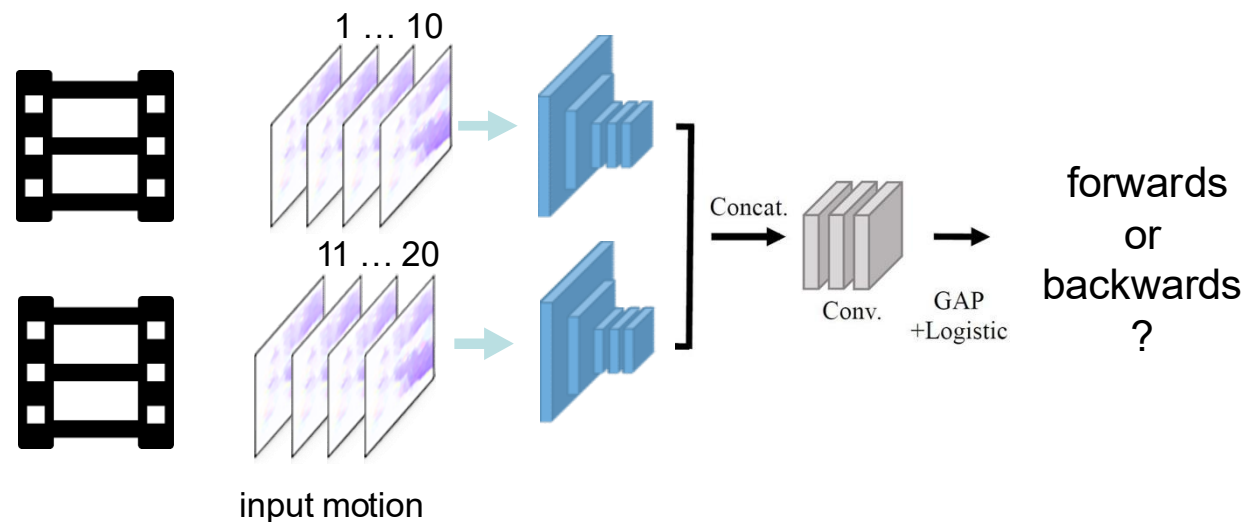


Figure 1: Seeing these ordered frames from videos, can you tell whether each video is playing forward or backward? (answer below¹). Depending on the video, solving the task may require (a) low-level understanding (e.g. physics), (b) high-level reasoning (e.g. semantics), or (c) familiarity with very subtle effects or with (d) camera conventions. In this work, we learn and exploit several types of knowledge to predict the arrow of time automatically with neural network models trained on large-scale video datasets.

Temporal Class-Activation Map Network



T-CAM Model:

Input: optical flow in two chunks

Final layer: global average pooling to allow class activation map (CAM)

Wei, D., Lim, J. J., Zisserman, A., & Freeman, W. T. (2018). Learning and using the arrow of time. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 8052-8060).

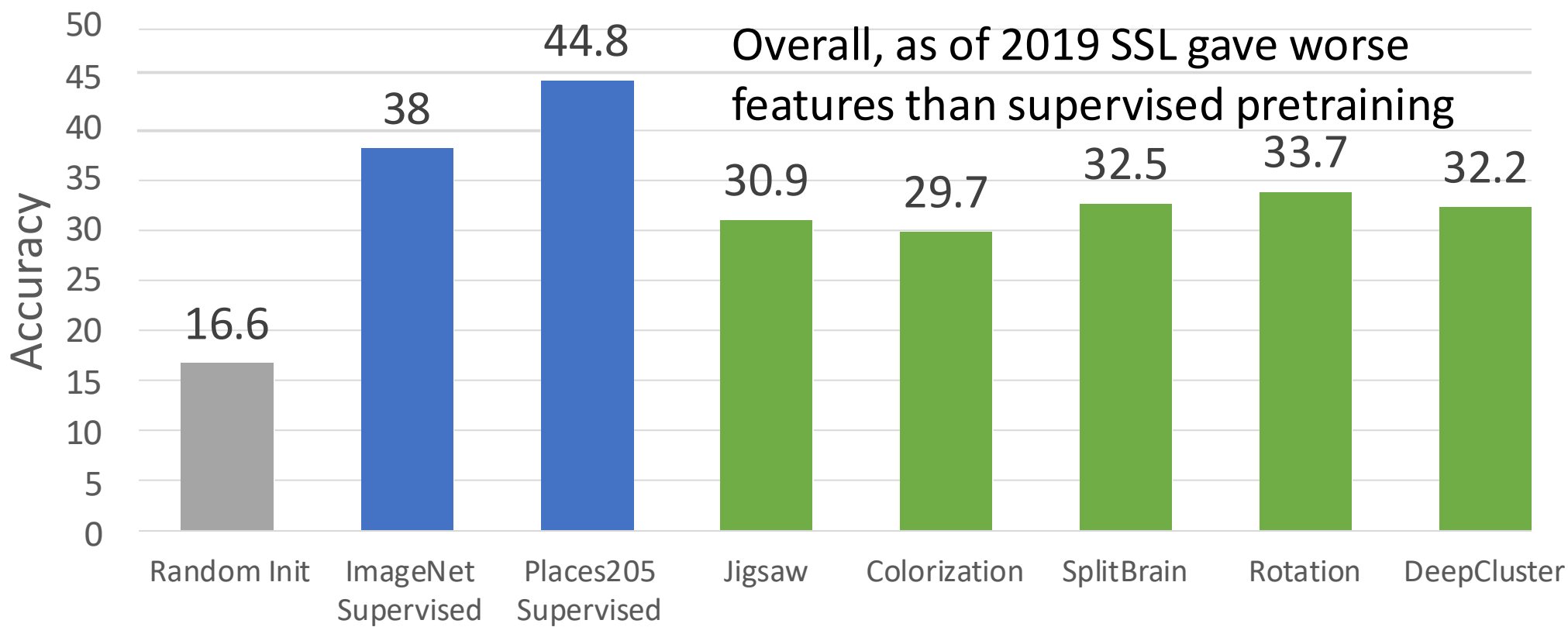
Results – Action Recognition

Dataset	Initialization	Mean Classification Accuracy
UCF101	Random	38.6
	Shuffle & Learn	50.2
	Colorization	52.1
	Sorting Sequences	56.3
	Odd-One-Out	60.3
	Arrow of Time	<u>68.9</u>
	ImageNet pre-trained	67.1

Which SSL Method is best?

Some papers have tried to perform fair comparisons of many SSL methods

Places205 Linear Classification from AlexNet conv5



Goyal et al, "Scaling and Benchmarking Self-Supervised Visual Representation Learning", ICCV 2019

Summary: pretext tasks from transformations

- Pretext tasks focus on “visual common sense”, e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images/videos, e.g., semantic representation of an object category or region correspondences in order to solve the pretext tasks.
- We often do not care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- **Problems:**
 - 1) coming up with individual pretext tasks is tedious
 - 2) the learned representations may not be general.

Pretext tasks from image transformations

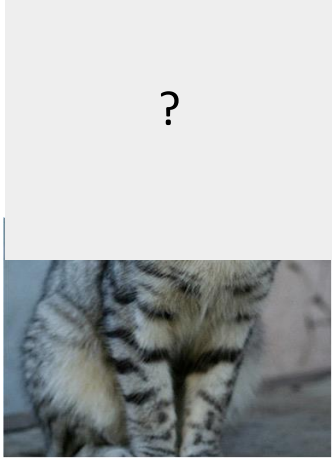
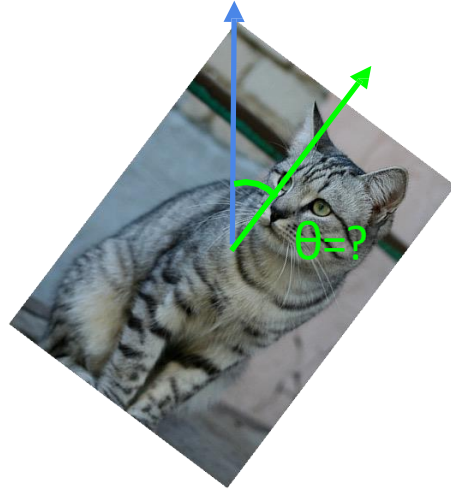


image completion



rotation prediction



“jigsaw puzzle”



colorization

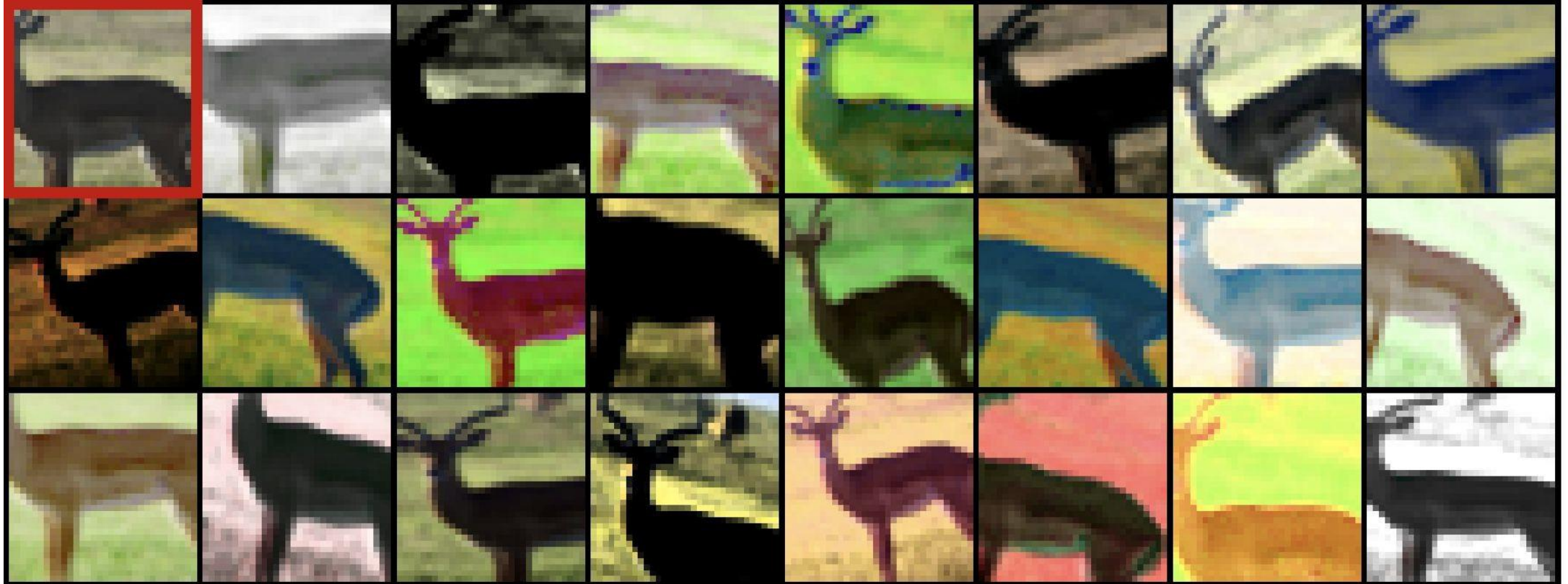
Learned representations may be tied to a specific pretext task!

Can we come up with a more general pretext task?

Contrastive Approaches

Invariance-Based Learning

Learn representations that are invariant to data transformations



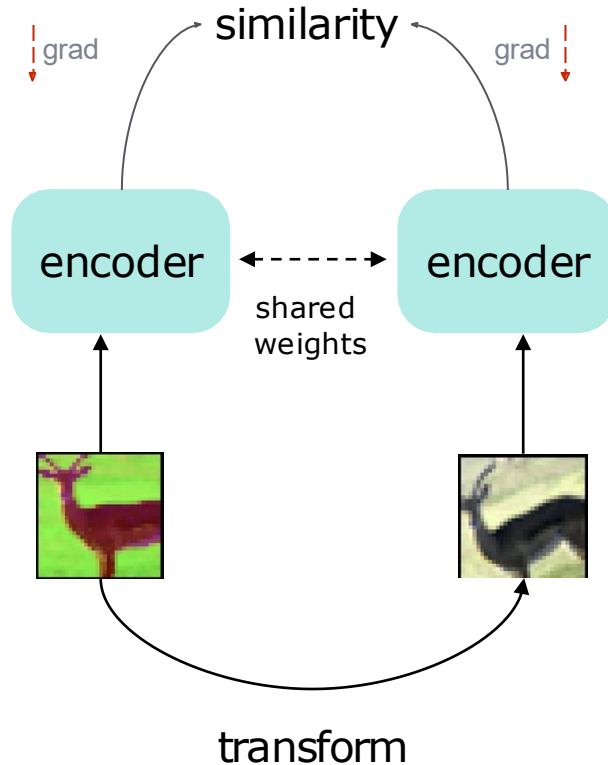
Learn features such that:

$$f_{\theta}(I) = f_{\theta}(\text{augment}(I))$$

Figure from Dosovitskiy et al., 2014

Simple Approach

$$f_{\theta}(I) = f_{\theta}(\text{augment}(I))$$



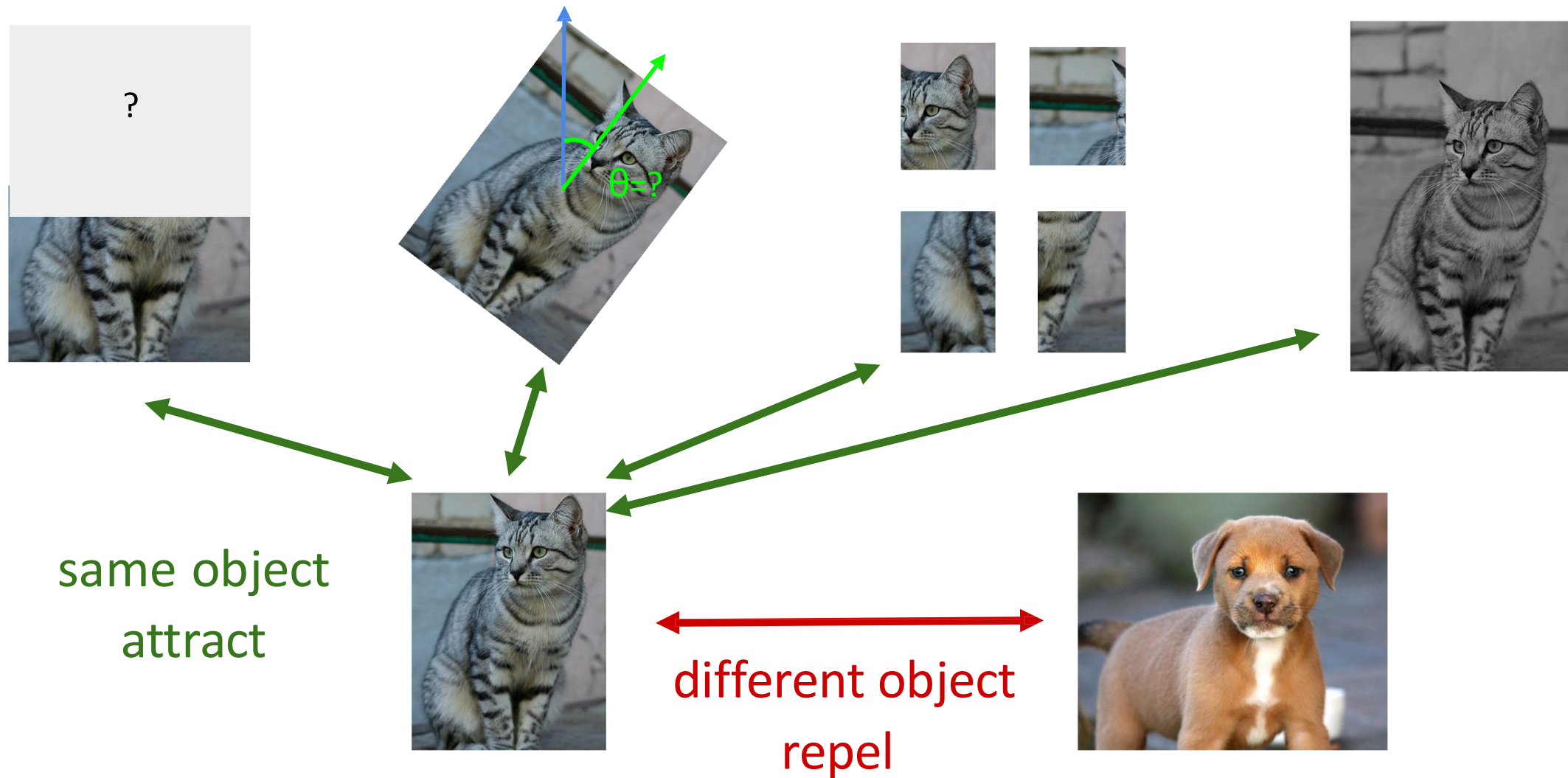
Trivial Solution

$$f_{\theta}(I) = \text{constant}$$

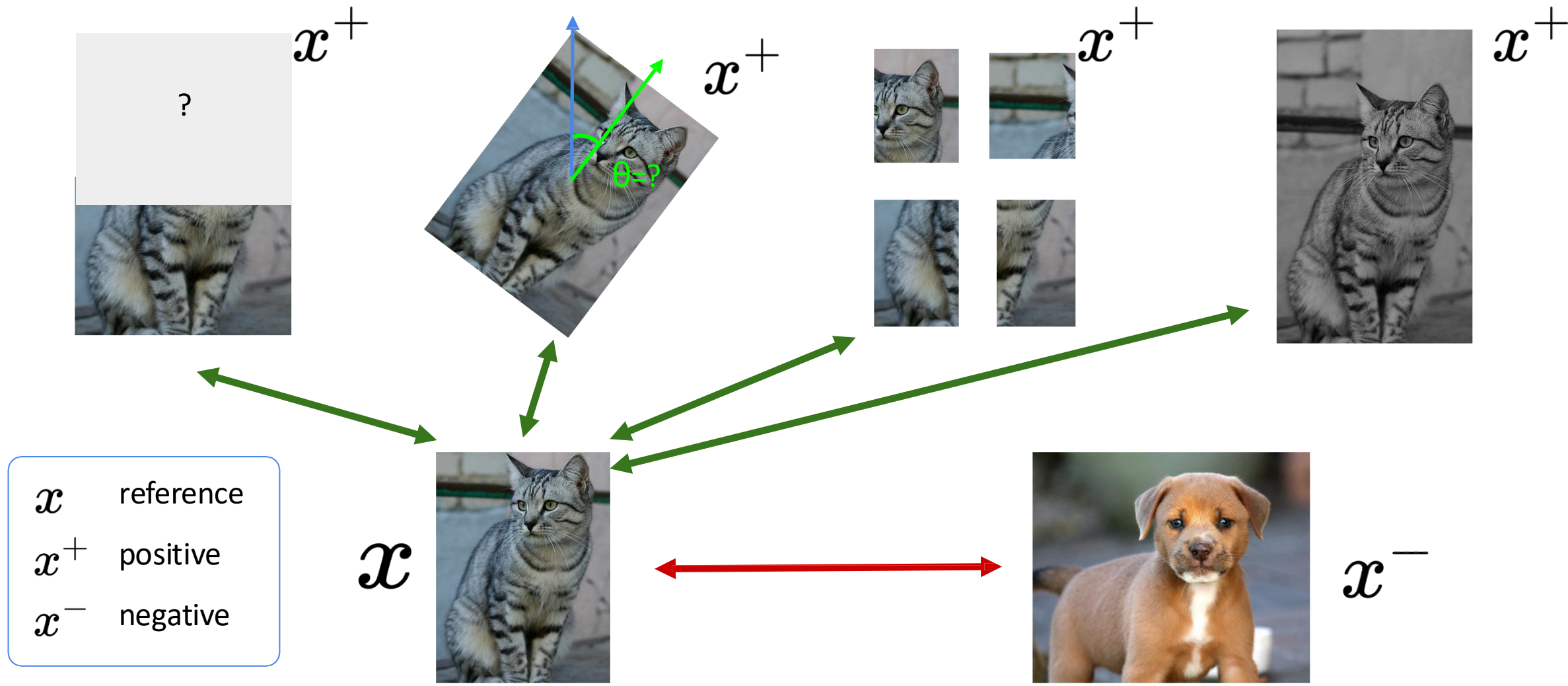
Satisfies the invariance property, but not useful
Often referred to as «representation collapse»

Contrastive learning methods allow to avoid this trivial solution

A more general pretext task?



Contrastive Representation Learning



A formulation of contrastive learning

What we want:

$$\text{score}(f(x), f(x^+)) \gg \text{score}(f(x), f(x^-))$$

x : reference sample; x^+ positive sample; x^- negative sample

Given a chosen score function, we aim to learn an encoder function f that yields high score for positive pairs (x, x^+) and low scores for negative pairs (x, x^-) .

A formulation of contrastive learning

Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\overbrace{\exp(s(f(x), f(x^+)))}}{\underbrace{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))} } \right]$$



x

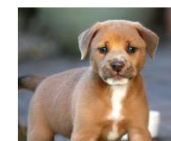


x^+

Score for the positive pair



x



x_1^-



x_2^-



x_3^-

Scores for the $\ddots$ N-1 negative pairs

This seems familiar ...

Cross entropy loss for a N-way softmax classifier!

I.e., learn to find the positive sample from the N samples

A formulation of contrastive learning

Loss function given 1 positive sample and $N - 1$ negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+)))}{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))} \right]$$

Commonly known as the InfoNCE loss ([van den Oord et al., 2018](#))

A lower bound on the mutual information between $f(x)$ and $f(x^+)$

$$MI[f(x), f(x^+)] - \log(N) \geq -L$$

The larger the negative sample size (N), the tighter the bound

Detailed derivation: [Poole et al., 2019](#)

SimCLR: A Simple Framework for Contrastive Learning

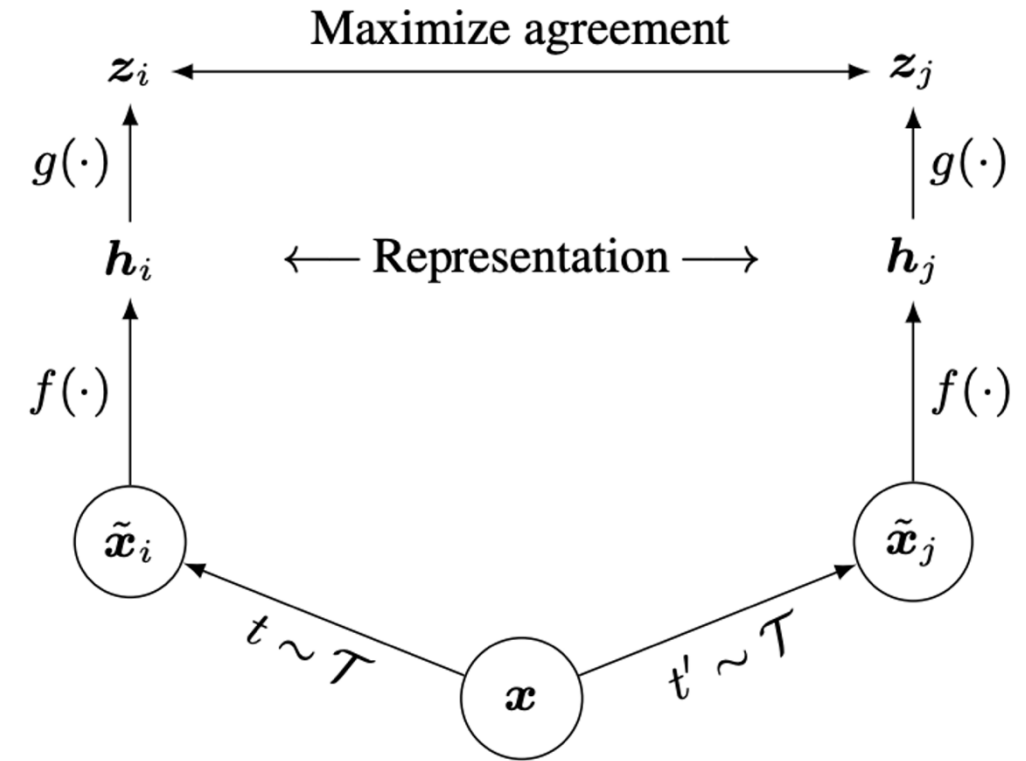
Cosine similarity as the score function:

$$s(u, v) = \frac{u^T v}{||u|| ||v||}$$

Use a projection network $g(\cdot)$ to project features to a space where contrastive learning is applied

Generate positive samples through data augmentation:

- random cropping, random color distortion, and random blur.



Source: [Chen et al., 2020](#)

SimCLR: generating positive samples from data augmentation



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



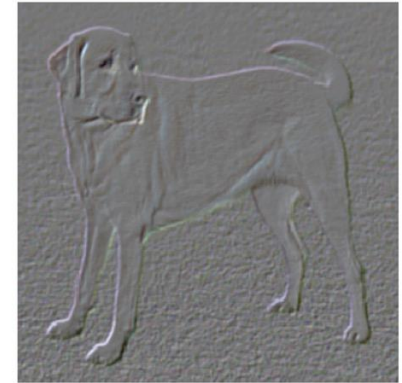
(g) Cutout



(h) Gaussian noise



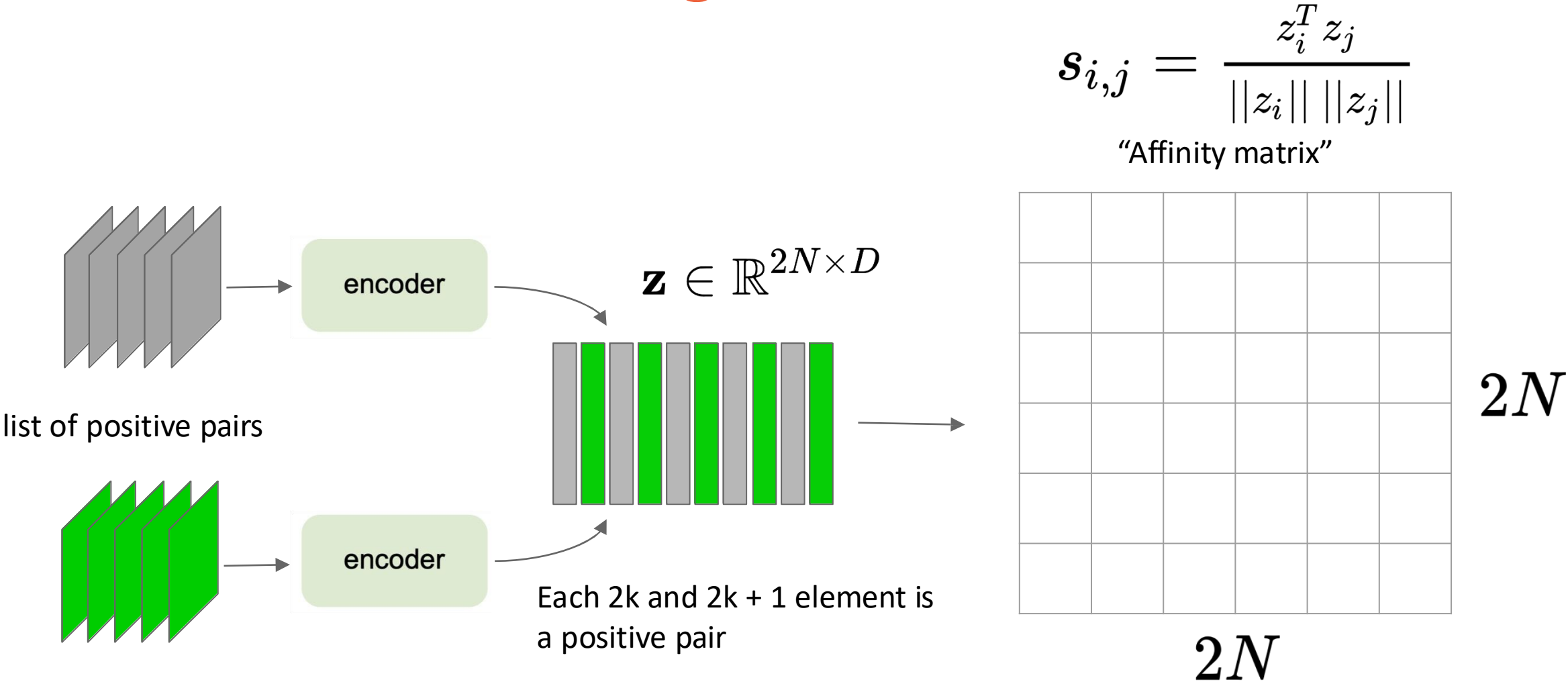
(i) Gaussian blur



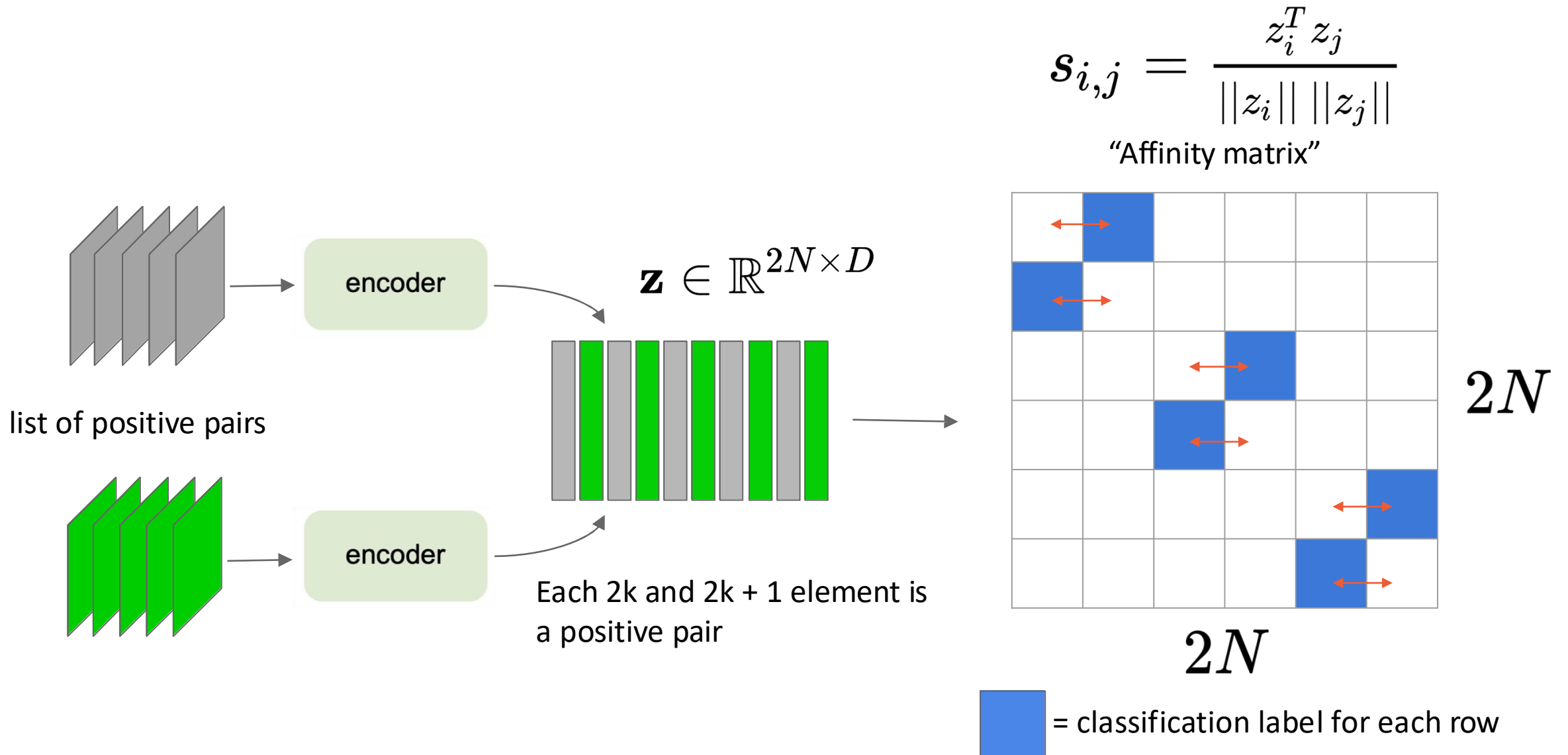
(j) Sobel filtering

Source: [Chen et al., 2020](#)

SimCLR: mini-batch training



SimCLR: mini-batch training



Semi-supervised learning on SimCLR features

Method	Architecture	Label fraction	
		1%	10%
		Top 5	
Supervised baseline	ResNet-50	48.4	80.4
<i>Methods using other label-propagation:</i>			
Pseudo-label	ResNet-50	51.6	82.4
VAT+Entropy Min.	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
<i>Methods using representation learning only:</i>			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(*)	77.9	91.2
SimCLR (ours)	ResNet-50	75.5	87.8
SimCLR (ours)	ResNet-50 (2×)	83.0	91.2
SimCLR (ours)	ResNet-50 (4×)	85.8	92.6

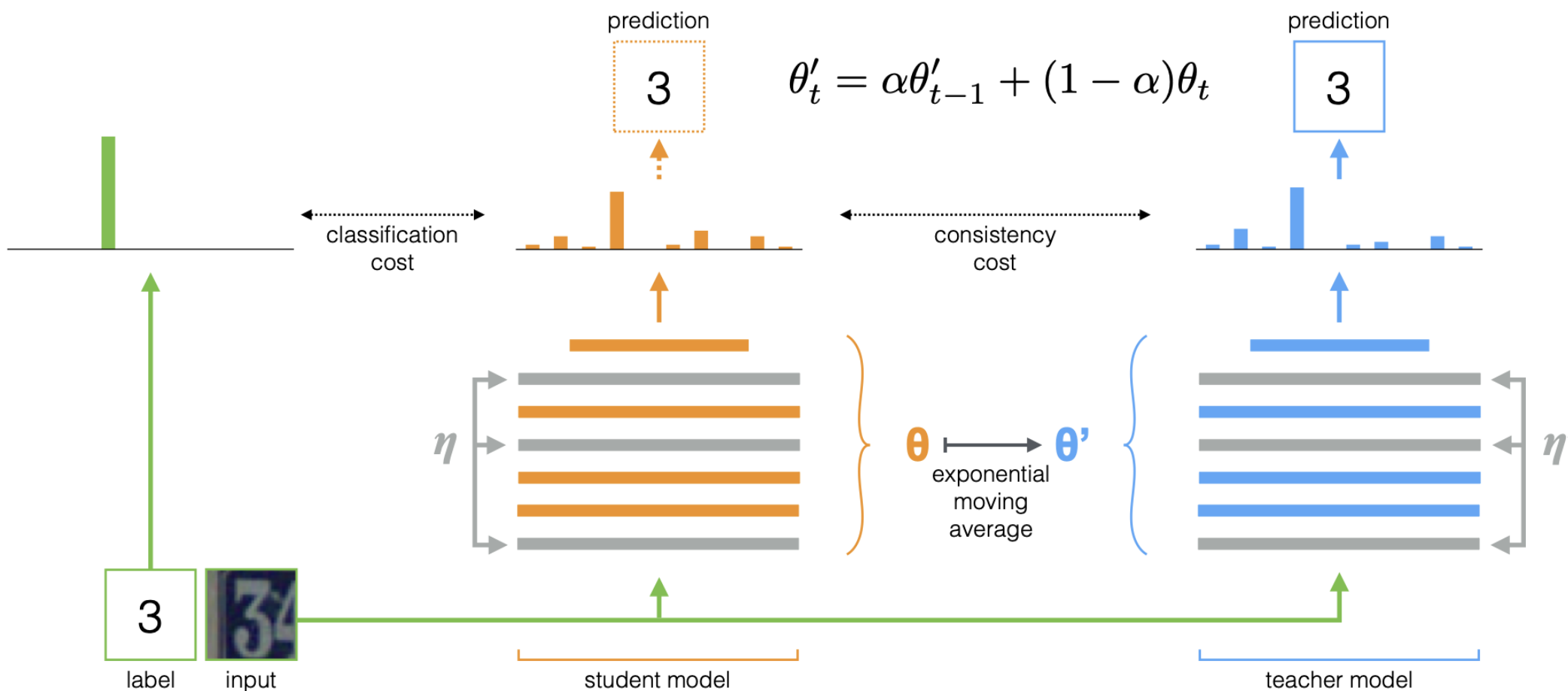
Train feature encoder on ImageNet (entire training set) using SimCLR.

Finetune the encoder with 1% / 10% of labeled data on ImageNet.

Table 7. ImageNet accuracy of models trained with few labels.

Source: [Chen et al., 2020](#)

Exponential Moving Average (EMA) and Mean Teachers



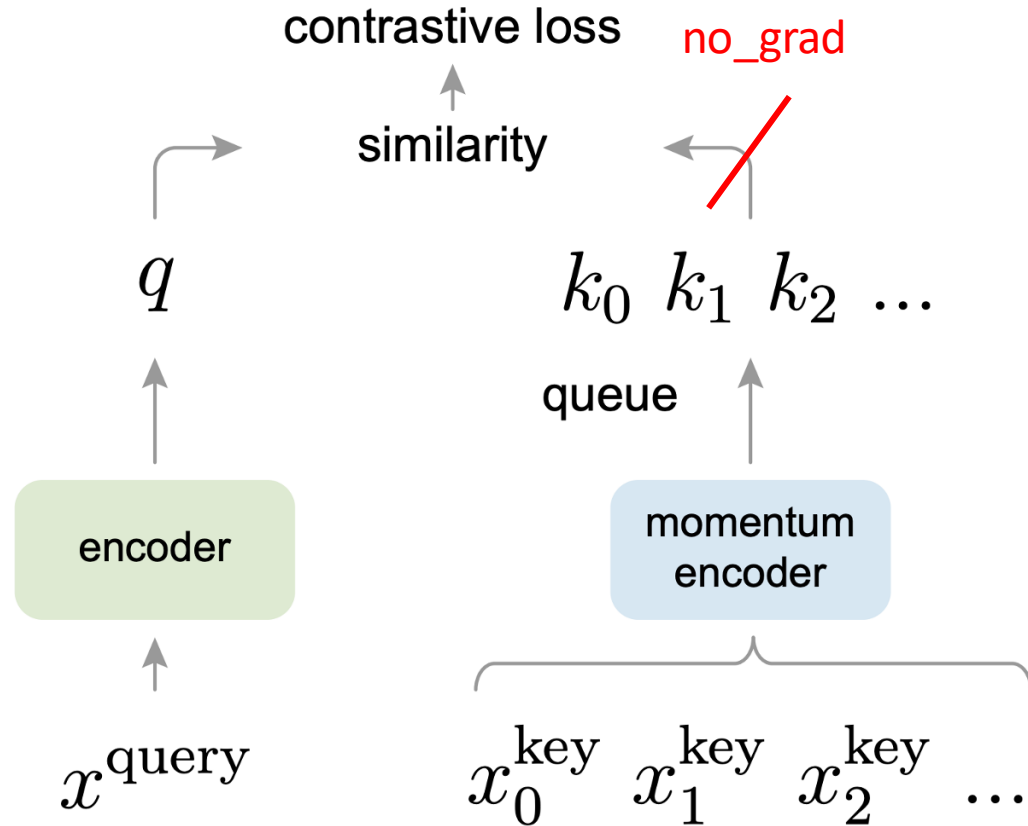
A form of distillation to create a «teacher» providing consistency.

At the end of training, we use the teacher (EMA weights)

This concept is used a lot in self-supervised learning.

Tarvainen, Antti, and Harri Valpola. "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results." *Advances in neural information processing systems* 30 (2017).

Momentum Contrastive Learning (MoCo)



Key differences to SimCLR:

- Keep a running **queue** of keys (negative samples).
- Compute gradients and update the encoder **only through the queries**.
- Decouple min-batch size with the number of keys: can support **a large number of negative samples**.
- The key encoder is **slowly progressing** through the momentum update rules:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

Source: [He et al., 2020](#)

Improved Baselines with Momentum Contrastive Learning

Xinlei Chen Haoqi Fan Ross Girshick Kaiming He
Facebook AI Research (FAIR)

A hybrid of ideas from SimCLR and MoCo:

- From SimCLR: non-linear projection head and strong data augmentation.
- From MoCo: momentum-updated queues that allow training on a large number of negative samples.

Source: [Chen et al., 2020](#)

An Empirical Study of Training Self-Supervised Vision Transformers

“This paper does not describe a novel method.”

Xinlei Chen* Saining Xie* Kaiming He
Facebook AI Research (FAIR)

Code: <https://github.com/facebookresearch/moco-v3>

Abstract

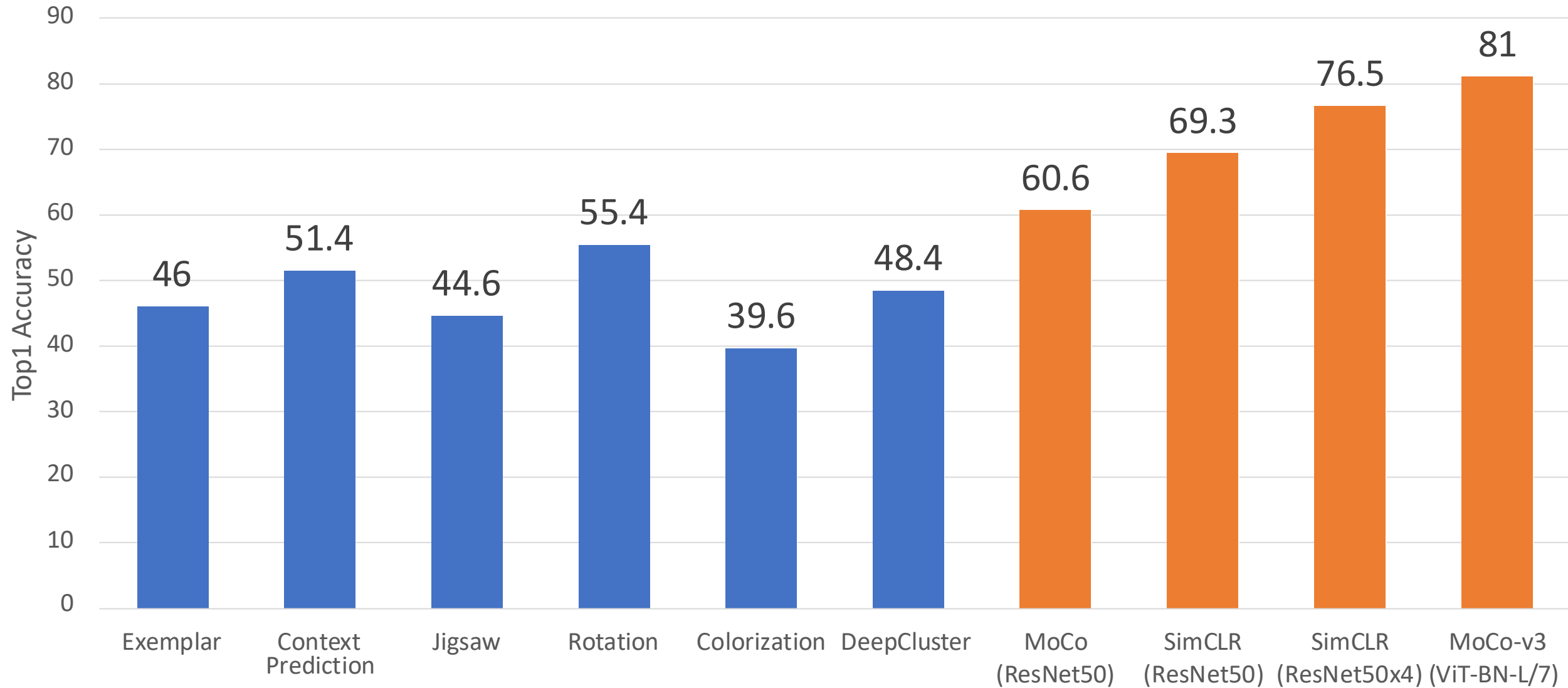
This paper does not describe a novel method. Instead, it studies a straightforward, incremental, yet must-know baseline given the recent progress in computer vision: self-supervised learning for Vision Transformers (ViT). While the training recipes for standard convolutional networks have been highly mature and robust, the recipes for ViT are yet to be built, especially in the self-supervised scenarios where training becomes more challenging. In this work, we go back to basics and investigate the effects of several fundamental components for training self-supervised ViT. We observe that instability is a major issue that degrades accuracy, and it can be hidden by apparently good results. We reveal that these results are indeed partial failure, and they can be improved when training is made more stable. We benchmark ViT results in MoCo v3 and several other self-supervised frameworks, with ablations in various aspects. We discuss the currently positive evidence as well as challenges and open questions. We hope that this work will provide useful data points and experience for future research.

framework	model	params	acc. (%)
<i>linear probing:</i>			
iGPT [9]	iGPT-L	1362M	69.0
iGPT [9]	iGPT-XL	6801M	72.0
MoCo v3	ViT-B	86M	76.7
MoCo v3	ViT-L	304M	77.6
MoCo v3	ViT-H	632M	78.1
MoCo v3	ViT-BN-H	632M	79.1
MoCo v3	ViT-BN-L/7	304M	81.0
<i>end-to-end fine-tuning:</i>			
masked patch pred. [16]	ViT-B	86M	79.9 [†]
MoCo v3	ViT-B	86M	83.2
MoCo v3	ViT-L	304M	84.1

Table 1. **State-of-the-art Self-supervised Transformers** in ImageNet classification, evaluated by linear probing (top panel) or end-to-end fine-tuning (bottom panel). Both iGPT [9] and masked patch prediction [16] belong to the masked auto-encoding paradigm. MoCo v3 is a contrastive learning method that compares two (224×224) crops. ViT-B, -L, -H are the Vision Transformers proposed in [16]. ViT-BN is modified with BatchNorm, and “/7” denotes a patch size of 7×7 . [†]: pre-trained in JFT-300M.

Chen et al., An Empirical Study of Training Self-Supervised Vision Transformers, FAIR

ImageNet Linear Classification from SSL Features



He et al, "Momentum Contrast for Unsupervised Visual Representation Learning", CVPR 2020
Chen et al, "A Simple Framework for Contrastive Learning of Visual Representations", ICML 2020
Chen et al, "An Empirical Study of Training Self-Supervised Vision Transformers", ICCV 2021

(Lots of caveats here ... different architectures, etc)

Negative-free Contrastive Methods

The problem with negatives...

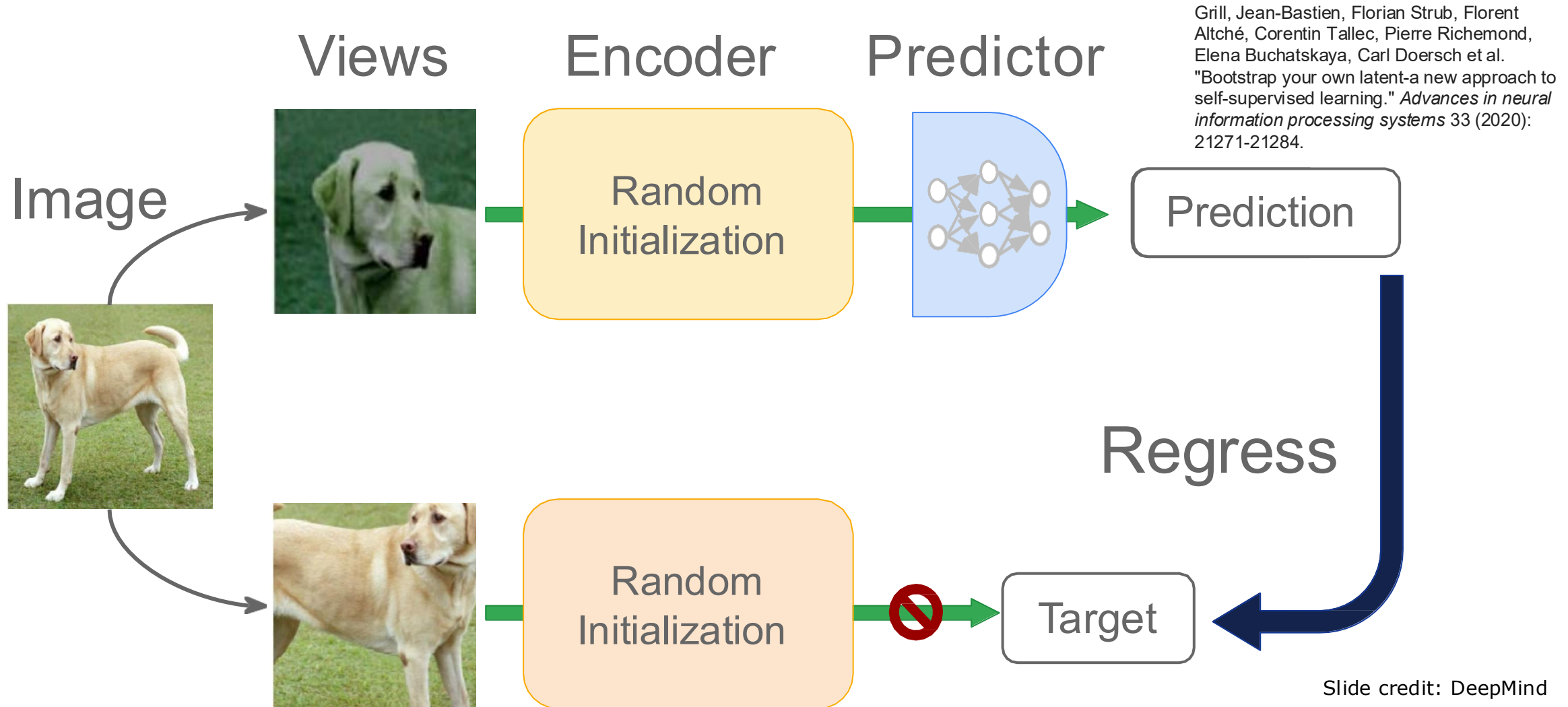
The use of negatives in contrastive learning is powerful, but it comes with problems and trade-offs:

- Sampling Bias:
 - Not all negatives are truly dissimilar
 - In large datasets, negatives may include samples from the same class or semantically similar content
 - This causes false negatives, which hurt learning – the model is penalized for pulling together things that should be similar
 - *Example:* Two clips of people swimming — sampled from different videos but semantically the same action — are treated as negatives.
- Requires Large Batches or Memory Banks
 - InfoNCE converges to the Mutual Information lowerbound for large N
 - This leads to the need for large batches
 - If there are not enough negatives, representations may collapse
- Negatives make methods difficult to tune
 - Methods are sensitive to the augmentation strategy used to produce positives

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

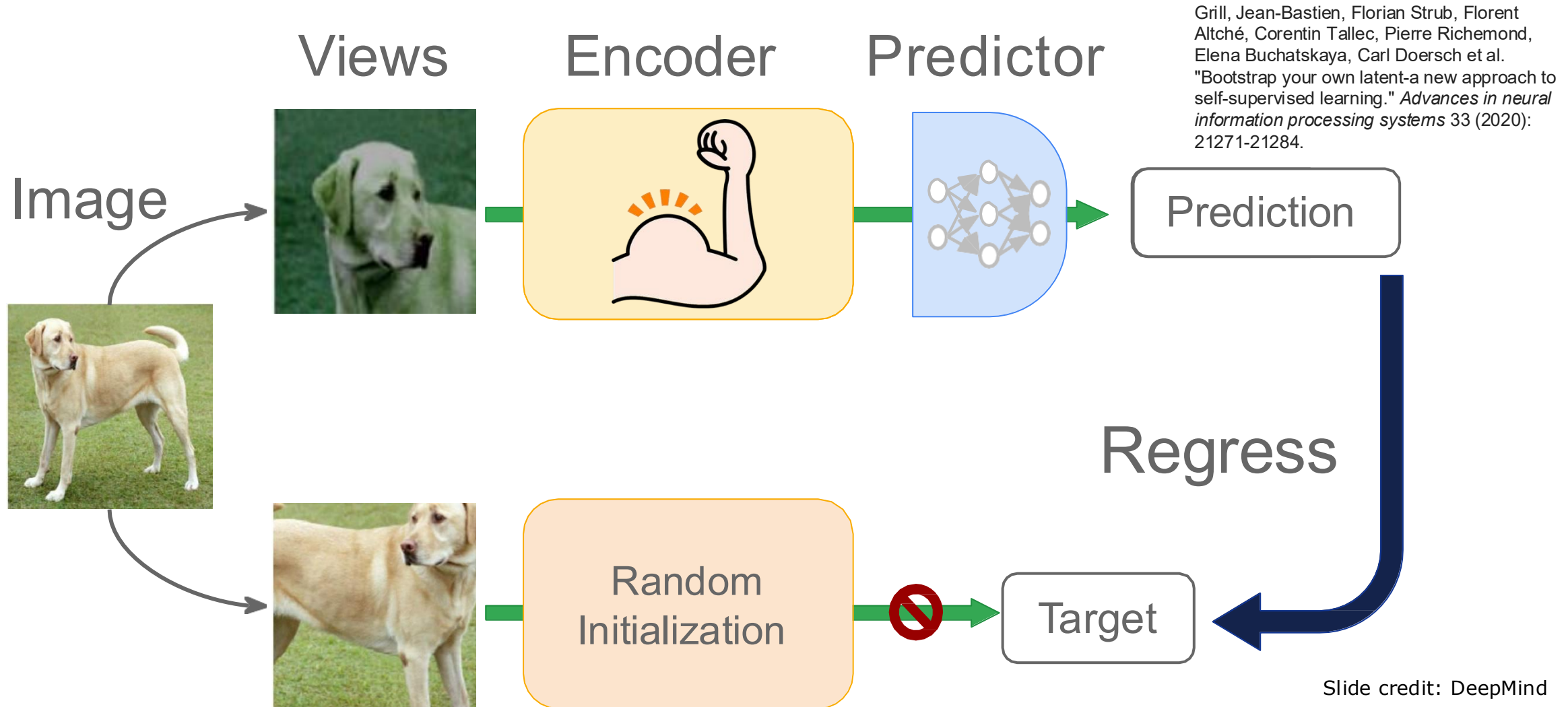


Slide credit: DeepMind

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames



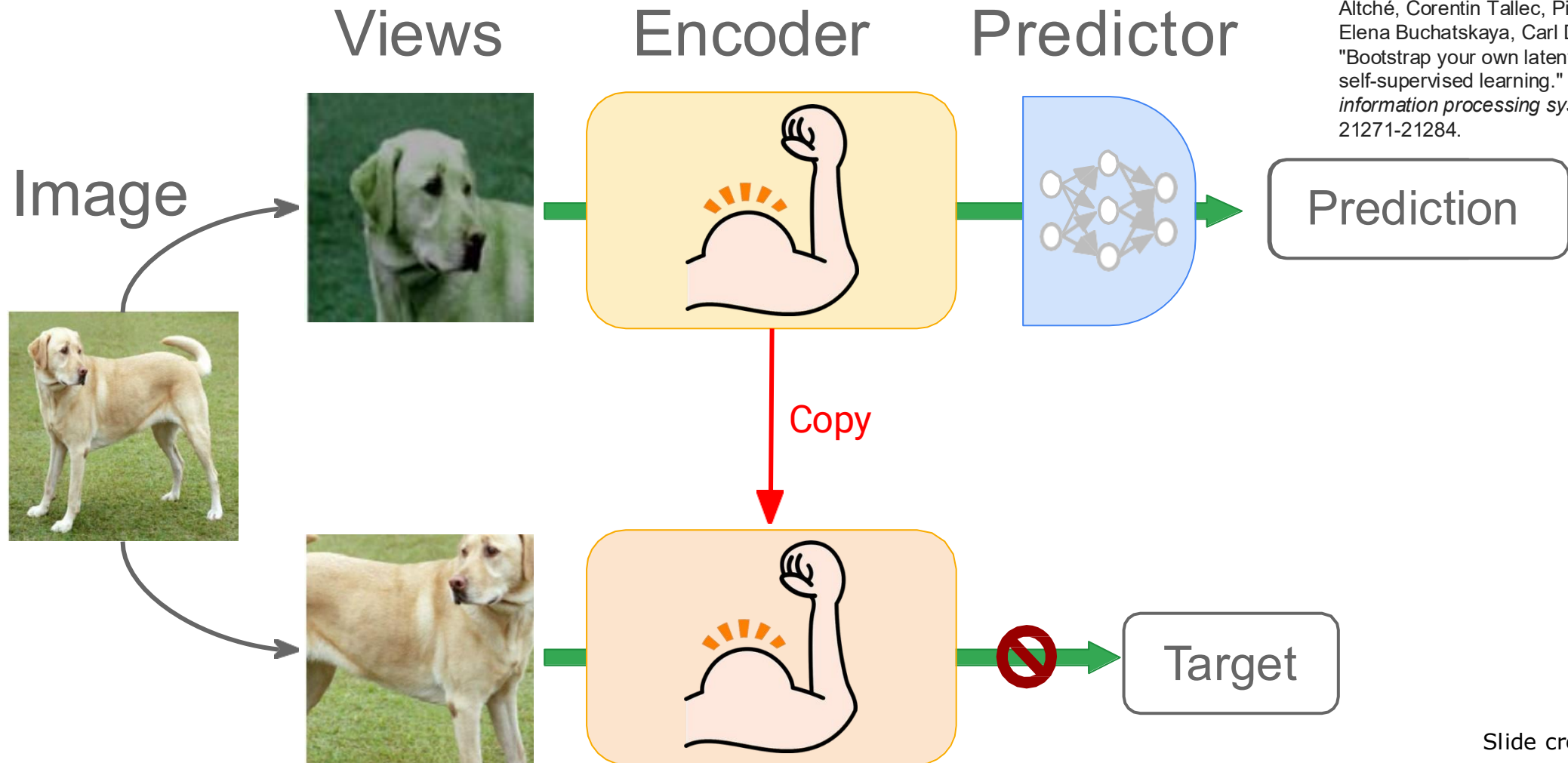
Slide credit: DeepMind

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

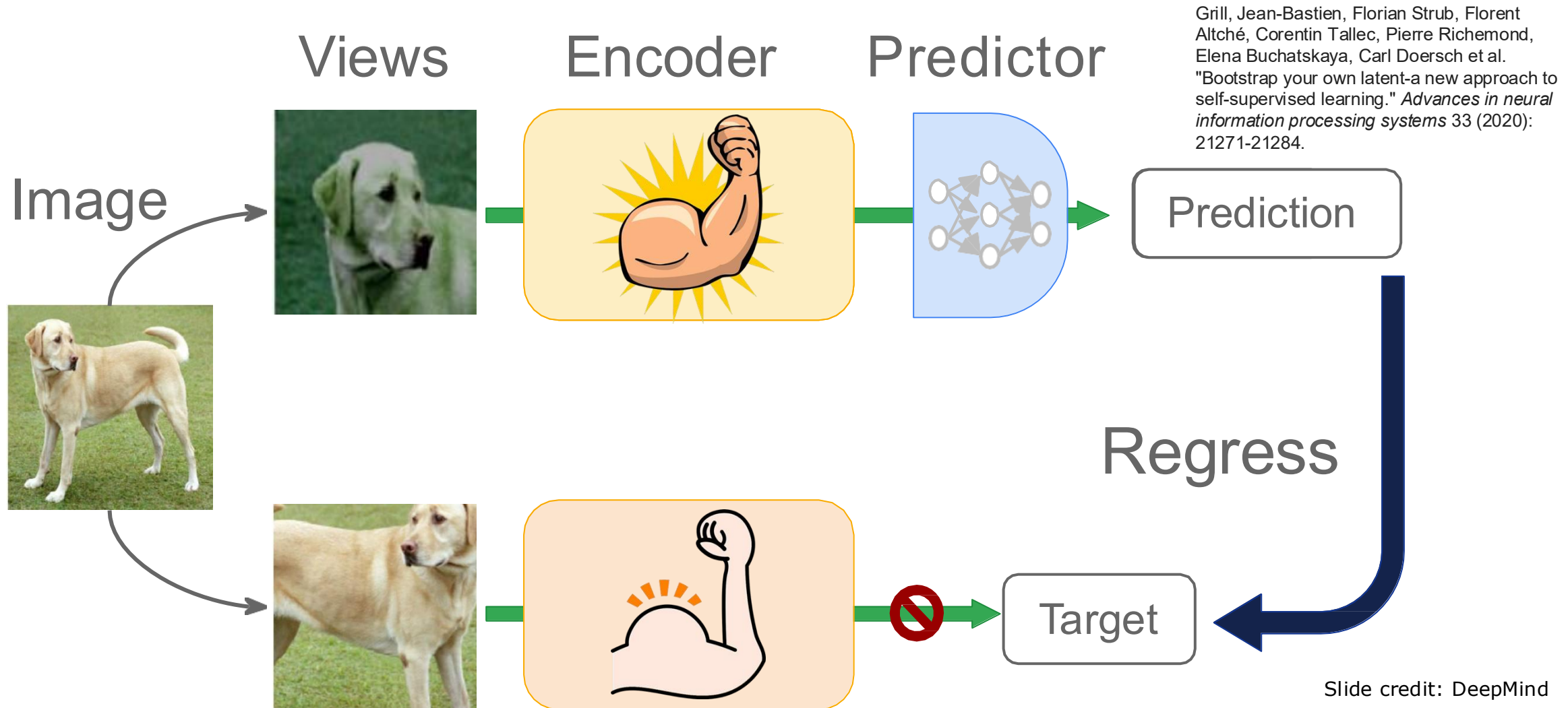
Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al.
"Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.



Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames



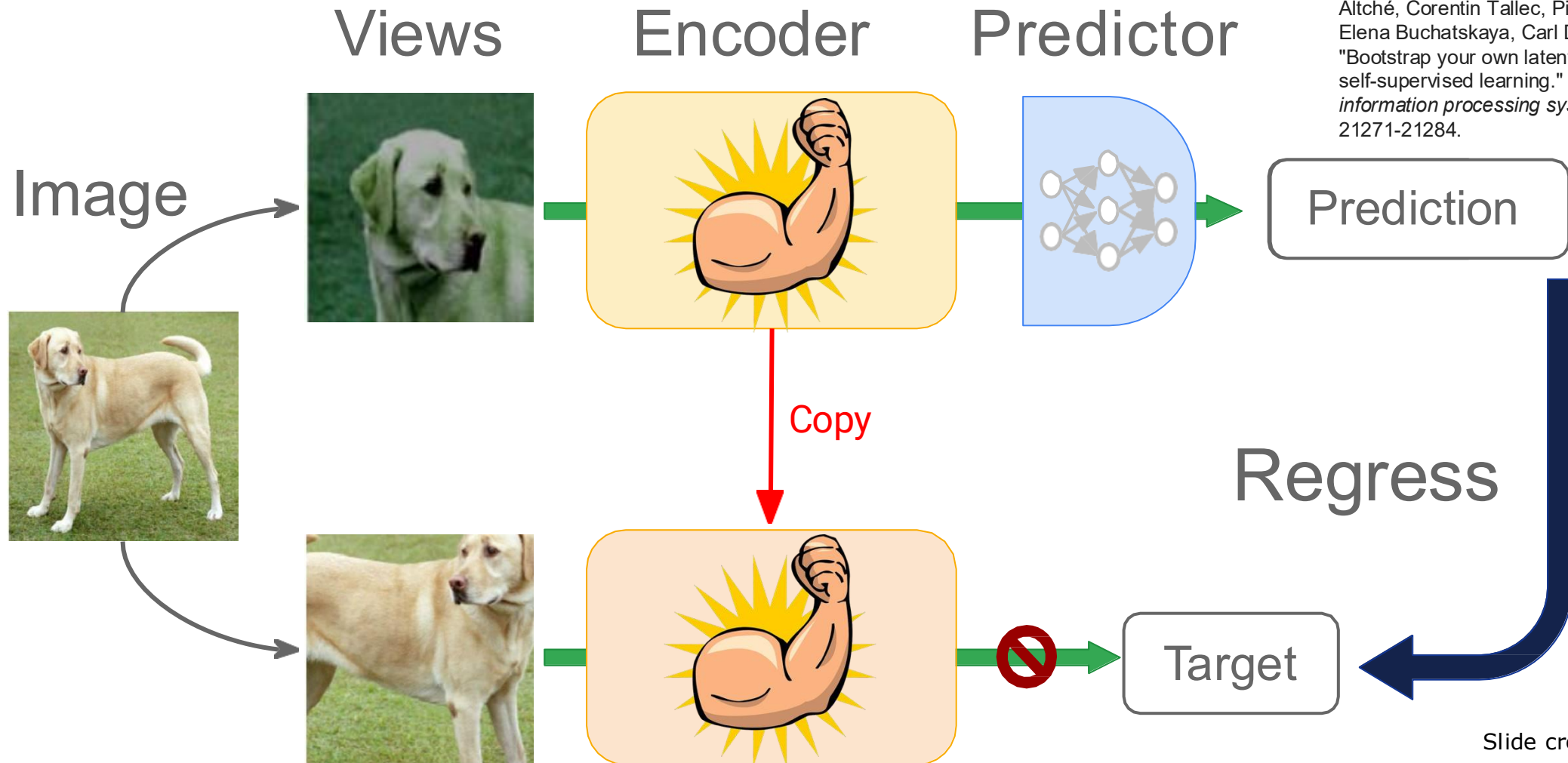
Slide credit: DeepMind

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al.
"Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.

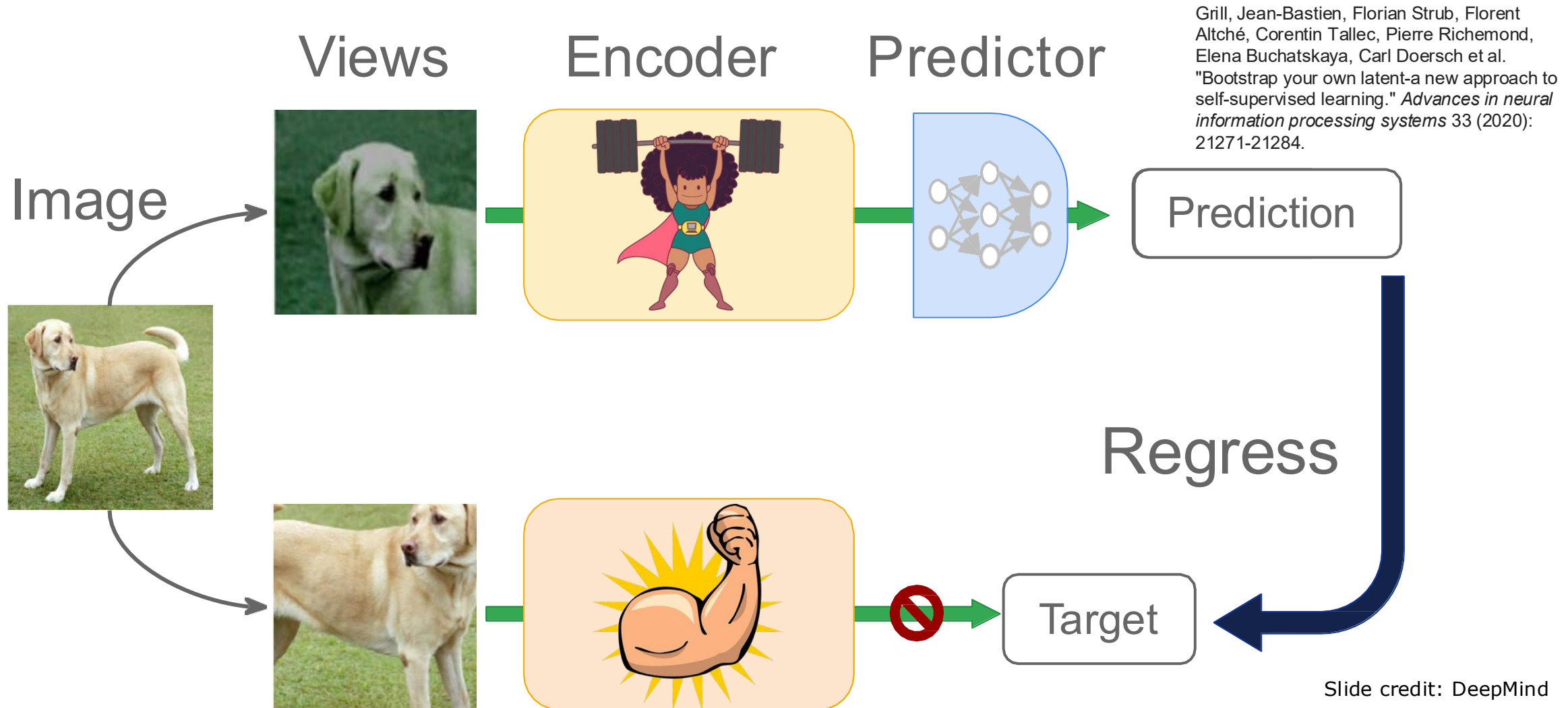


Slide credit: DeepMind

Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

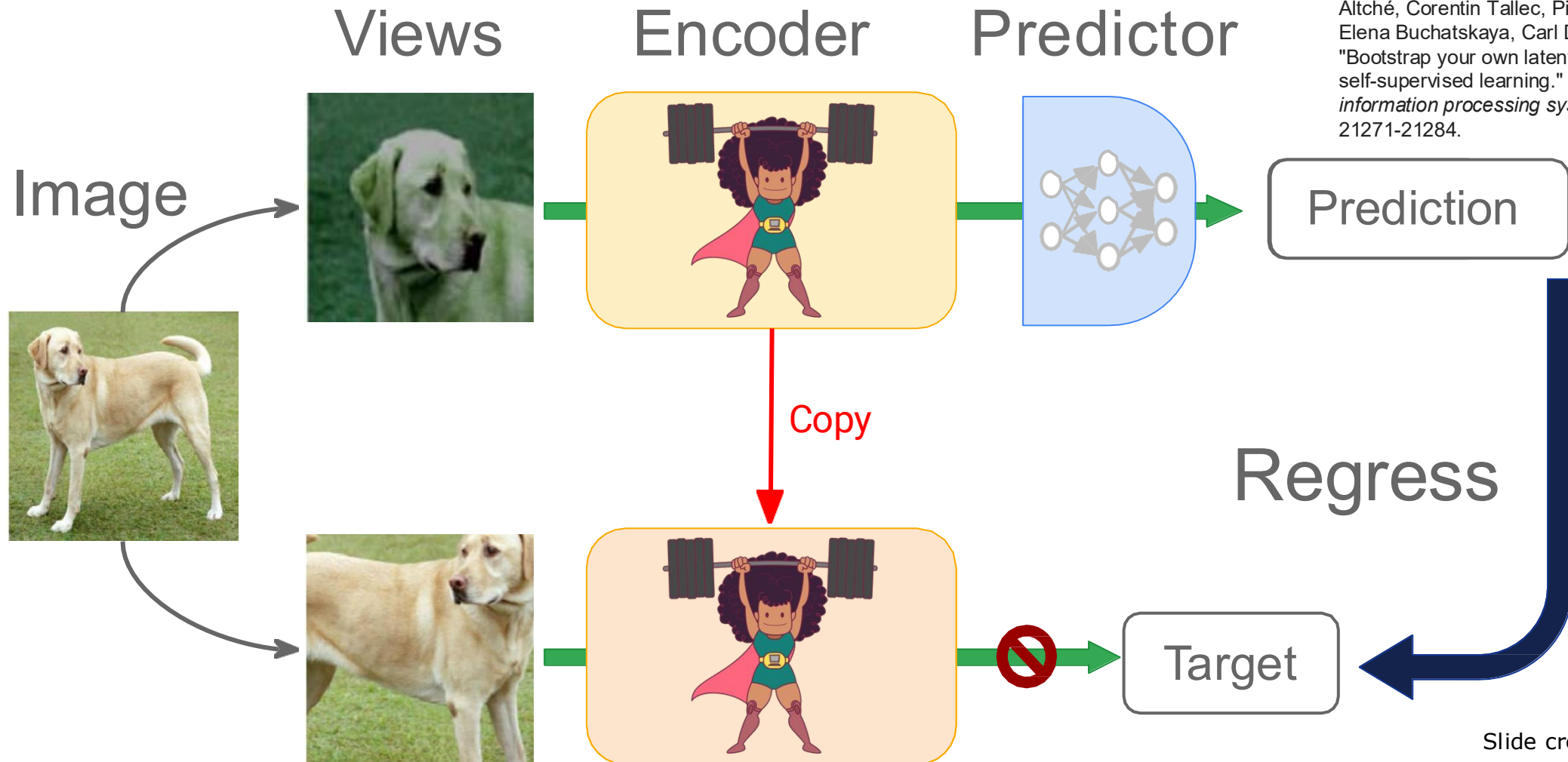


Bootstrap Your Own Latent (BYOL)

Idea: train two separate encoders independently with no negatives

Since the encoders are not trained jointly, they will not collapse to same representation for all frames

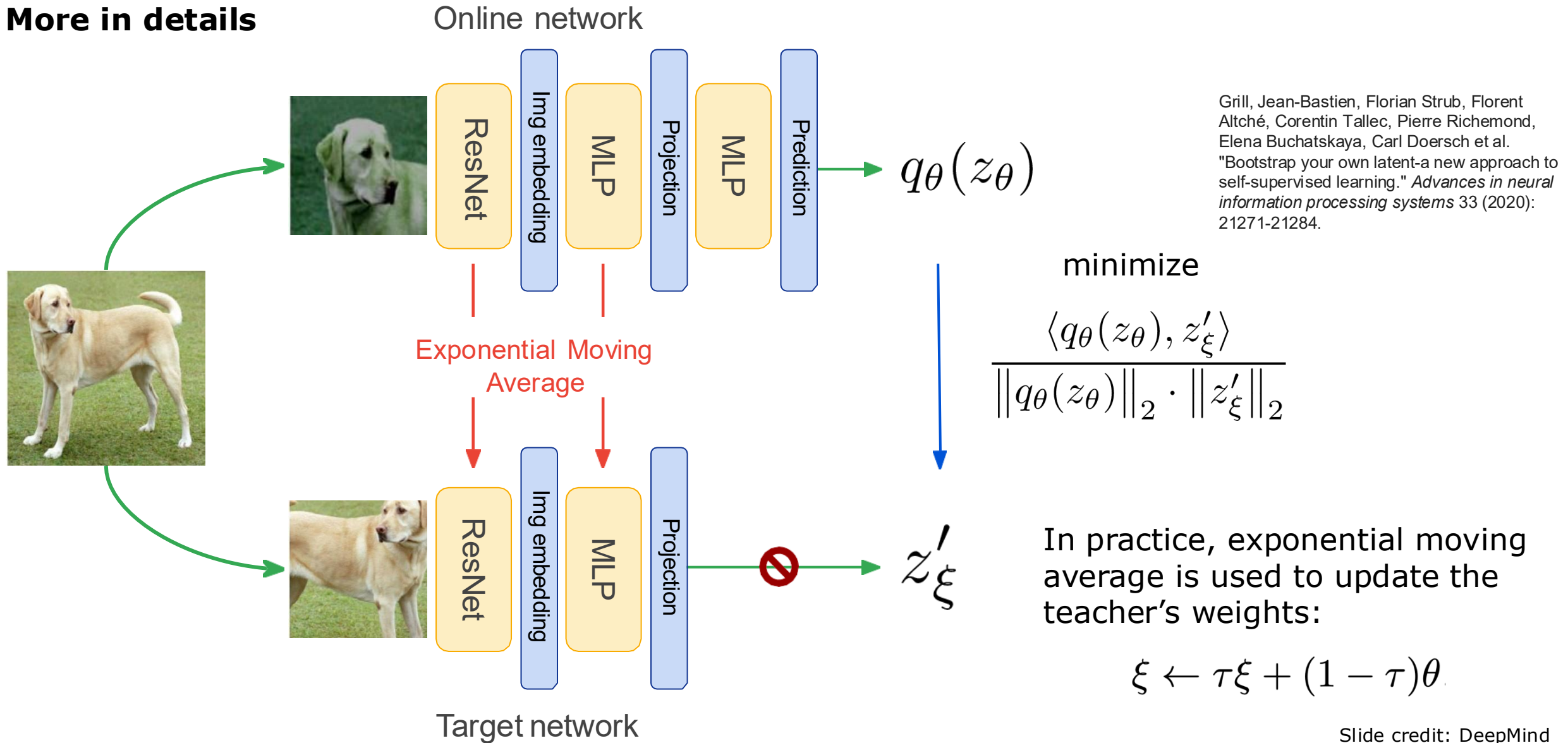
Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al.
"Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.



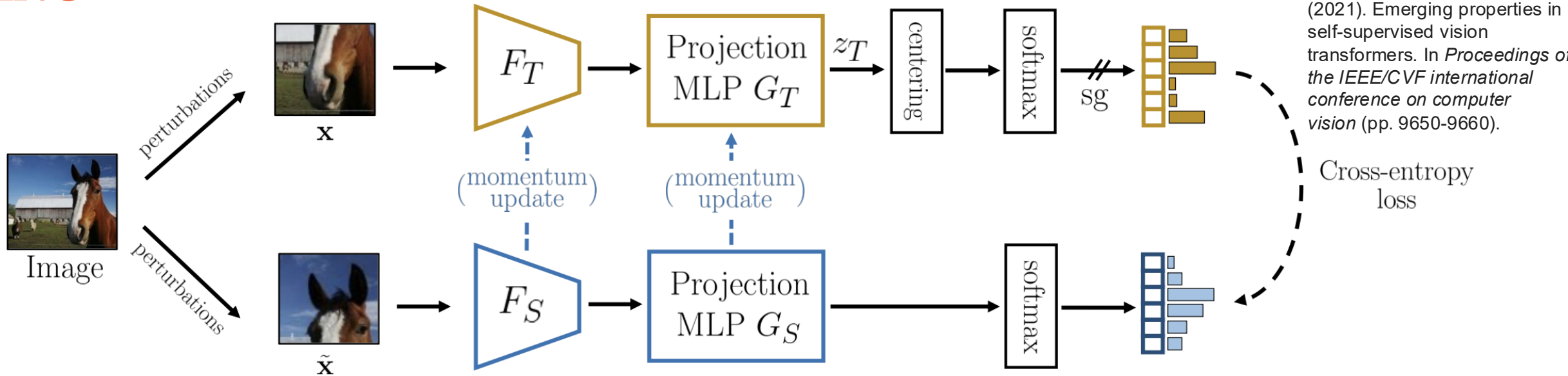
Slide credit: DeepMind

BYOL Architecture

More in details



Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch et al. "Bootstrap your own latent-a new approach to self-supervised learning." *Advances in neural information processing systems* 33 (2020): 21271-21284.



Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9650-9660).

DINO extends over BYOL and previous works with some key changes:

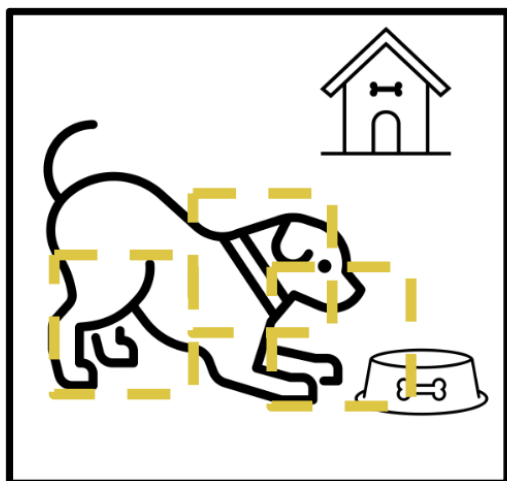
- No prediction head post-processing
- Features are centered (subtract running mean of teacher's output) and passed through a softmax with low temperature – this creates a sharp distribution which is used as a pseudo-label
- A cross-entropy loss is used to train the student, rather than MSE

Method	Mom.	Loss	Pred.	k -NN	Lin.
DINO	✓	CE	✗	72.8	76.1
	✗	CE	✗	0.1	0.1
	✓	MSE	✗	52.6	62.4
	✓	CE	✓	71.8	75.6
BYOL	✓	MSE	✓	66.6	71.4

DINO Details – Multi-Crop Strategy

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 9650-9660).

Local crops



Global crops



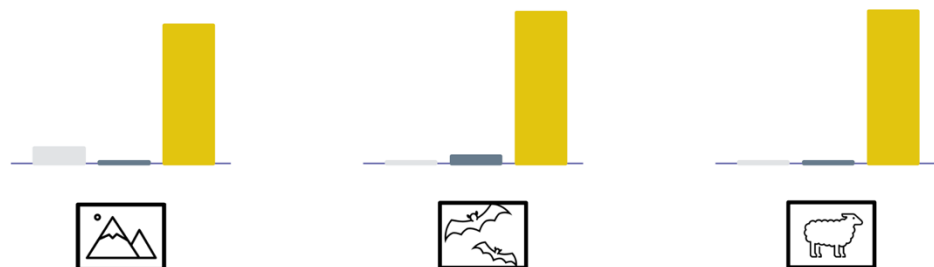
Local predict the pseudo-label of global

Local-to-global matching

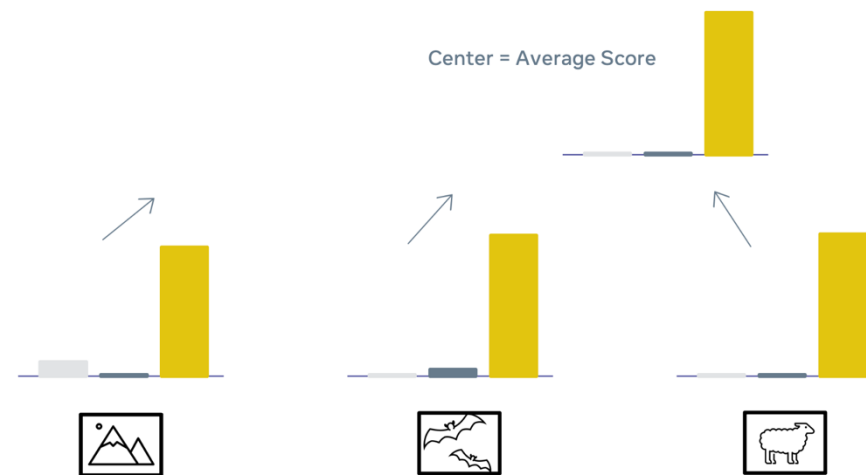
A multi-crop strategy is applied, where a «local-to-global» matching is learned.

DINO Details – Centering and Sharpening

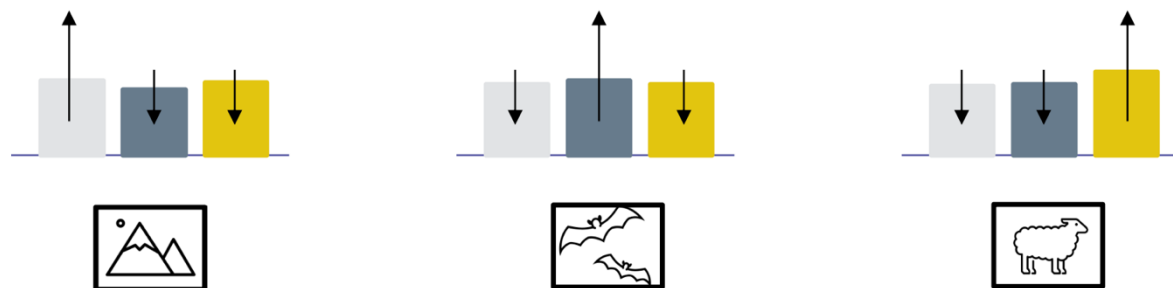
Some dimensions in the embedding space might have a wider scale, hence dominating the distribution of activations.



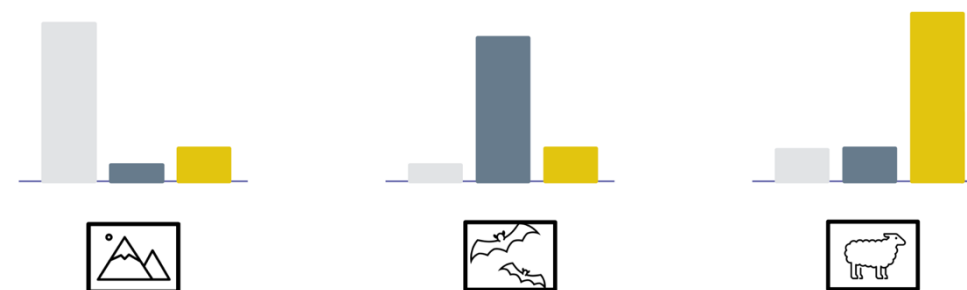
1) Original Features



2) Computing Average (running)



3) Centered Features (average subtracted)



4) Sharpening (low temperature softmax)

DINOv2: Learning Robust Visual Features without Supervision

Maxime Oquab**, Timothée Darcet**, Théo Moutakanni**,
Huy V. Vo*, Marc Szafraniec*, Vasil Khalidov*, Pierre Fernandez, Daniel Haziza,
Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba,
Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat,
Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal¹,
Patrick Labatut*, Armand Joulin*, Piotr Bojanowski*

Meta AI Research ¹*Inria*

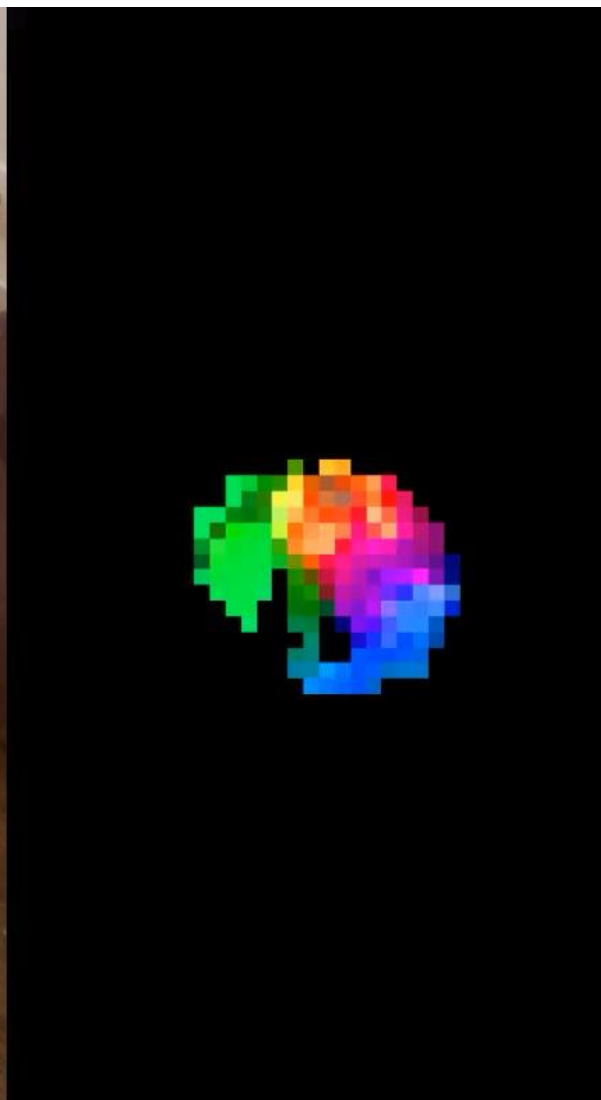
*core team **equal contribution

Reviewed on OpenReview: <https://openreview.net/forum?id=a68SUt6zFt>

Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez et al. "Dinov2: Learning robust visual features without supervision." *arXiv preprint arXiv:2304.07193* (2023).

DINOv2 – Emerging Properties

<https://dinov2.metademolab.com>



DINOv2 – Emerging Properties

<https://dinov2.metademolab.com>



Dense Matching

DINOv3

Oriane Siméoni* Huy V. Vo* Maximilian Seitzer* Federico Baldassarre* Maxime Oquab*
Cijo Jose Vasil Khalidov Marc Szafraniec Seungeun Yi Michaël Ramamonjisoa
Francisco Massa Daniel Haziza Luca Wehrstedt Jianyuan Wang
Timothée Darcet Théo Moutakanni Leonel Sentana Claire Roberts
Andrea Vedaldi Jamie Tolan John Brandt¹ Camille Couprie
Julien Mairal² Hervé Jégou Patrick Labatut Piotr Bojanowski

Meta AI Research ¹*WRI* ²*Inria*

*corresponding authors: {osimeoni,huyvvo,seitzer,baldassarre,qas}@meta.com

<https://ai.meta.com/research/dinov3/>

Reconstructive Approaches

Masked Auto-Encoding

- Very simple method, but highly effective
- BERT-like masked modeling objective, but with crucial design changes for computer vision

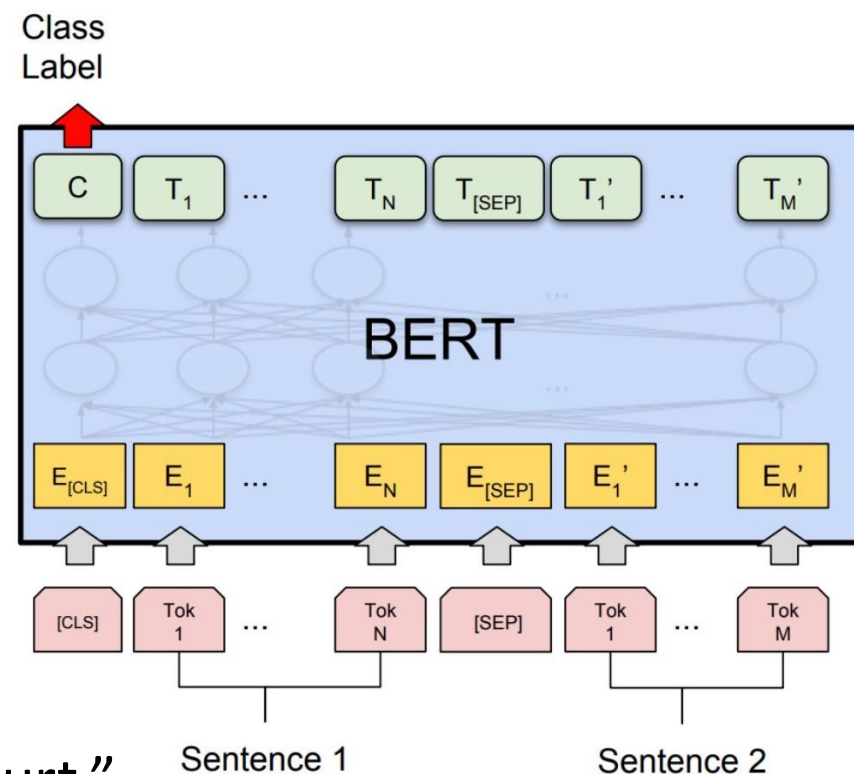
men

basketball

“Two
Uni
ct

playing a game of

on an outside court.”



Masked Auto-Encoding

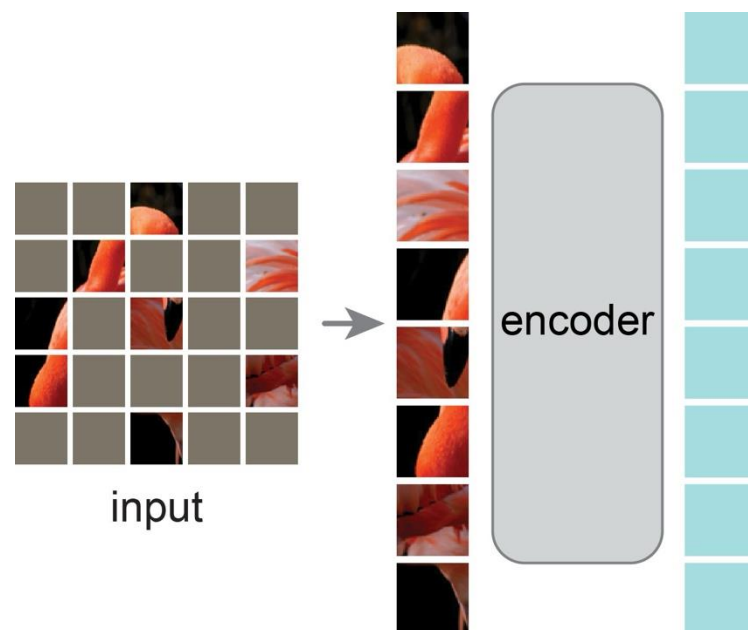
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Random masking

Masked Auto-Encoding

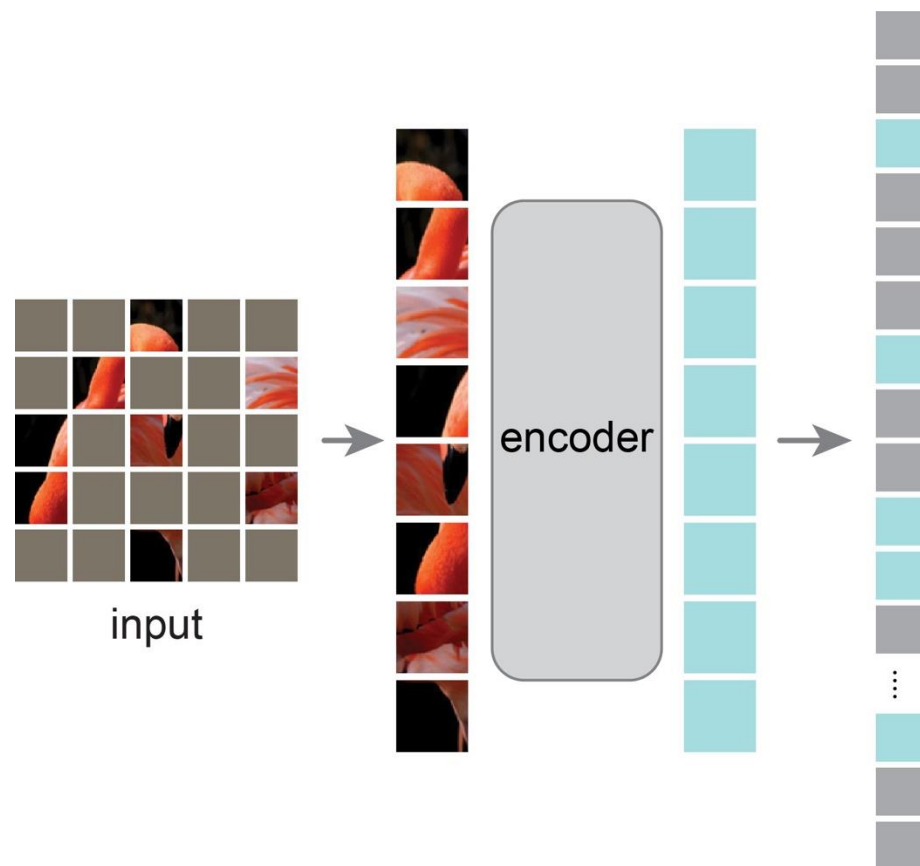
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Encode visible patches

Masked Auto-Encoding

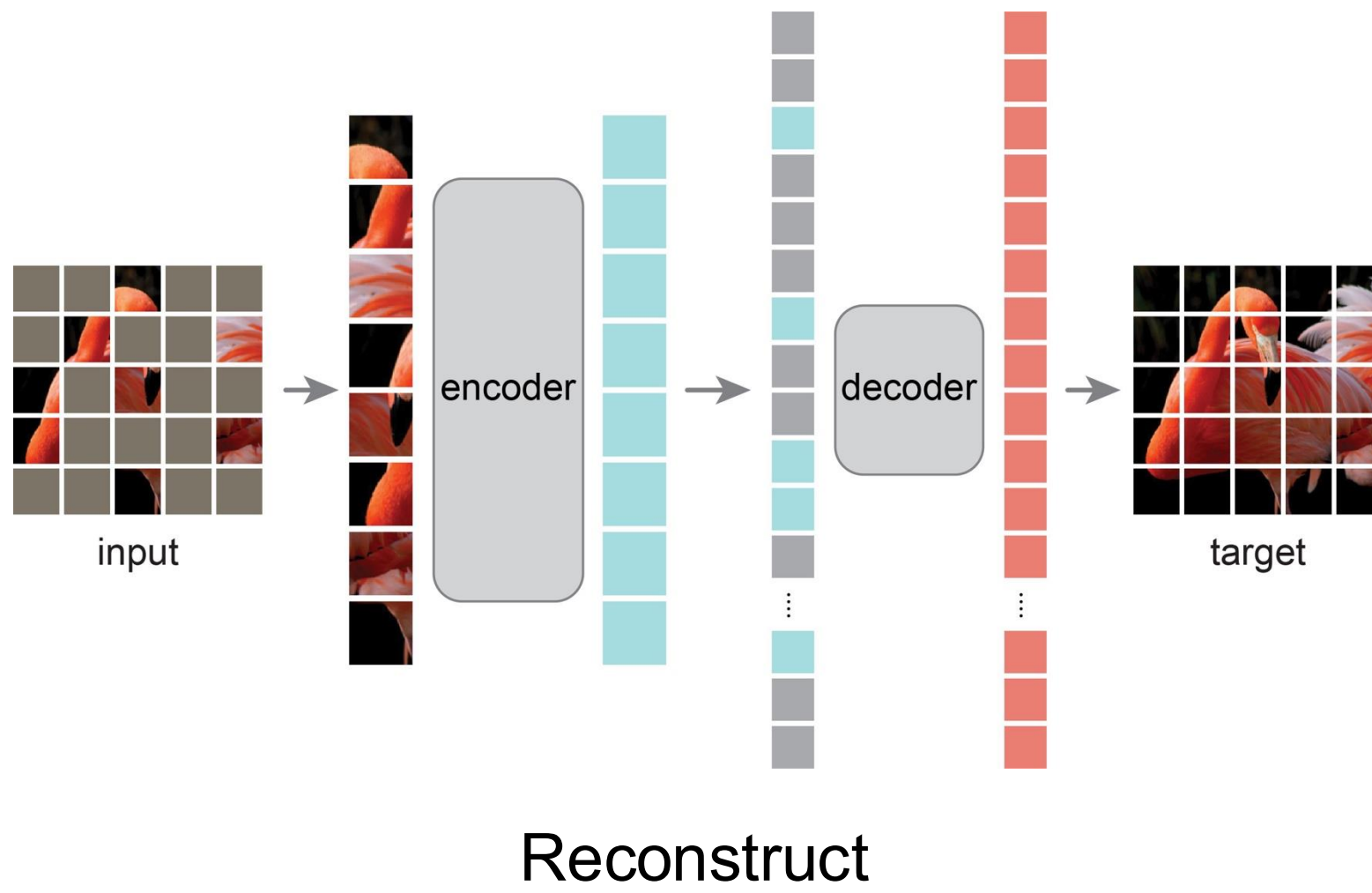
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Add mask tokens

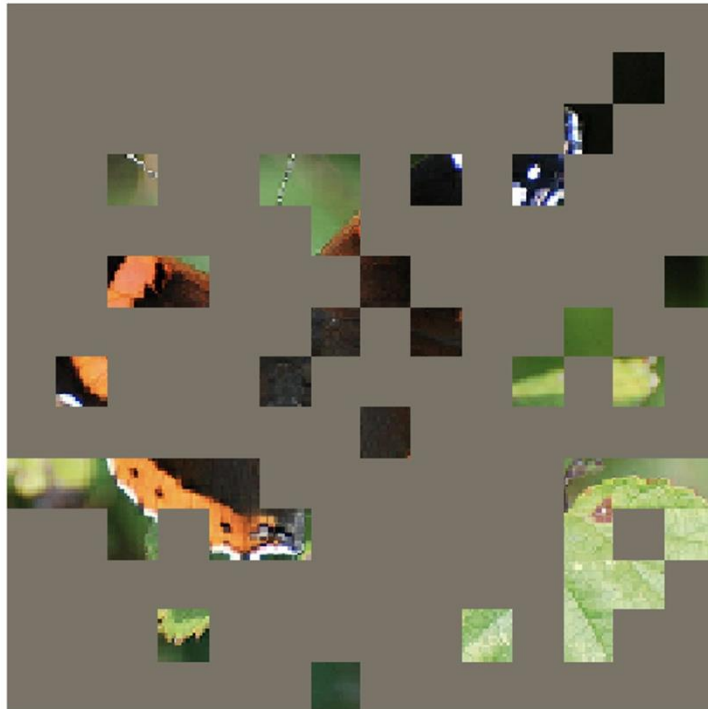
Masked Auto-Encoding

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).

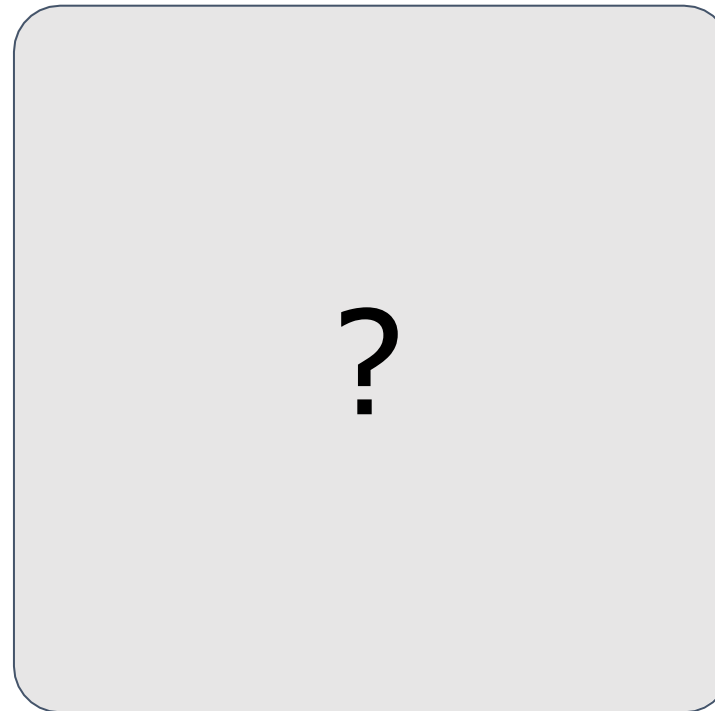


Reconstruction Error

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



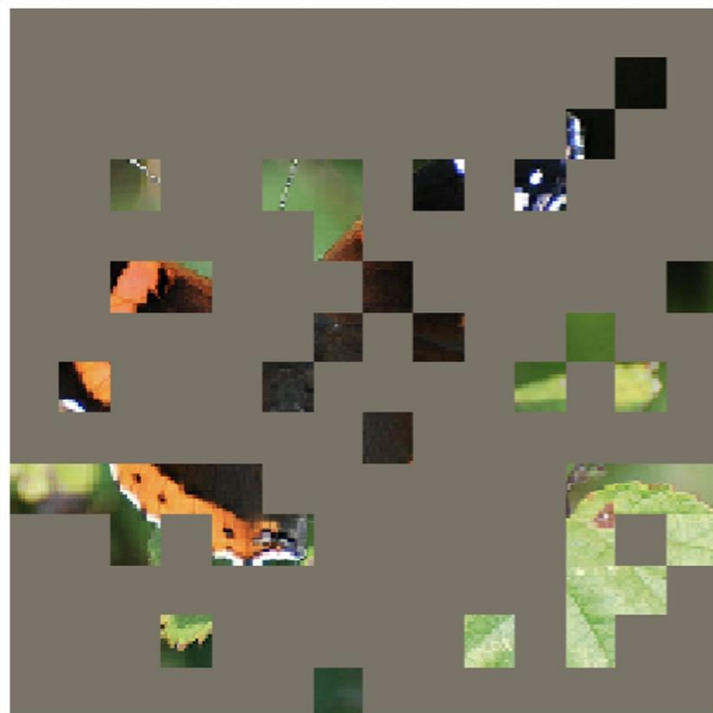
Masked input: 80%



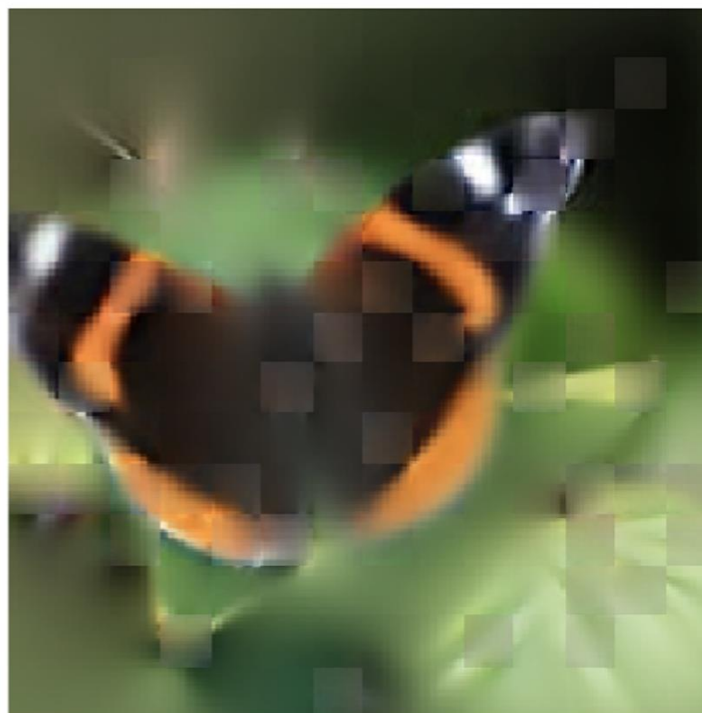
You guess?

Reconstruction Error

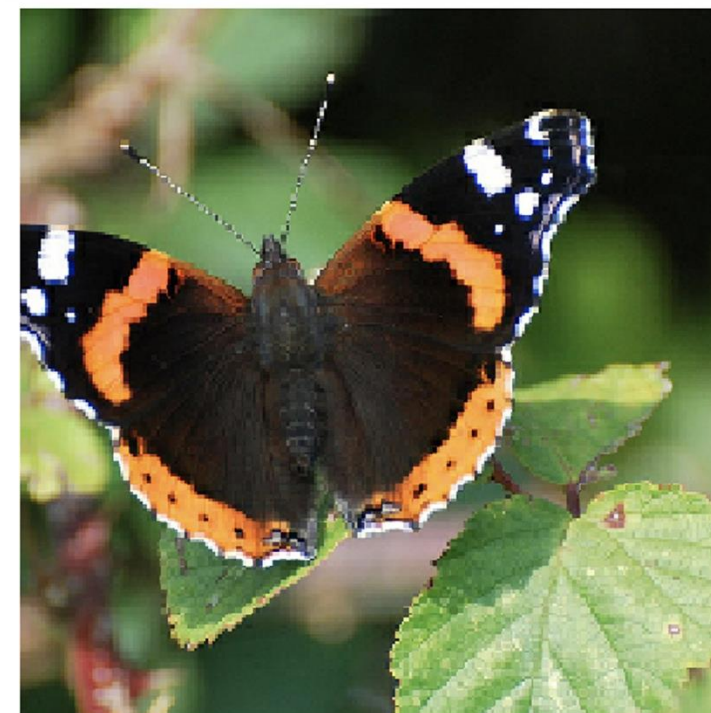
He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16000-16009).



Masked input: 80%



MAE's guess

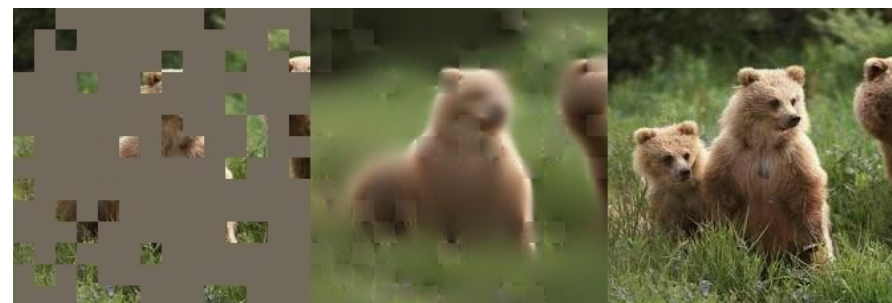
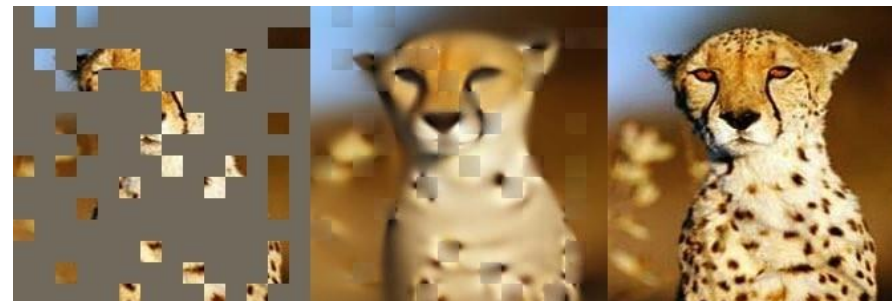
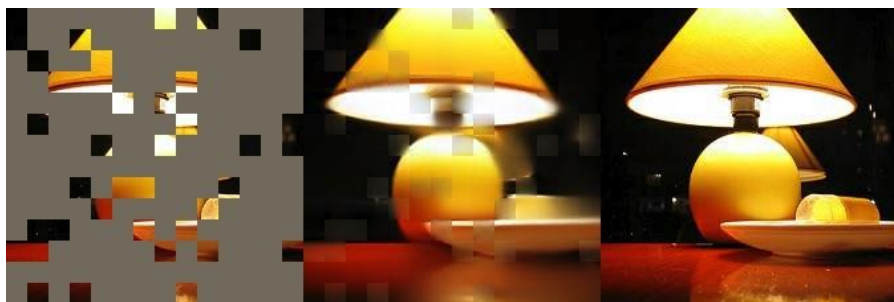
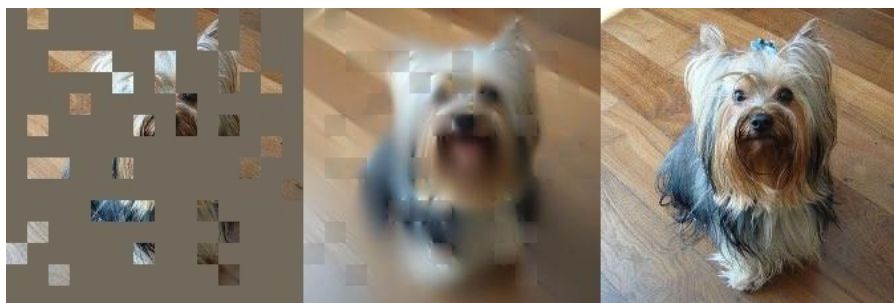


Ground truth

Reconstruction Error

ImageNet val set (unseen)

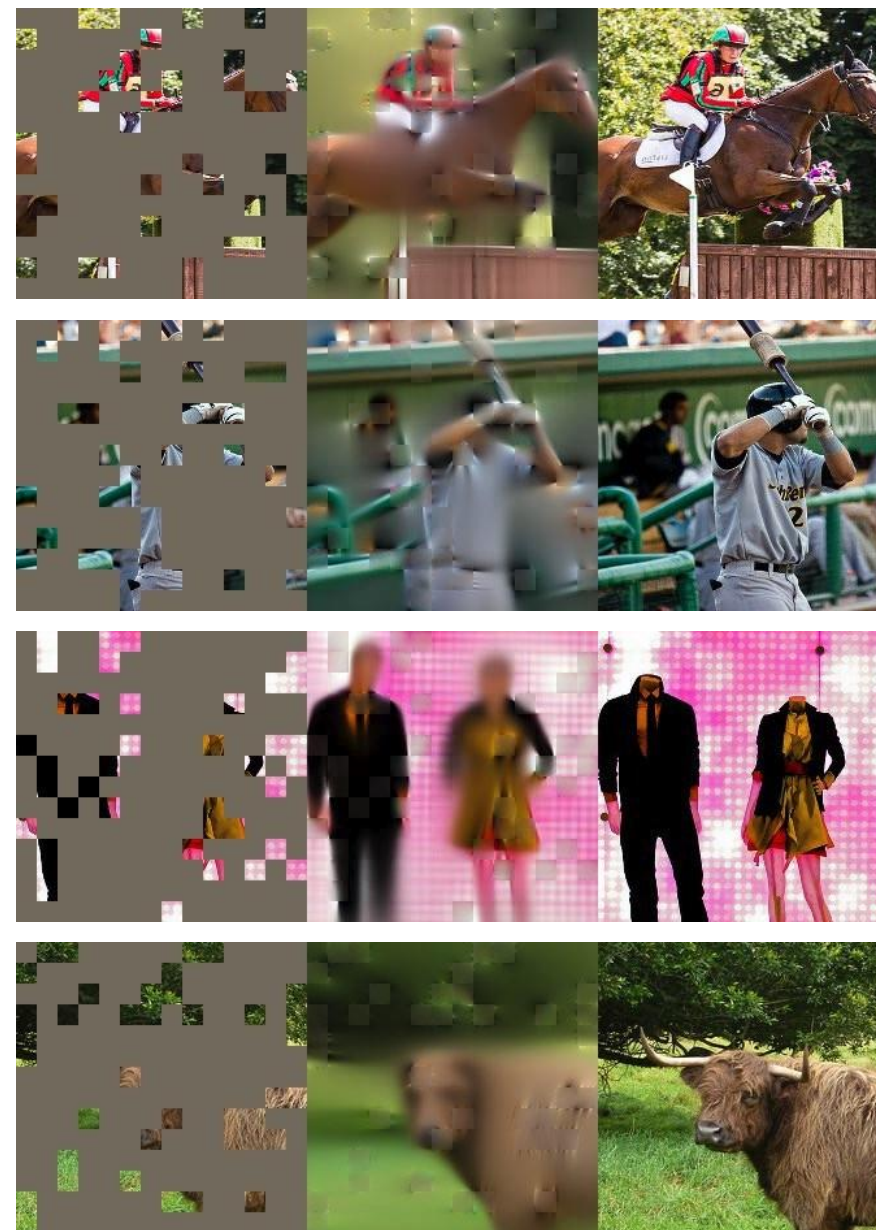
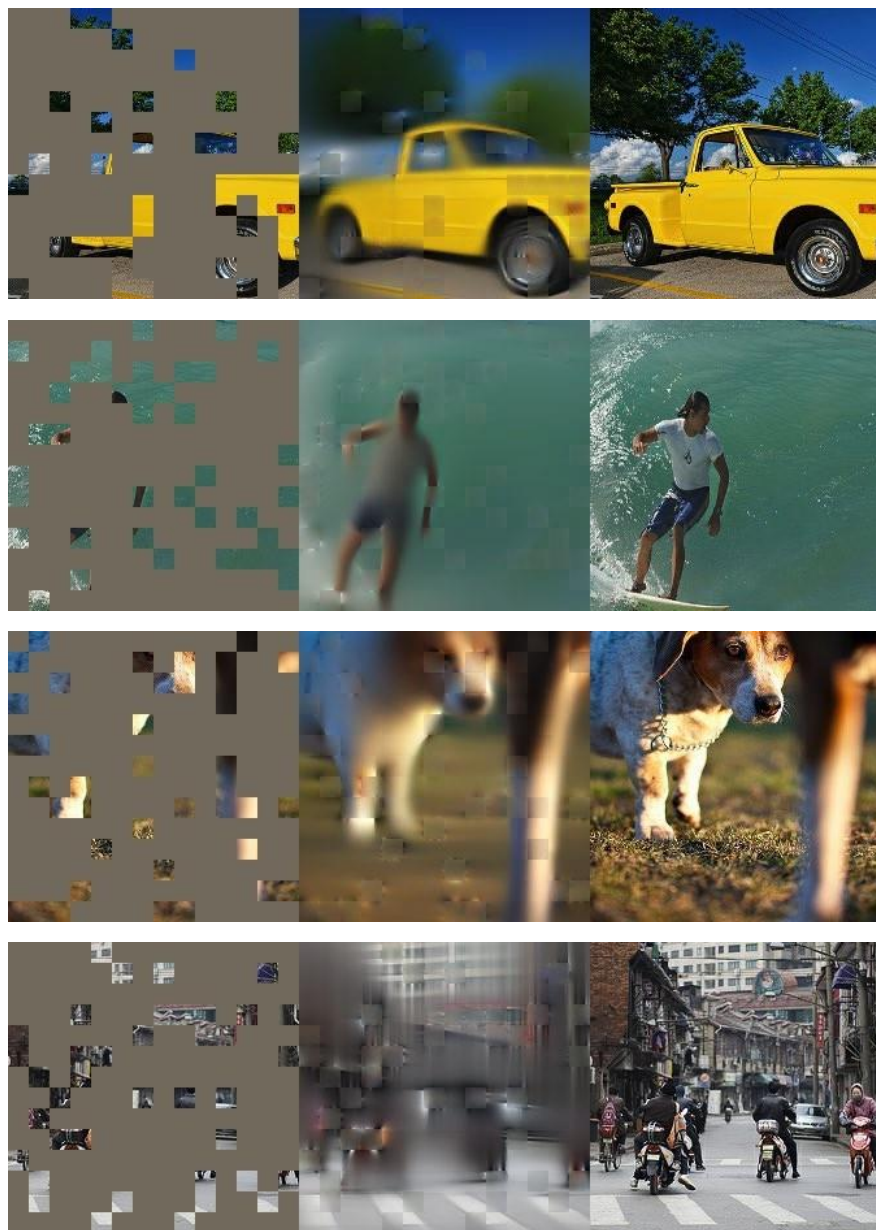
Uni
ct



Reconstruction Error

ImageNet val set (unseen)

Uni
ct



MAE Can Generalize

75% mask

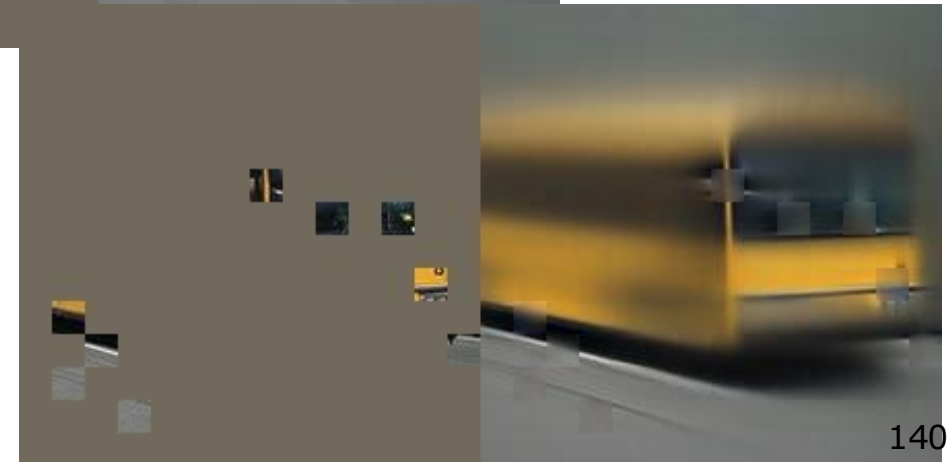


original

85% mask

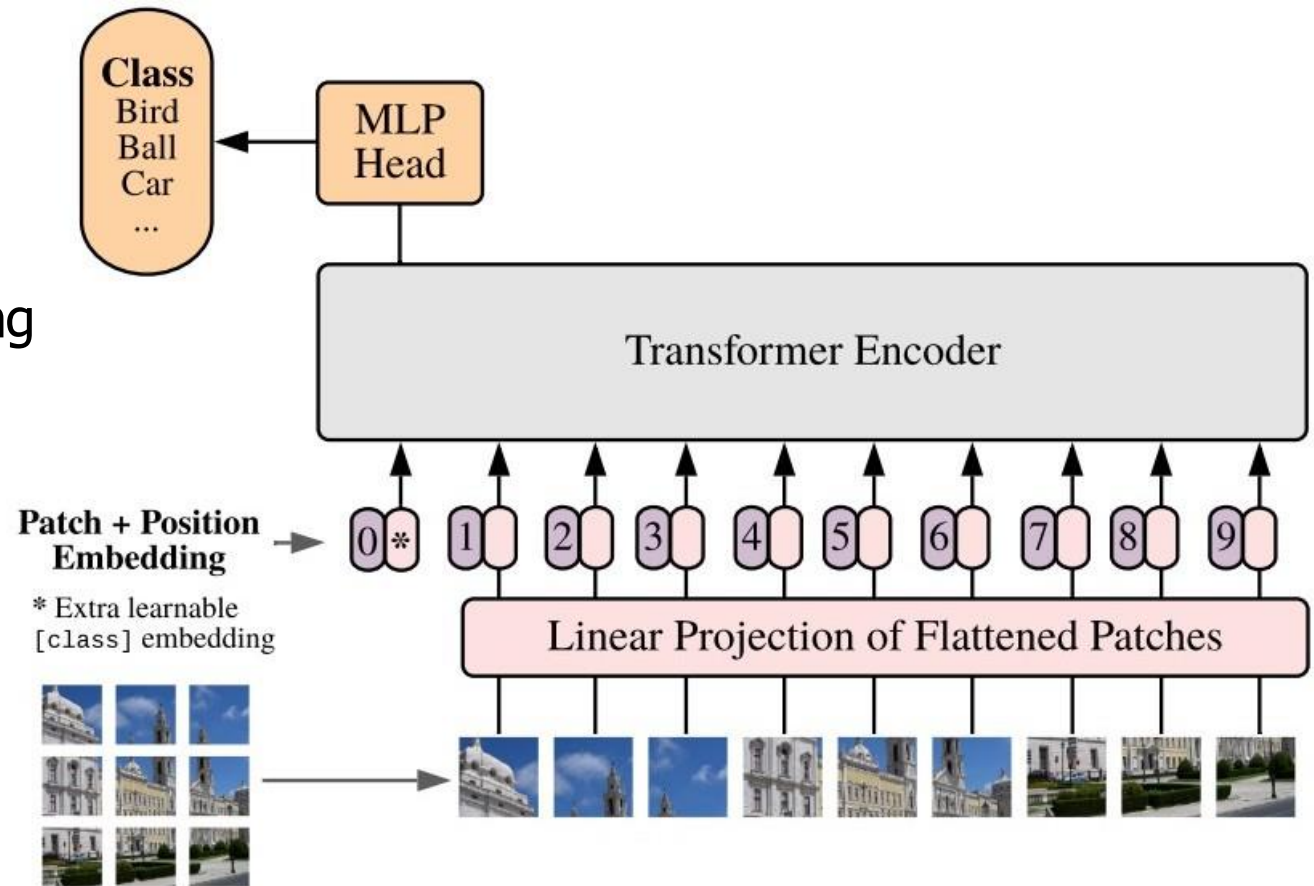


95% mask



BERT-Like: Transformers

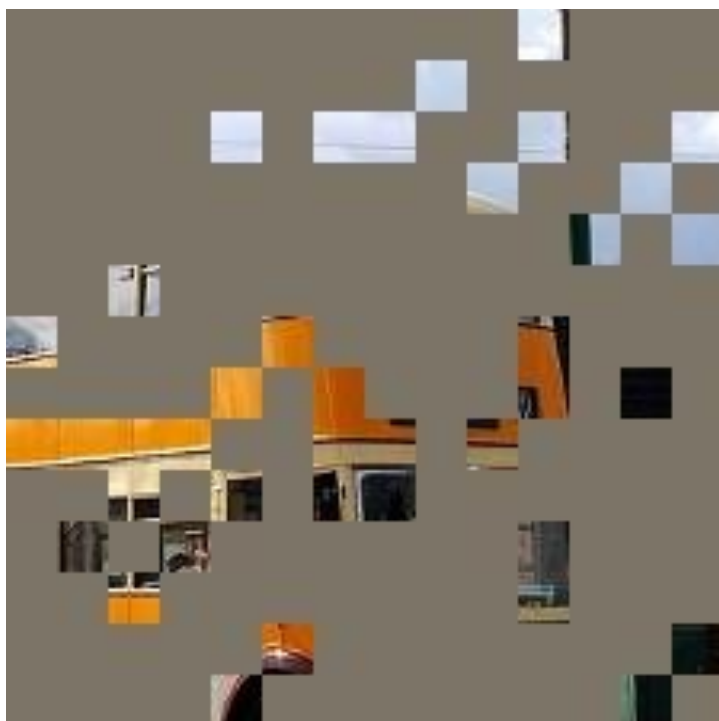
- Vision Transformer (ViT)
 - Less inductive bias
 - Non-overlapping tokenization
 - Easier for masked auto-encoding
- *Scalable*
 - with larger models
 - on larger datasets



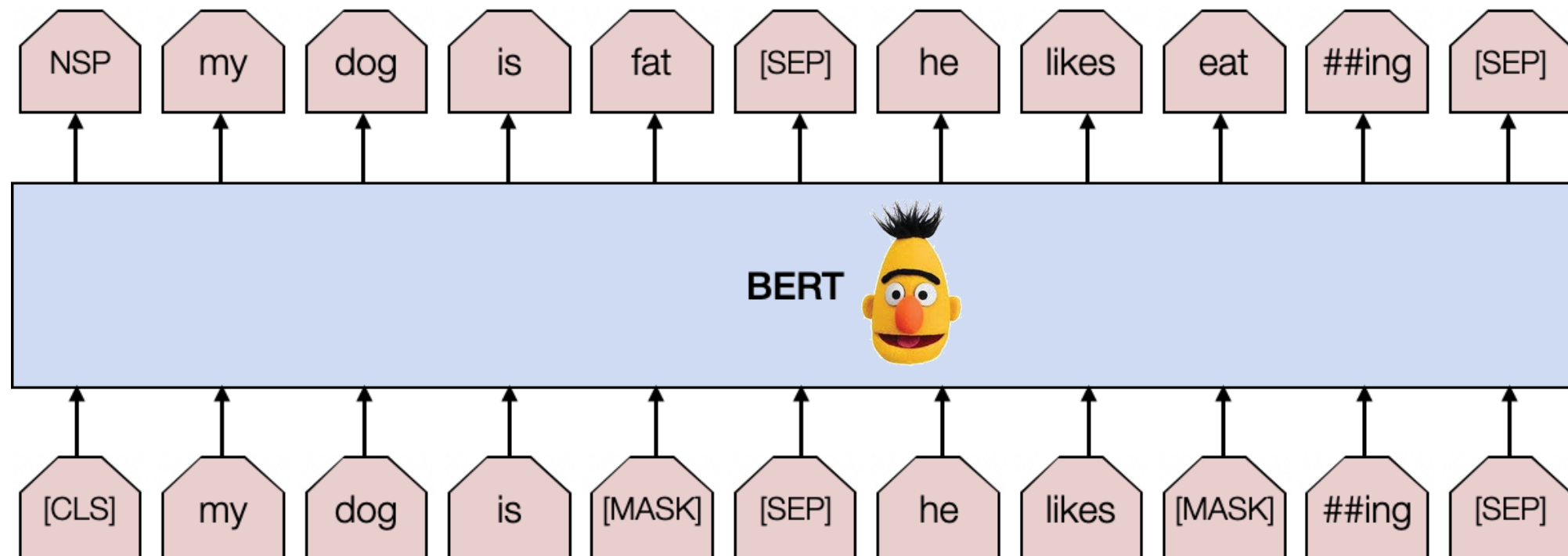
BERT-Unlike: Mask Ratio

[Devlin et al, NAACL 2019]

- BERT: 15% is enough to create a challenging task
- MAE: a high ratio of 75% - 80% to be meaningful



- BERT: encoder-*only* pre-training

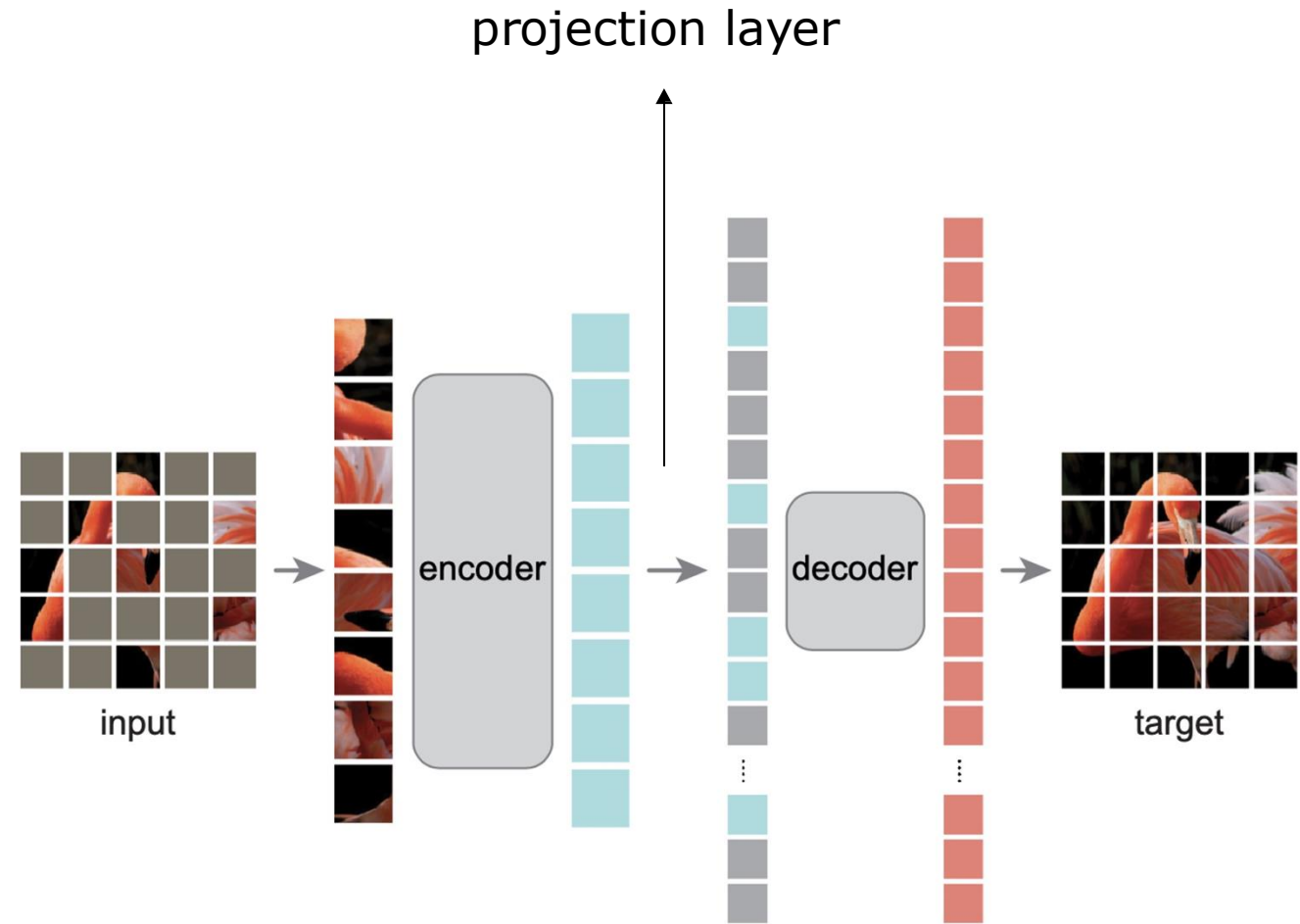


BERT-*unlike*: Encoder-Decoder

MAE:

- *Large* encoder on *visible* tokens
- Small decoder on *all* tokens
- Projection layer to connect the two

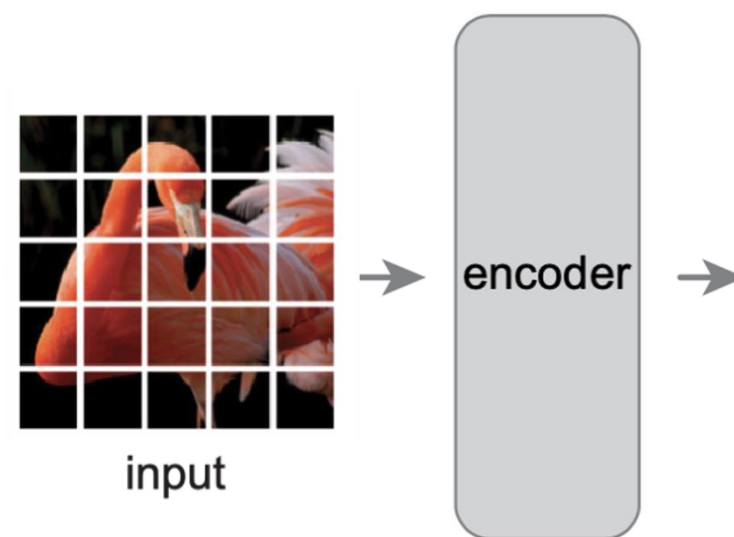
Very efficient when coupled with
high mask ratio (75%)

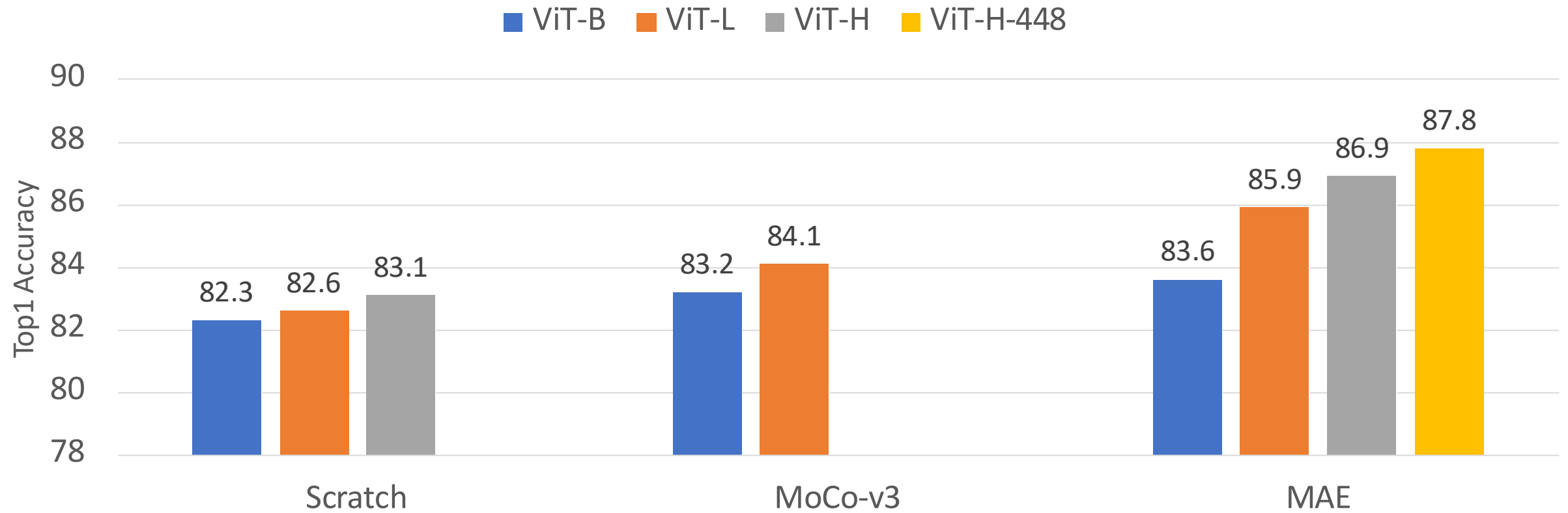


[Devlin et al, NAACL 2019]

MAE for Downstream Tasks: *Encoder Only*

- After MAE pre-training, just *throw away* the decoder
- Encoder is used for representations with *full-sequence* input

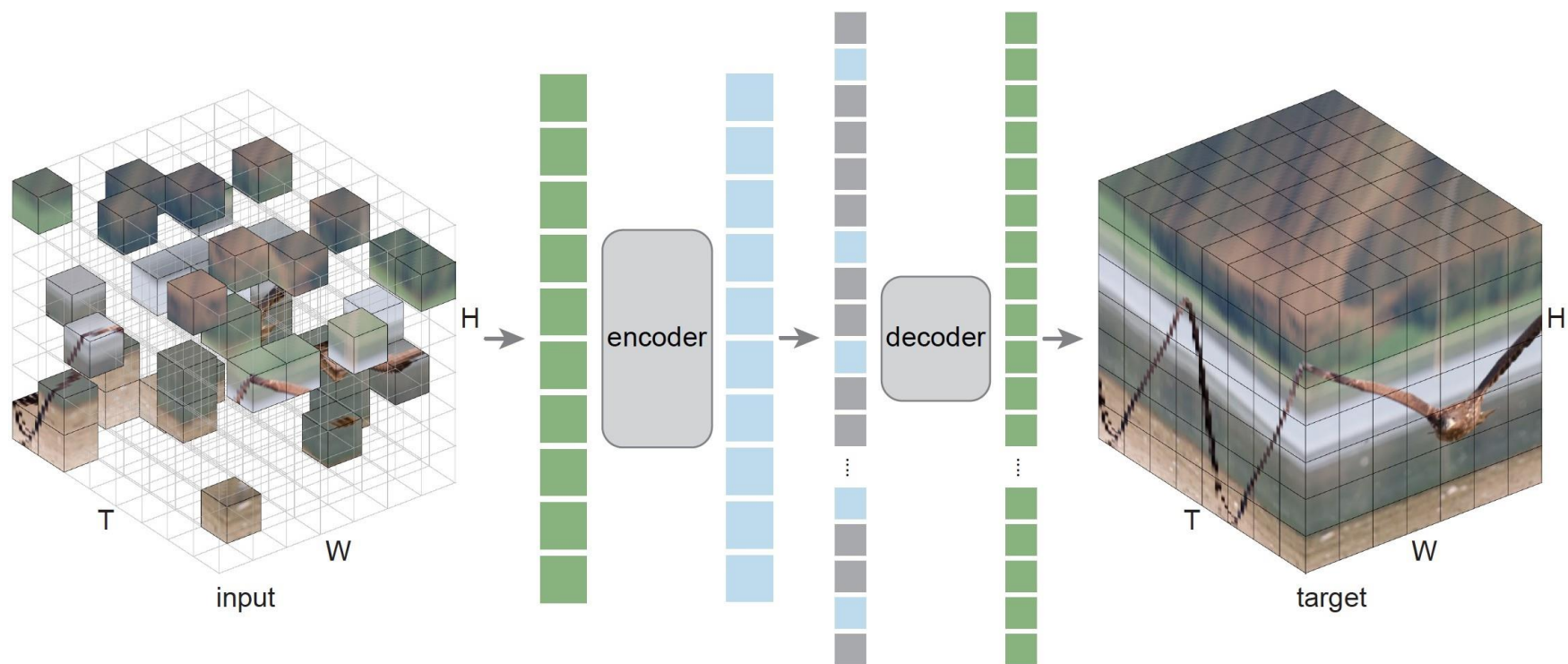




MAE Pretraining outperforms training from scratch, and allows scaling to larger ViT models

Reconstructive Approaches (Video Based)

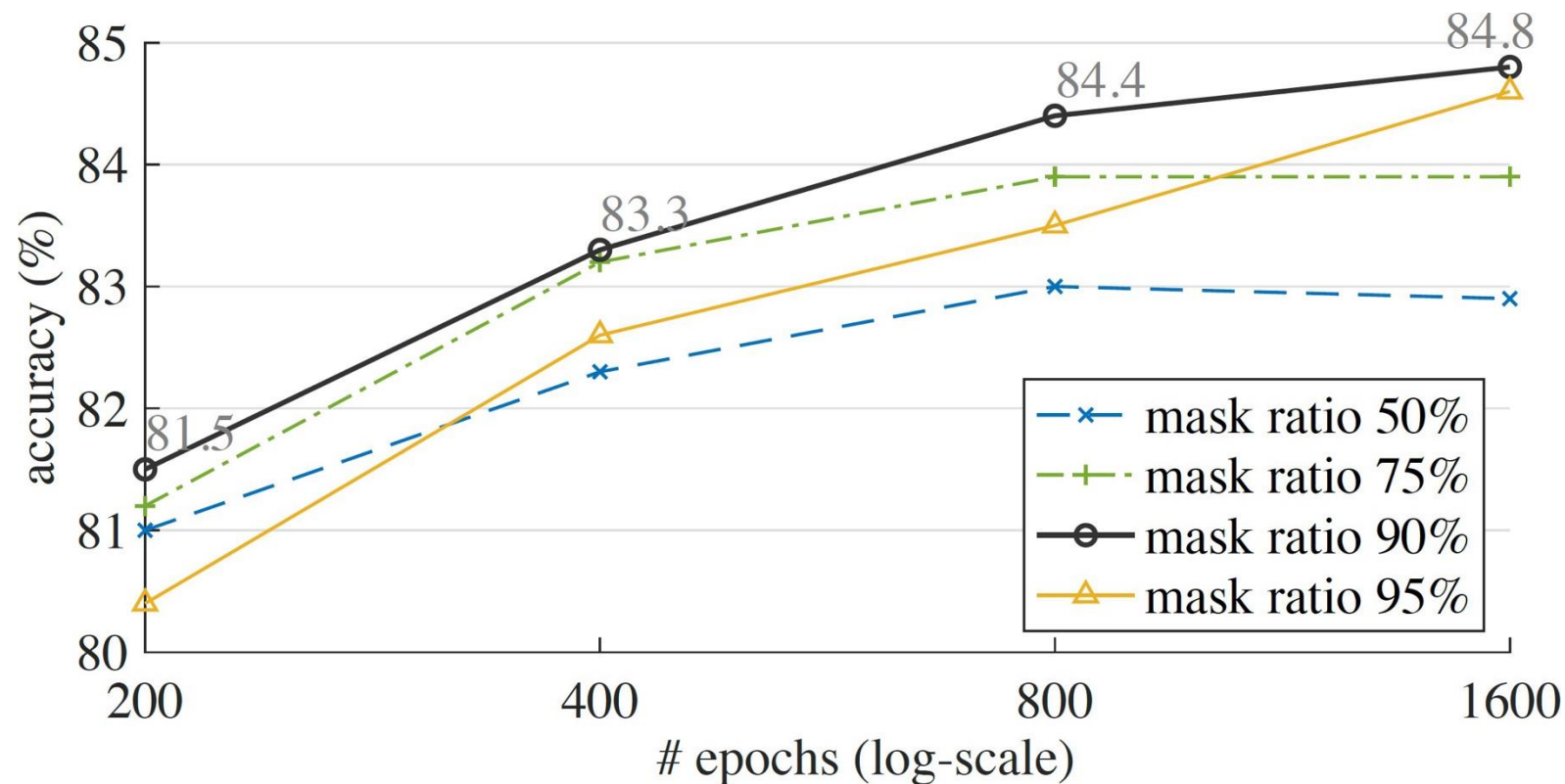
Masked Autoencoders as spatiotemporal learners



- Masking of random patches in spacetime
- Encoder operates on the set of visible patches
- A small decoder on encoded patches and mask tokens reconstruct input
- Except for patch and positional embeddings, no inductive bias

Masking ratio can be extremely high

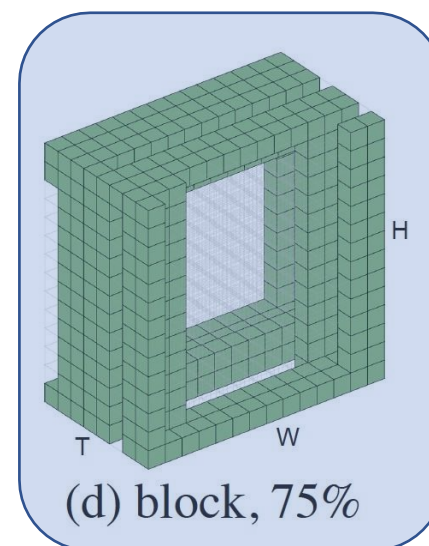
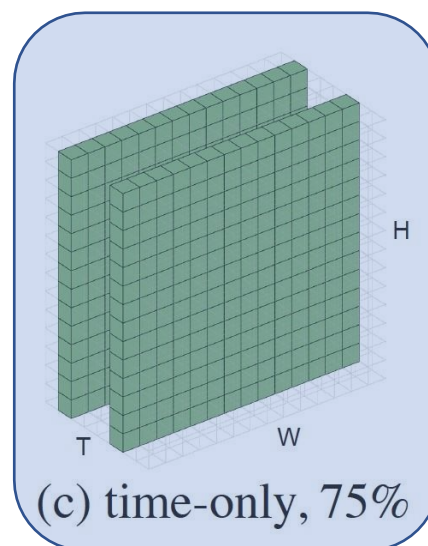
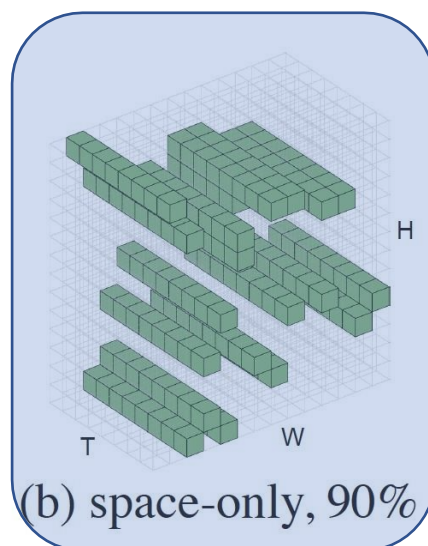
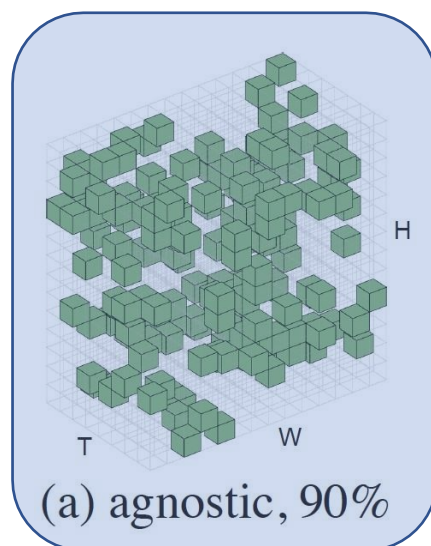
- Task:
Kinetics-400 (K400)
video classification
- Metric: accuracy (acc.)
- Model: ViT-L
- Pre-train: 200-1600 epochs
- Fine-tune: 100 epochs
- Training from scratch: 71.4%



- For image classification, 75% is the optimal value, but for video 90% is

Uni
ct considerably better

Masking can be agnostic in spacetime



case	ratio	acc.
agnostic	90	84.4
space-only	90	83.5
time-only	75	79.1
block	75	83.2

MAE visualizations

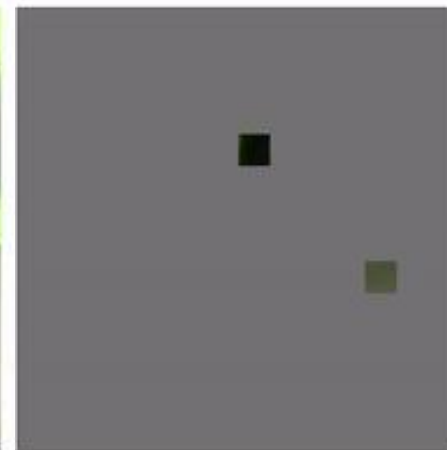
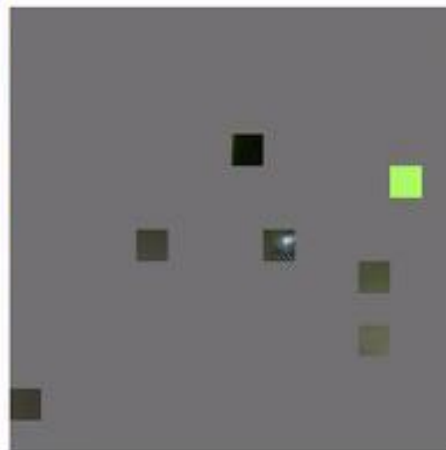
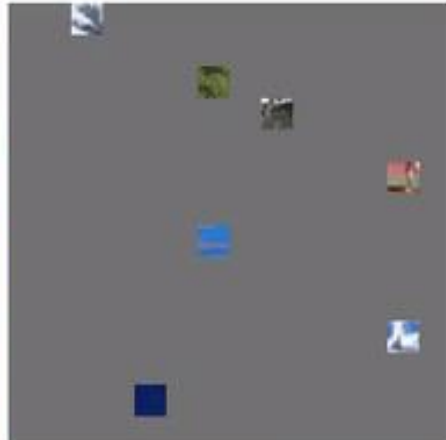
original

input 95% masked

output 95%

input 98% masked

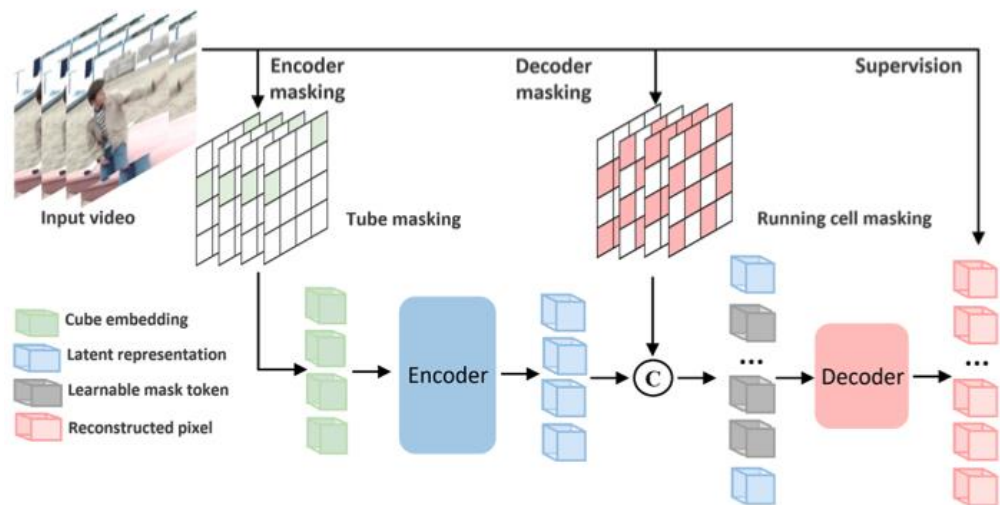
output 98%



VideoMAE v2

VideoMAE V2: Scaling Video Masked Autoencoders with Dual Masking

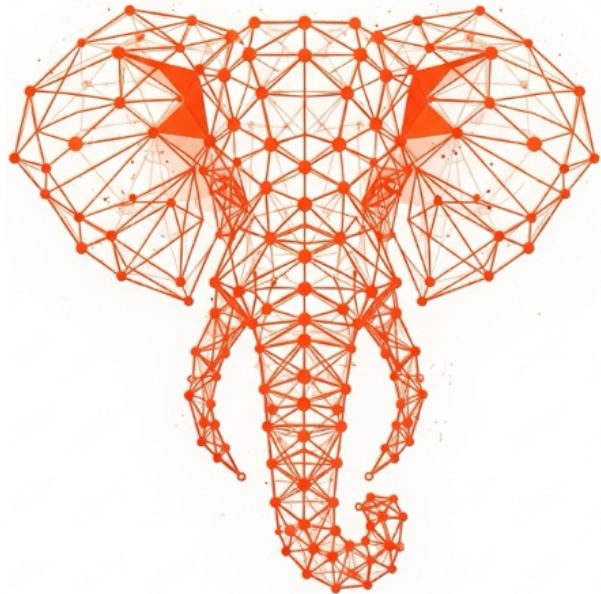
Limin Wang^{1,2,*} Bingkun Huang^{1,2,*} Zhiyu Zhao^{1,2} Zhan Tong¹
Yinan He² Yi Wang² Yali Wang^{3,2} Yu Qiao^{2,3}
¹ State Key Laboratory for Novel Software Technology, Nanjing University, China
² Shanghai AI Lab, China ³ Shenzhen Institute of Advanced Technology, CAS, China



- Trains on more data
- Introduces dual masking – adding masking to the decoder – to reduce computation
- Refines training techniques
- Trains a ViT with 1Billion parameters
- Reported results on the left are on Kinetics-400

method	pre-train data	data size	epoch	ViT-B	ViT-L	ViT-H	ViT-g
MAE-ST [28]	Kinetics400	0.24M	1600	81.3	84.8	85.1	-
MAE-ST [28]	IG-uncurated	1M	1600	-	84.4	-	-
VideoMAE V1 [90]	Kinetics400	0.24M	1600	81.5	85.2	86.6	-
VideoMAE V2	UnlabeledHybrid	1.35M	1200	81.5 (77.0)	85.4 (81.3)	86.9 (83.2)	87.2 (83.9)
$\Delta Acc.$ with V1	-	-	-	+ 0%	+ 0.2%	+ 0.3%	-

Wang, L., Huang, B., Zhao, Z., Tong, Z., He, Y., Wang, Y., ... & Qiao, Y. (2023). Videomae v2: Scaling video masked autoencoders with dual masking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14549-14560).



Uni
ct **DEEP LEARNING**
ADVANCED MODELS AND METHODS

Deep Learning

Advanced Models & Methods

Prof. Antonino Furnari (antonino.furnari@unict.it)

Corso di Laurea Magistrale in Informatica

Dip. di Matematica e Informatica

Università di Catania

The End